# THE EXTREMAL HARD EDGE: MINIMAL LEAST SINGULAR VALUES OF RANDOM SUBMATRICES, WITH UNIVERSALITY, LARGE DEVIATIONS, AND PHASE-RETRIEVAL STABILITY

YITZCHAK SHMALO

ABSTRACT. For a tall random matrix $A \in \mathbb{F}^{N\times m}$ we study the smallest least singular value over all of its square row submatrices, $M_m = \min_{|T|=m} \sigma_{\min}(A_T)$, and its generalizations. Building on the Gaussian theorem that fixes the exponential base $M_m = c_\gamma^{-m/d_{\mathbb{F}}+o_{\mathbb{P}}(m)}$ when $N/m \to \gamma > 1$, we isolate the mechanism — the competition between the combinatorial entropy of the submatrix family and the small-ball exponent of a single least singular value — as an abstract first/second moment principle. This gives a rigorous distribution-free lower side, including finite-entropy moving-aspect and summable-budget almost-sure forms, whenever a one-matrix least-singular-value tail is available. This covers real and complex iid entries with bounded or subexponentially growing scalar density bounds, entrywise density profiles, independent rows with directional anti-concentration, including fixed-excess lower sides, real isotropic log-concave rows on the lower side, and smoothed deterministic frames, including row-dependent smoothing budgets, spherical-direction ensembles with controlled radial weights, together with row- and column-normalized Gaussian variants, and deterministic Gram/sample-covariance transfers. The matching non-Gaussian upper side, for iid bounded-density entries, is reduced to a single decorrelation estimate for random normals and then established through an elementary off-support resolvent estimate, proved self-contained by a leave-one-out argument (Hanson–Wright and the Marčenko–Pastur fixed point; the anisotropic local law gives an alternative route), with full error control. The lower-side estimates have summable failure probabilities and hence yield almost-sure lower envelopes on any common coupling, while Gaussian row-count Poissonization and square row/right-factor conditioning are handled by monotone and pathwise squeezes. We also record fixed-excess rectangular and two-sided templates, correct the scale for proportional rectangular excess, prove the fixed-$k$ Gaussian higher-singular-value lower side, its fixed-excess extension, the quaternion fixed-excess fixed-$k$ lower-side analogue, and their Gram-eigenvalue transfers with finite-entropy budgets, extend the subexponential deterministic conditioning transfer to these fixed-excess fixed-$k$ lower sides, record the endpoint aspect asymptotics of the entropy base, isolate the atomic first-moment obstruction without claiming a high-probability discrete dichotomy, prove a deterministic finite-alphabet projective-pair obstruction whose sign case gives failure of critical real sign frames, and derive the corresponding lower-side Balan–Wang statement for continuous critical real frames, including normalized Gaussian measurements and diagonal coordinate preconditionings. The final section lists open problems on universality, fluctuations, atomic full spark, complex phase retrieval, and concentration for the extremal stability constant.

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*Date*: June 29, 2026.

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## 1. Introduction

Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, let $d_{\mathbb{F}} = \dim_{\mathbb{R}} \mathbb{F}$, and let

$$h(\gamma) = \gamma \log \gamma - (\gamma - 1) \log(\gamma - 1), \qquad c_\gamma = e^{h(\gamma)} = \frac{\gamma^\gamma}{(\gamma - 1)^{\gamma - 1}}.$$

Here $\gamma$ denotes the row aspect ratio $N_m/m$, and throughout this paper $T$ denotes a selected row index set. The source theorem of [15] says that if $A_m \in \mathbb{F}^{N_m \times m}$ has independent standard Gaussian entries and $N_m/m \to \gamma > 1$, then

$$M_m^{\mathbb{F}}(A_m) := \min_{\substack{T \subset [N_m] \\ |T| = m}} \sigma_{\min}(A_{m,T}) = c_\gamma^{-m/d_{\mathbb{F}} + o_{\mathbb{P}}(m)}.$$

Equivalently,

$$\frac{1}{m} \log M_m^{\mathbb{F}}(A_m) \xrightarrow{\mathbb{P}} -\frac{h(\gamma)}{d_{\mathbb{F}}}.$$

For $N_m = \gamma m + O(1)$ the same paper gives an $O_\varepsilon(m^{-1})$ probability rate for deviations of $(1/m) \log M_m^{\mathbb{F}}$ from this limit.

The phase-retrieval application is the real critical case $N = 2m - 1$. For a full-spark matrix $A \in \mathbb{R}^{(2m-1) \times m}$, the Balan–Wang stability parameter [1, 2, 4] is

$$\omega(A) = \min_{\substack{T \subset [2m-1] \\ |T| = m}} \sigma_{\min}(A_T).$$

Thus [15] proves

$$\frac{1}{m} \log \omega(A_m) \xrightarrow{\mathbb{P}} -\log 4, \qquad \omega(A_m) = 4^{-m + o_{\mathbb{P}}(m)}$$

for iid real Gaussian measurements. This answers Randomstrasse Open Problem 21 at the exponential-base level: the problem asks for the value of $\omega(A)$ and the exponential base in the Balan–Wang conjecture for iid Gaussian measurements [3]. The same base remains true after normalizing every Gaussian measurement vector to unit Euclidean norm: the random row scalings have subexponential extrema over $O(m)$ rows, so they do not affect the $m^{-1}$ log scale. The same pathwise

transfer also covers positive diagonal column preconditioning with subexponential logarithmic extrema, including column-normalized Gaussian matrices; in the phase-retrieval interpretation this is a coordinate preconditioning of the unknown signal rather than measurement-vector normalization.

The continuous assumption is essential for the full-spark reduction. In the opposite atomic direction, Proposition 4.3 gives a deterministic projective-pair obstruction for finite alphabets, and Proposition 4.4 shows in particular that every real sign matrix with $2m-1$ rows and $m \geq 2$ columns has a singular $m$-row submatrix. Thus critical Rademacher frames fail the real phase-retrieval injectivity condition deterministically, not merely with high probability.

The present note separates the mechanism from the Gaussian input. The mechanism is the competition

$$\text{number of tested submatrices} \quad \text{versus} \quad \text{small-ball exponent of one least singular value.}$$

If $|\mathcal{T}_m| = \exp((H + o(1))m)$ and a typical variable $Y_T$ has $\mathbb{P}\{Y_T \leq t\} \approx t^a$ at exponential scales, then the predicted extremal scale is

$$\min_{T \in \mathcal{T}_m} Y_T = \exp\{-(H/a + o_{\mathbb{P}}(1))m\}.$$

The lower bound is usually a union bound. The upper bound is the real work: one must prove that enough of the events $\{Y_T \leq t\}$ decorrelate. The Gaussian paper proves that decorrelation for square row submatrices by a second-moment argument. Below we state the abstract principle, prove the distribution-free lower side under explicit one-matrix small-ball hypotheses, including density-rate and smoothed deterministic-frame variants, and reduce the non-Gaussian second-moment step to a single decorrelation estimate for random normals, which we then establish for all iid bounded-density entries through an elementary off-support resolvent estimate (Appendix B), proved self-contained by a leave-one-out argument (Hanson–Wright and the Marčenko–Pastur fixed point), with the anisotropic local law an optional citation and full error control. Because the lower-side failure probabilities are exponentially summable once an $\varepsilon$-slack is fixed, the same union-bound estimates also give almost-sure lower envelopes by the first Borel–Cantelli lemma. This is only a lower-envelope statement; matching almost-sure exact laws would require summable upper-side or pair-correlation estimates. At the finite-budget level, a chosen summable budget $\sum_m e^{-u_m} < \infty$ gives an eventual almost-sure lower bound at the full displayed threshold, including the polynomial hard-edge factors and the $u_m$-slack. The Gaussian exact base is also stable under Poissonization or other random row-count models with $N_m/m \to \gamma > 1$, by a monotone squeeze between deterministic row counts. On the square row-submatrix scale, the same exact base is stable under positive real prefix row weights and one global all-column right factor with subexponential absolute singular-value scale; positive column diagonals and full-prefix Gaussian column normalization are special cases. For independent rows with directional anti-concentration, the lower-side method also extends from square row submatrices to fixed-excess tall submatrices: a cofactor row-block certificate replaces the square negative-second-moment identity and exposes the $r+1$ scalar constraints responsible for the fixed-excess exponent.

No Toeplitz, Hankel, circulant, or structured-symbol matrices are used here. All extensions stay in the row-submatrix and random-frame setting of the source paper.

## 2. The abstract extremal small-ball principle

For each $m$, let $\mathcal{T}_m$ be a finite index set and let $(Y_{m,T})_{T \in \mathcal{T}_m}$ be nonnegative random variables. Put

$$Y_m^* = \min_{T \in \mathcal{T}_m} Y_{m,T}.$$

**Definition 2.1** (Exponential-scale small-ball data)**.** Fix $H > 0$ and $a > 0$. We say that the family has upper small-ball exponent $a$ and entropy $H$ if

$$\log |\mathcal{T}_m| \leq (H + o(1))m$$

and, for every fixed $s > 0$,

$$\sup_{T \in \mathcal{T}_m} \mathbb{P}\{Y_{m,T} \le e^{-sm}\} \le \exp\{-(as + o(1))m\}.$$

It has a second-moment lower small-ball principle below $H/a$ if, for every fixed $s < H/a$, the counting variable

$$Z_m(s) = \sum_{T \in \mathcal{T}_m} \mathbf{1}_{\{Y_{m,T} \le e^{-sm}\}}$$

satisfies

$$\mathbb{E}Z_m(s) \to \infty, \qquad \frac{\operatorname{Var} Z_m(s)}{(\mathbb{E}Z_m(s))^2} \to 0.$$

**Theorem 2.2** (Extremal small-ball principle)**.** *Assume the upper small-ball exponent and entropy condition above. Then, for every $\varepsilon > 0$,*

$$\mathbb{P}\left\{Y_m^* \le \exp\left[-\left(\frac{H}{a} + \varepsilon\right) m\right]\right\} \longrightarrow 0.$$

*If, in addition, the second-moment lower small-ball principle holds below $H/a$, then*

$$\frac{1}{m} \log Y_m^* \xrightarrow{\mathbb{P}} -\frac{H}{a}.$$

*Proof.* Let $s > H/a$. By the union bound,

$$\mathbb{P}\{Y_m^* \le e^{-sm}\} \le |\mathcal{T}_m| \sup_{T \in \mathcal{T}_m} \mathbb{P}\{Y_{m,T} \le e^{-sm}\} \le \exp\{(H - as + o(1))m\},$$

which tends to zero. This proves the lower bound on $Y_m^*$.

Let $s < H/a$. If the second-moment hypothesis holds, then Chebyshev gives

$$\mathbb{P}\{Z_m(s) = 0\} \le \frac{\operatorname{Var} Z_m(s)}{(\mathbb{E}Z_m(s))^2} \to 0.$$

Thus, with probability tending to one, some $T$ has $Y_{m,T} \le e^{-sm}$. This says

$$\mathbb{P}\left\{Y_m^* \le \exp\left[-\left(\frac{H}{a} - \varepsilon\right) m\right]\right\} \to 1$$

after taking $s = H/a - \varepsilon$. The two estimates are exactly convergence in probability of $m^{-1} \log Y_m^*$ to $-H/a$. □

**Corollary 2.3** (Polynomial small-ball version)**.** *Suppose*

$$|\mathcal{T}_m| \le \exp((H + o(1))m)$$

*and for some constants $C, B, a > 0$,*

$$\sup_T \mathbb{P}\{Y_{m,T} \le t\} \le C m^B t^a, \qquad 0 < t \le 1.$$

*Then for every $\varepsilon > 0$,*

$$\mathbb{P}\left\{Y_m^* \le \exp\left[-\left(\frac{H}{a} + \varepsilon\right) m\right]\right\} \to 0.$$

*If the matching second-moment condition holds below $H/a$, then $(1/m) \log Y_m^* \to -H/a$ in probability.*

**Corollary 2.4** (Finite-budget lower side)**.** *Let $\mathcal{T}_m$ be finite and suppose that, for some $C_m \ge 1$ and $a_m > 0$,*

$$\sup_{T \in \mathcal{T}_m} \mathbb{P}\{Y_{m,T} \le t\} \le C_m t^{a_m}, \qquad 0 < t \le 1.$$

*Then, for every $u_m > 0$,*

$$\mathbb{P}\left\{Y_m^* \le \exp\left[-\frac{\log |\mathcal{T}_m| + \log C_m + u_m}{a_m}\right]\right\} \le e^{-u_m}.$$

*Proof.* Put $t_m = \exp[-(\log|\mathcal{T}_m| + \log C_m + u_m)/a_m]$. Since $|\mathcal{T}_m|, C_m \geq 1$, one has $0 < t_m \leq 1$. The union bound gives

$$\mathbb{P}\{Y_m^* \leq t_m\} \leq |\mathcal{T}_m| C_m t_m^{a_m} = e^{-u_m}.$$

□

**Corollary 2.5** (Nonuniform finite-budget lower side)**.** *Let $\mathcal{T}_m$ be nonempty and finite. Suppose that, for deterministic numbers $a_m > 0$ and $C_{m,T} \geq 0$,*

$$\mathbb{P}\{Y_{m,T} \leq t\} \leq C_{m,T} t^{a_m}, \qquad 0 < t \leq 1, \quad T \in \mathcal{T}_m.$$

*If $B_m \geq 1 \vee \sum_{T\in\mathcal{T}_m} C_{m,T}$, then, for every $u_m > 0$,*

$$\mathbb{P}\left\{Y_m^* \leq \exp\left[-\frac{\log B_m + u_m}{a_m}\right]\right\} \leq e^{-u_m}.$$

*Proof.* The displayed threshold is at most one. Hence the union bound gives

$$\mathbb{P}\{Y_m^* \leq t\} \leq \sum_{T\in\mathcal{T}_m} C_{m,T} t^{a_m} \leq B_m t^{a_m}, \qquad 0 < t \leq 1.$$

Substitute $t = \exp[-(\log B_m + u_m)/a_m]$. □

**Corollary 2.6** (Almost-sure nonuniform finite-budget lower side)**.** *Assume that the random variables for all $m$ are realized on a common probability space. Suppose that the index sets $\mathcal{T}_m$ are nonempty, finite, and deterministic, and that $a_m > 0$, $u_m > 0$, $C_{m,T} \geq 0$, and $B_m \geq 1 \vee \sum_{T\in\mathcal{T}_m} C_{m,T}$ are deterministic. If*

$$\mathbb{P}\{Y_{m,T} \leq t\} \leq C_{m,T} t^{a_m}, \qquad 0 < t \leq 1, \quad T \in \mathcal{T}_m,$$

*and*

$$\sum_{m=1}^{\infty} e^{-u_m} < \infty,$$

*then, almost surely, for all sufficiently large $m$,*

$$Y_m^* > \exp\left[-\frac{\log B_m + u_m}{a_m}\right].$$

*Proof.* Apply Corollary 2.5 at the displayed threshold and then use the first Borel–Cantelli lemma. No independence between different values of $m$, or between the events inside one $\mathcal{T}_m$, is used. □

**Corollary 2.7** (Almost-sure finite-budget lower side)**.** *Assume that the random variables for all $m$ are realized on a common probability space. Suppose that the index sets $\mathcal{T}_m$ are nonempty, finite, and deterministic, and that the numbers $C_m \geq 1$, $a_m > 0$, and $u_m > 0$ are deterministic. If*

$$\sup_{T\in\mathcal{T}_m} \mathbb{P}\{Y_{m,T} \leq t\} \leq C_m t^{a_m}, \qquad 0 < t \leq 1,$$

*and*

$$\sum_{m=1}^{\infty} e^{-u_m} < \infty,$$

*then, almost surely, for all sufficiently large $m$,*

$$Y_m^* > \exp\left[-\frac{\log|\mathcal{T}_m| + \log C_m + u_m}{a_m}\right].$$

*Proof.* Put

$$t_m = \exp\left[-\frac{\log|\mathcal{T}_m| + \log C_m + u_m}{a_m}\right].$$

Then $0 < t_m \leq 1$, and the proof of Corollary 2.4 gives

$$\mathbb{P}\{Y_m^* \leq t_m\} \leq e^{-u_m}.$$

The first Borel–Cantelli lemma gives $\mathbb{P}\{Y_m^* \le t_m \text{ infinitely often}\} = 0$. Hence $Y_m^* > t_m$ eventually almost surely. No independence between different values of $m$, or between the events inside one $\mathcal{T}_m$, is used. □

**Corollary 2.8** (Almost-sure lower envelope)**.** *Assume that the random variables for all $m$ are realized on a common probability space. Suppose there is a constant $c \ge 0$ such that, for every $\varepsilon > 0$,*

$$\sum_{m=1}^{\infty} |\mathcal{T}_m| \sup_{T \in \mathcal{T}_m} \mathbb{P}\{Y_{m,T} \le e^{-(c+\varepsilon)m}\} < \infty.$$

*Then*

$$\liminf_{m\to\infty} \frac{1}{m} \log Y_m^* \ge -c \qquad \textit{almost surely.}$$

*In particular, the upper small-ball exponent and entropy hypothesis in Theorem 2.2 imply the almost-sure lower envelope with $c = H/a$.*

*Proof.* For fixed $\varepsilon > 0$, the event

$$E_m(\varepsilon) = \{Y_m^* \le e^{-(c+\varepsilon)m}\}$$

has probability bounded by the summand in the hypothesis. The first Borel–Cantelli lemma gives $\mathbb{P}\{E_m(\varepsilon) \text{ infinitely often}\} = 0$. Intersecting the resulting probability-one events over $\varepsilon = 1, 1/2, 1/3, \dots$ gives the displayed liminf. No independence between different $m$ is used.

For the last assertion, take $s = H/a + \varepsilon$. The union bound in Theorem 2.2 gives

$$\mathbb{P}\{Y_m^* \le e^{-sm}\} \le \exp\{(-a\varepsilon + o(1))m\},$$

which is summable. □

The Gaussian source theorem is the case

$$\mathcal{T}_m = \{T \subset [N_m] : |T| = m\}, \qquad H = h(\gamma), \qquad a = d_{\mathbb{F}},$$

with the second-moment condition proved in [15].

## 3. Universality across entry distributions

The first robust part of the Gaussian theorem is the lower bound on how small the extremal singular value can be. It only uses a one-matrix lower-tail estimate.

**Definition 3.1** (Exponential least-singular-value tail)**.** Let $A_m \in \mathbb{F}^{N_m \times m}$, $N_m/m \to \gamma > 1$. We say that the square row submatrices have exponent $a > 0$ if, for every fixed $s > 0$,

$$\sup_{\substack{T \subset [N_m] \\ |T| = m}} \mathbb{P}\{\sigma_{\min}(A_{m,T}) \le e^{-sm}\} \le \exp\{-(as + o(1))m\}.$$

**Theorem 3.2** (Universal lower side)**.** *Assume $N_m/m \to \gamma > 1$ and the exponent-$a$ tail condition in Definition 3.1. Then for every $\varepsilon > 0$,*

$$\mathbb{P}\left\{ \min_{\substack{T \subset [N_m] \\ |T| = m}} \sigma_{\min}(A_{m,T}) \le \exp\left[ -\left( \frac{h(\gamma)}{a} + \varepsilon \right) m \right] \right\} \to 0.$$

*Proof.* Use Theorem 2.2 with $\mathcal{T}_m = \{T \subset [N_m] : |T| = m\}$. Stirling's formula gives

$$\frac{1}{m} \log \binom{N_m}{m} \to h(\gamma).$$

□

**Proposition 3.3** (Real and complex bounded-density one-matrix tail)**.** *Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, let $d_{\mathbb{F}} = \dim_{\mathbb{R}} \mathbb{F}$, and let $G \in \mathbb{F}^{m\times m}$ have independent entries whose $d_{\mathbb{F}}$-dimensional Lebesgue densities are bounded by $K$. Then there is a constant $C_{\mathbb{F},K}$ such that, for $0 < t \le 1$,*

$$\mathbb{P}\{\sigma_{\min}(G) \le t\} \le C_{\mathbb{F},K} m^{d_{\mathbb{F}}+1} t^{d_{\mathbb{F}}}.$$

*Proof.* Let $R_i$ be the rows of $G$, and let

$$d_i = \operatorname{dist}(R_i, \operatorname{span}\{R_j : j \ne i\}).$$

The negative second-moment identity, over $\mathbb{F}$, gives

$$\sum_{k=1}^{m} \sigma_k(G)^{-2} = \sum_{i=1}^{m} d_i^{-2}.$$

Thus $\sigma_{\min}(G) \le t$ implies $d_i \le \sqrt{m}\, t$ for some $i$. Condition on all rows except $R_i$. If the other rows span a hyperplane with unit normal $u$, then $d_i = |\langle R_i, u\rangle|$. Some coordinate of the unit normal has modulus at least $m^{-1/2}$. Conditioning on all other coordinates of $R_i$, the remaining scalar is an affine image of one entry by the factor $u_j$. Its density on $\mathbb{F} \simeq \mathbb{R}^{d_{\mathbb{F}}}$ is bounded by $K|u_j|^{-d_{\mathbb{F}}} \le K m^{d_{\mathbb{F}}/2}$. Therefore, with $\tau = \sqrt{m}\, t$,

$$\mathbb{P}\{d_i \le \sqrt{m}\, t \mid R_j,\ j \ne i\} \le K m^{d_{\mathbb{F}}/2} \operatorname{vol}_{d_{\mathbb{F}}}(B_{\mathbb{F}}(0,\tau)) \le C_{\mathbb{F},K} m^{d_{\mathbb{F}}} t^{d_{\mathbb{F}}}.$$

Taking a union bound over $i$ gives the claim. This is Proposition 6.1 of [15] in the real case; the complex case is the same argument with real codimension two. □

**Corollary 3.4** (Bounded-density entries: lower side)**.** *Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. Let $A_m \in \mathbb{F}^{N_m\times m}$, $N_m/m \to \gamma > 1$, have independent entries whose $d_{\mathbb{F}}$-dimensional Lebesgue densities are uniformly bounded by $K$. Then, for every $\varepsilon > 0$,*

$$\mathbb{P}\left\{\min_{|T|=m} \sigma_{\min}(A_{m,T}) \le \exp[-(h(\gamma)/d_{\mathbb{F}} + \varepsilon)m]\right\} \to 0.$$

*Proof.* Combine Proposition 3.3 with Corollary 2.3. □

**Corollary 3.5** (Finite-entropy square lower side)**.** *Let $N_m \ge m$ be arbitrary. Let $A_m \in \mathbb{F}^{N_m\times m}$, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, have independent entries whose $d_{\mathbb{F}}$-dimensional Lebesgue densities are bounded by $K_m \ge 1$, uniformly in their position. Put*

$$L_m = \log \binom{N_m}{m}.$$

*Then, for every $\varepsilon > 0$,*

$$\mathbb{P}\left\{\min_{|T|=m} \sigma_{\min}(A_{m,T}) \le \exp\left[-\left(\frac{L_m + \log K_m}{d_{\mathbb{F}} m} + \varepsilon\right) m\right]\right\} \longrightarrow 0.$$

*More quantitatively, for every $u_m > 0$,*

$$\mathbb{P}\left\{\min_{|T|=m} \sigma_{\min}(A_{m,T}) \le \exp\left[-\frac{L_m + \log K_m + (d_{\mathbb{F}}+1)\log m + O_{\mathbb{F}}(1) + u_m}{d_{\mathbb{F}}}\right]\right\} \le e^{-u_m}.$$

*Moreover, on any common coupling, if $\sum_m e^{-u_m} < \infty$, then the displayed bad event occurs only finitely often almost surely, with the $O_{\mathbb{F}}(1)$ interpreted as one fixed constant from the one-matrix tail bound.*

*Proof.* The proof of Proposition 3.3 gives the explicit estimate

$$\mathbb{P}\{\sigma_{\min}(G) \le t\} \le C_{\mathbb{F}} K_m m^{d_{\mathbb{F}}+1} t^{d_{\mathbb{F}}}, \qquad 0 < t \le 1,$$

for each selected square row submatrix. Apply Corollary 2.4 with

$$|\mathcal{T}_m| = \binom{N_m}{m}, \qquad C_m = C_{\mathbb{F}} K_m m^{d_{\mathbb{F}}+1}, \qquad a_m = d_{\mathbb{F}}.$$

The first displayed convergence follows by taking $u_m = d_{\mathbb{F}} \varepsilon m$. □

**Corollary 3.6** (Row-wise heterogeneous-density square budget)**.** *Let $N_m \geq m$ be arbitrary. Let $A_m \in \mathbb{F}^{N_m \times m}$, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, have independent entries. Assume that for each row $i \in [N_m]$ there is a deterministic number $K_{m,i} \geq 1$ such that every entry in row $i$ has $d_{\mathbb{F}}$-dimensional Lebesgue density bounded by $K_{m,i}$. Put*

$$\Lambda_m^{\mathrm{row}} = \log \binom{N_m - 1}{m-1} + \log \left( \sum_{i=1}^{N_m} K_{m,i} \right).$$

*Then, for every $\varepsilon > 0$,*

$$\mathbb{P}\left\{ \min_{|T|=m} \sigma_{\min}(A_{m,T}) \leq \exp\left[ -\left( \frac{\Lambda_m^{\mathrm{row}}}{d_{\mathbb{F}} m} + \varepsilon \right) m \right] \right\} \longrightarrow 0.$$

*More quantitatively, for every $u_m > 0$,*

$$\mathbb{P}\left\{ \min_{|T|=m} \sigma_{\min}(A_{m,T}) \leq \exp\left[ -\frac{\Lambda_m^{\mathrm{row}} + d_{\mathbb{F}} \log m + O_{\mathbb{F}}(1) + u_m}{d_{\mathbb{F}}} \right] \right\} \leq e^{-u_m}.$$

*Moreover, on any common coupling, if $\sum_m e^{-u_m} < \infty$, then the displayed bad event occurs only finitely often almost surely, with the $O_{\mathbb{F}}(1)$ interpreted as one fixed constant.*

*Proof.* Fix a row set $T$ with $|T| = m$. For $i \in T$, let

$$d_i = \operatorname{dist}(R_i, \operatorname{span}\{R_\ell : \ell \in T,\ \ell \neq i\}),$$

where $R_i$ is the $i$-th row of $A_m$. The negative second-moment identity gives that $\sigma_{\min}(A_{m,T}) \leq t$ implies $d_i \leq \sqrt{m}\, t$ for some $i \in T$. Conditioning on all other rows and on all but one coordinate of $R_i$, choose a coordinate of the resulting unit normal with modulus at least $m^{-1/2}$. The remaining scalar density is bounded by $K_{m,i} m^{d_{\mathbb{F}}/2}$, so for $0 < t \leq 1$,

$$\mathbb{P}\{\sigma_{\min}(A_{m,T}) \leq t\} \leq C_{\mathbb{F}} m^{d_{\mathbb{F}}} t^{d_{\mathbb{F}}} \sum_{i \in T} K_{m,i}.$$

Hence the nonuniform budget over all square row sets is

$$C_{\mathbb{F}} m^{d_{\mathbb{F}}} \sum_{|T|=m} \sum_{i \in T} K_{m,i} = C_{\mathbb{F}} m^{d_{\mathbb{F}}} \binom{N_m - 1}{m-1} \sum_{i=1}^{N_m} K_{m,i}.$$

Apply Corollary 2.5. The convergence-in-probability statement follows by taking $u_m = d_{\mathbb{F}} \varepsilon m$, since the polynomial and constant terms are $o(m)$. □

**Corollary 3.7** (Independent rows with directional anti-concentration)**.** *Let $N_m \geq m$, let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, let $d_{\mathbb{F}} = \dim_{\mathbb{R}} \mathbb{F}$, and let $A_m \in \mathbb{F}^{N_m \times m}$ have independent rows $R_{m,i}$. Assume that there are deterministic numbers $L_{m,i} \geq 1$ such that, for every unit vector $u \in \mathbb{F}^m$, every $z \in \mathbb{F}$, and every $0 < r \leq 1$,*

$$\mathbb{P}\{|\langle R_{m,i}, u \rangle - z| \leq r\} \leq L_{m,i} r^{d_{\mathbb{F}}}.$$

*Put*

$$\Lambda_m^{\mathrm{dir}} = \log \binom{N_m - 1}{m-1} + \log \left( \sum_{i=1}^{N_m} L_{m,i} \right).$$

*Then, for every $u_m > 0$,*

$$\mathbb{P}\left\{\min_{|T|=m} \sigma_{\min}(A_{m,T}) \le \exp\left[-\frac{\Lambda_m^{\mathrm{dir}} + \frac{d_{\mathbb{F}}}{2}\log m + u_m}{d_{\mathbb{F}}}\right]\right\} \le e^{-u_m}.$$

*Moreover, on any common coupling, if $\sum_m e^{-u_m} < \infty$, then the displayed bad event occurs only finitely often almost surely. If $N_m/m \to \gamma > 1$ and*

$$\kappa_{\mathrm{dir}} = \limsup_{m\to\infty} \frac{1}{m}\log\left(\sum_{i=1}^{N_m} L_{m,i}\right) < \infty,$$

*then for every $\varepsilon > 0$,*

$$\mathbb{P}\left\{\min_{|T|=m} \sigma_{\min}(A_{m,T}) \le \exp\left[-\left(\frac{h(\gamma) + \kappa_{\mathrm{dir}}}{d_{\mathbb{F}}} + \varepsilon\right) m\right]\right\} \to 0.$$

*Proof.* Fix $T \subset [N_m]$ with $|T| = m$. For $i \in T$, let

$$d_i = \operatorname{dist}(R_{m,i}, \operatorname{span}\{R_{m,\ell} : \ell \in T,\ \ell \ne i\}).$$

The negative second-moment identity gives that $\sigma_{\min}(A_{m,T}) \le t$ implies $d_i \le \sqrt{m}\, t$ for some $i \in T$. Condition on all rows except $R_{m,i}$, and choose a measurable unit normal $u$ to the span of the other selected rows. Row independence makes this normal independent of $R_{m,i}$. If $r = \sqrt{m}\, t \le 1$, the directional hypothesis gives

$$\mathbb{P}\{d_i \le \sqrt{m}\, t \mid R_{m,\ell},\ \ell \ne i\} \le L_{m,i} m^{d_{\mathbb{F}}/2} t^{d_{\mathbb{F}}}.$$

If $r > 1$, the same bound is trivial because $L_{m,i} \ge 1$. Thus, for $0 < t \le 1$,

$$\mathbb{P}\{\sigma_{\min}(A_{m,T}) \le t\} \le m^{d_{\mathbb{F}}/2} t^{d_{\mathbb{F}}} \sum_{i\in T} L_{m,i}.$$

Summing this nonuniform budget over all row sets gives

$$m^{d_{\mathbb{F}}/2}\binom{N_m - 1}{m-1}\sum_{i=1}^{N_m} L_{m,i}.$$

Apply Corollaries 2.5 and 2.6. The fixed-aspect statement follows by taking $u_m = d_{\mathbb{F}}\varepsilon m$ and absorbing the $o(m)$ polynomial term. □

**Corollary 3.8** (Directional densities and real log-concave rows)**.** *If each projection $\langle R_{m,i}, u\rangle$, $u \in \mathbb{F}^m$ unit, has $d_{\mathbb{F}}$-dimensional Lebesgue density bounded by $K_{m,i}$, uniformly in $u$, then Corollary 3.7 applies with $L_{m,i} = v_{\mathbb{F}} K_{m,i}$, where $v_{\mathbb{R}} = 2$ and $v_{\mathbb{C}} = \pi$. In particular, iid real isotropic log-concave rows satisfy*

$$\mathbb{P}\left\{\min_{|T|=m} \sigma_{\min}(A_{m,T}) \le e^{-(h(\gamma)+\varepsilon)m}\right\} \to 0, \qquad N_m/m \to \gamma > 1,$$

*because their one-dimensional marginals have universally bounded densities [6]. The matching exact law for log-concave rows still requires a pair-correlation theorem and is not claimed.*

**Corollary 3.9** (Entrywise density-profile square budget)**.** *Let $N_m \ge m$, and let $A_m \in \mathbb{F}^{N_m\times m}$, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, have independent entries. Assume that entry $(i,j)$ has $d_{\mathbb{F}}$-dimensional density bounded by a deterministic number $K_{m,i,j} \ge 1$, and put*

$$K_{m,i}^* = \max_{1\le j\le m} K_{m,i,j}, \qquad \Lambda_m^{\mathrm{ent}} = \log\binom{N_m - 1}{m-1} + \log\left(\sum_{i=1}^{N_m} K_{m,i}^*\right).$$

*Then, for every $u_m > 0$,*

$$\mathbb{P}\left\{\min_{|T|=m} \sigma_{\min}(A_{m,T}) \le \exp\left[-\frac{\Lambda_m^{\mathrm{ent}} + d_{\mathbb{F}} \log m + O_{\mathbb{F}}(1) + u_m}{d_{\mathbb{F}}}\right]\right\} \le e^{-u_m}.$$

*Moreover, on any common coupling, if $\sum_m e^{-u_m} < \infty$, then the displayed bad event occurs only finitely often almost surely, with the $O_{\mathbb{F}}(1)$ interpreted as one fixed constant. If $N_m/m \to \gamma > 1$ and*

$$\kappa_* = \limsup_{m\to\infty} \frac{1}{m} \log\left(\sum_{i=1}^{N_m} K_{m,i}^*\right) < \infty,$$

*then for every $\varepsilon > 0$,*

$$\mathbb{P}\left\{\min_{|T|=m} \sigma_{\min}(A_{m,T}) \le \exp\left[-\left(\frac{h(\gamma) + \kappa_*}{d_{\mathbb{F}}} + \varepsilon\right) m\right]\right\} \to 0.$$

*Proof.* For each row $i$, every entry density is bounded by $K_{m,i}^*$. Apply Corollary 3.6. The exponential-scale statement follows from $m^{-1} \log \binom{N_m - 1}{m-1} \to h(\gamma)$. □

**Theorem 3.10** (Density-rate lower side)**.** *Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, let $d_{\mathbb{F}} = \dim_{\mathbb{R}} \mathbb{F}$, and let $A_m \in \mathbb{F}^{N_m \times m}$, $N_m/m \to \gamma > 1$, have independent entries. Assume that every entry has $d_{\mathbb{F}}$-dimensional Lebesgue density bounded by a deterministic number $K_m \ge 1$, uniformly in its position, and put*

$$\kappa = \limsup_{m\to\infty} \frac{1}{m} \log K_m < \infty.$$

*Then, for every $\varepsilon > 0$,*

$$\mathbb{P}\left\{\min_{|T|=m} \sigma_{\min}(A_{m,T}) \le \exp\left[-\left(\frac{h(\gamma) + \kappa}{d_{\mathbb{F}}} + \varepsilon\right) m\right]\right\} \to 0.$$

*In particular, subexponential density growth $K_m = e^{o(m)}$ preserves the bounded-density lower-side base $h(\gamma)/d_{\mathbb{F}}$.*

*Proof.* The proof of Proposition 3.3 gives the explicit one-matrix bound

$$\mathbb{P}\{\sigma_{\min}(G) \le t\} \le C_{\mathbb{F}} K_m m^{d_{\mathbb{F}}+1} t^{d_{\mathbb{F}}}, \qquad 0 < t \le 1,$$

for every $m \times m$ matrix $G$ whose entries are independent and have densities bounded by $K_m$. Indeed, after all rows except one are fixed, the selected scalar coordinate has density at most $K_m |u_j|^{-d_{\mathbb{F}}} \le K_m m^{d_{\mathbb{F}}/2}$, and the $d_{\mathbb{F}}$-dimensional ball has volume $C_{\mathbb{F}}(\sqrt{m}\, t)^{d_{\mathbb{F}}}$.

Thus, with $M_m = \min_{|T|=m} \sigma_{\min}(A_{m,T})$,

$$\mathbb{P}\{M_m \le e^{-sm}\} \le \binom{N_m}{m} C_{\mathbb{F}} K_m m^{d_{\mathbb{F}}+1} e^{-d_{\mathbb{F}} sm}.$$

The logarithm divided by $m$ has limsup at most $h(\gamma) + \kappa - d_{\mathbb{F}} s$. This is negative when $s > (h(\gamma) + \kappa)/d_{\mathbb{F}}$, which gives the claim. □

**Corollary 3.11** (Smoothed deterministic frames: lower side)**.** *Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, $N_m/m \to \gamma > 1$, and let $F_m \in \mathbb{F}^{N_m \times m}$ be arbitrary deterministic matrices. Let $G_m$ have iid standard $\mathbb{F}$-Gaussian entries, let $\sigma_m > 0$, and put*

$$\eta = \limsup_{m\to\infty} \frac{1}{m} \log \sigma_m^{-1} < \infty.$$

*For $A_m = F_m + \sigma_m G_m$, every $\varepsilon > 0$ satisfies*

$$\mathbb{P}\left\{\min_{|T|=m} \sigma_{\min}((A_m)_T) \le \exp\left[-\left(\frac{h(\gamma)}{d_{\mathbb{F}}} + \eta + \varepsilon\right) m\right]\right\} \to 0.$$

*If $\sigma_m = e^{-o(m)}$, the Gaussian lower-side base $h(\gamma)/d_{\mathbb{F}}$ is therefore recovered for every deterministic background frame $F_m$.*

*Proof.* Translation by $F_m$ does not change density suprema. Scaling a standard $\mathbb{F}$-Gaussian scalar by $\sigma_m$ gives a $d_{\mathbb{F}}$-dimensional density bounded by $C_{\mathbb{F}}\sigma_m^{-d_{\mathbb{F}}}$. Theorem 3.10 applies with $\kappa \le d_{\mathbb{F}}\eta$. □

**Corollary 3.12** (Row-dependent smoothing budget)**.** *Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, $N_m/m \to \gamma > 1$, and let $F_m \in \mathbb{F}^{N_m \times m}$ be deterministic. Let $G_m$ have iid standard $\mathbb{F}$-Gaussian entries, let $\sigma_{m,i} > 0$ be deterministic row noise levels, and put*

$$A_m = F_m + D_m G_m, \qquad D_m = \operatorname{diag}(\sigma_{m,1}, \dots, \sigma_{m,N_m}).$$

*Define the row smoothing rate*

$$\eta_{\mathrm{row}} = \limsup_{m\to\infty} \frac{1}{d_{\mathbb{F}} m} \log \left( \sum_{i=1}^{N_m} \sigma_{m,i}^{-d_{\mathbb{F}}} \right) < \infty.$$

*Then, for every $\varepsilon > 0$,*

$$\mathbb{P}\left\{ \min_{|T|=m} \sigma_{\min}((A_m)_T) \le \exp\left[ -\left( \frac{h(\gamma)}{d_{\mathbb{F}}} + \eta_{\mathrm{row}} + \varepsilon \right) m \right] \right\} \to 0.$$

*More quantitatively, for every $u_m > 0$, with*

$$\Lambda_m^{\mathrm{sm,row}} = \log \binom{N_m - 1}{m-1} + \log \left( \sum_{i=1}^{N_m} \sigma_{m,i}^{-d_{\mathbb{F}}} \right),$$

$$\mathbb{P}\left\{ \min_{|T|=m} \sigma_{\min}((A_m)_T) \le \exp\left[ -\frac{\Lambda_m^{\mathrm{sm,row}} + d_{\mathbb{F}} \log m + O_{\mathbb{F}}(1) + u_m}{d_{\mathbb{F}}} \right] \right\} \le e^{-u_m}.$$

*Thus a small set of poorly smoothed rows is charged through the weighted row budget, not through the worst value of $\sigma_{m,i}$ alone. When $\sigma_{m,i} = \sigma_m$ for all $i$, this recovers Corollary 3.11.*

*Proof.* Translation by $F_m$ does not change density suprema. In row $i$, each entry of $\sigma_{m,i} G_m$ has $d_{\mathbb{F}}$-dimensional density bounded by $C_{\mathbb{F}}\sigma_{m,i}^{-d_{\mathbb{F}}}$. Corollary 3.6 therefore applies with $K_{m,i} = C_{\mathbb{F}}\sigma_{m,i}^{-d_{\mathbb{F}}}$. The constant $C_{\mathbb{F}}$ is absorbed into the $O_{\mathbb{F}}(1)$ term. Since

$$\frac{1}{m} \log \binom{N_m - 1}{m-1} \to h(\gamma)$$

and the polynomial term is $o(m)$, the convergence-in-probability statement follows by taking $u_m = d_{\mathbb{F}} \varepsilon m$ and then absorbing a fixed fraction of the slack. □

**Corollary 3.13** (Almost-sure square lower envelopes)**.** *Assume the matrices for all $m$ are realized on a common probability space. For the bounded-density ensemble of Corollary 3.4,*

$$\liminf_{m\to\infty} \frac{1}{m} \log \min_{|T|=m} \sigma_{\min}(A_{m,T}) \ge -\frac{h(\gamma)}{d_{\mathbb{F}}} \qquad \text{almost surely.}$$

*For the density-rate ensemble of Theorem 3.10,*

$$\liminf_{m\to\infty} \frac{1}{m} \log \min_{|T|=m} \sigma_{\min}(A_{m,T}) \ge -\frac{h(\gamma) + \kappa}{d_{\mathbb{F}}} \qquad \text{almost surely.}$$

*For the smoothed ensemble $F_m + \sigma_m G_m$ of Corollary 3.11,*

$$\liminf_{m\to\infty} \frac{1}{m} \log \min_{|T|=m} \sigma_{\min}((F_m + \sigma_m G_m)_T) \ge -\frac{h(\gamma)}{d_{\mathbb{F}}} - \eta \qquad \text{almost surely.}$$

*For the row-dependent smoothing ensemble of Corollary 3.12,*

$$\liminf_{m\to\infty} \frac{1}{m} \log \min_{|T|=m} \sigma_{\min}((F_m + D_m G_m)_T) \ge -\frac{h(\gamma)}{d_{\mathbb{F}}} - \eta_{\mathrm{row}} \qquad \text{almost surely.}$$

*Proof.* In the bounded-density and density-rate cases, the union-bound estimates in the preceding proofs are exponentially summable at every threshold larger than the displayed constant. Corollary 2.8 applies.

For smoothing, write

$$F_m + \sigma_m G_m = \sigma_m(\sigma_m^{-1} F_m + G_m).$$

The shifted matrix in parentheses has standard Gaussian density bounds independent of the shift, so the square bounded-density almost-sure envelope applies to it. Since eventually $\sigma_m^{-1} \le e^{(\eta+\delta)m}$ for every $\delta > 0$, multiplying by $\sigma_m$ shifts the logarithmic lower envelope by at most $-\eta$.

For row-dependent smoothing, use the finite-budget estimate in Corollary 3.12 with $u_m = d_{\mathbb{F}}\varepsilon m$. The definition of $\eta_{\mathrm{row}}$ makes the resulting bad-event probabilities summable at every threshold larger than $h(\gamma)/d_{\mathbb{F}} + \eta_{\mathrm{row}}$, and Borel–Cantelli gives the claim. □

**Theorem 3.14** (Random number of Gaussian rows)**.** *For each $m$, let $(R_{im})_{i\ge 1}$ be an infinite sequence of independent standard $\mathbb{F}^m$-Gaussian rows. For an integer $n \ge m$, put*

$$M_m(n) = \min_{\substack{T\subset[n]\\ |T|=m}} \sigma_{\min}((R_{im})_{i\in T}).$$

*Let $N_m$ be integer-valued random variables, possibly dependent on the row array, such that*

$$\frac{N_m}{m} \xrightarrow{\mathbb{P}} \gamma > 1.$$

*Extend $M_m(N_m)$ by any fixed positive value on the event $N_m < m$. This event has probability tending to zero, and*

$$\frac{1}{m}\log M_m(N_m) \xrightarrow{\mathbb{P}} -\frac{h(\gamma)}{d_{\mathbb{F}}}.$$

*Proof.* The map $n \mapsto M_m(n)$ is nonincreasing, since increasing the number of available rows only enlarges the family of tested submatrices. Fix

$$1 < \gamma_- < \gamma < \gamma_+.$$

With probability tending to one,

$$\lfloor \gamma_- m \rfloor \le N_m \le \lceil \gamma_+ m \rceil.$$

On this event,

$$M_m(\lceil \gamma_+ m \rceil) \le M_m(N_m) \le M_m(\lfloor \gamma_- m \rfloor).$$

The deterministic Gaussian theorem applied to the two bracketed row counts gives

$$\frac{1}{m}\log M_m(\lceil \gamma_+ m \rceil) \xrightarrow{\mathbb{P}} -\frac{h(\gamma_+)}{d_{\mathbb{F}}}, \qquad \frac{1}{m}\log M_m(\lfloor \gamma_- m \rfloor) \xrightarrow{\mathbb{P}} -\frac{h(\gamma_-)}{d_{\mathbb{F}}}.$$

Since $h$ is continuous on $(1,\infty)$, letting $\gamma_- \uparrow \gamma$ and $\gamma_+ \downarrow \gamma$ squeezes $m^{-1}\log M_m(N_m)$ to $-h(\gamma)/d_{\mathbb{F}}$. No independence between $N_m$ and the row array is used. □

**Corollary 3.15** (Random row count: bounded-density lower side)**.** *Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. For each $m$, let $(R_{im})_{i\ge 1}$ be independent $\mathbb{F}^m$-valued rows with independent scalar entries whose $d_{\mathbb{F}}$-dimensional Lebesgue densities are bounded by $K_m \ge 1$, uniformly in the row and column positions. Let $N_m/m \to \gamma > 1$ in probability, possibly with $N_m$ dependent on the row array, and put*

$$\kappa = \limsup_{m\to\infty} \frac{1}{m}\log K_m < \infty.$$

*Let $M_m(n)$ denote the minimum over $m$-row subsets of the first $n$ rows, with any fixed positive convention when $n < m$. Then, for every $\varepsilon > 0$,*

$$\mathbb{P}\left\{ M_m(N_m) \le \exp\left[ -\left( \frac{h(\gamma)+\kappa}{d_{\mathbb{F}}} + \varepsilon \right) m \right] \right\} \to 0.$$

*If the row arrays are realized on a common probability space and $N_m/m \to \gamma$ almost surely, then*

$$\liminf_{m\to\infty} \frac{1}{m} \log M_m(N_m) \geq -\frac{h(\gamma)+\kappa}{d_{\mathbb{F}}} \qquad \text{almost surely.}$$

*Proof.* Fix $\gamma_+ > \gamma$. With probability tending to one, $N_m \leq \lceil \gamma_+ m \rceil$. By monotonicity,

$$M_m(N_m) \geq M_m(\lceil \gamma_+ m \rceil)$$

on this event. Apply Theorem 3.10 to the deterministic row count $\lceil \gamma_+ m \rceil$. This gives the lower-side threshold $(h(\gamma_+)+\kappa)/d_{\mathbb{F}}$. Let $\gamma_+ \downarrow \gamma$.

For the almost-sure statement, $N_m \leq \lceil \gamma_+ m \rceil$ eventually almost surely for every fixed $\gamma_+ > \gamma$. The deterministic almost-sure lower envelope of Corollary 3.13, applied at $\gamma_+$, gives the same bound and then $\gamma_+ \downarrow \gamma$. □

**Corollary 3.16** (Poissonization and independent thinning)**.** *The conclusion of Theorem 3.14 applies when $N_m$ is independent of the Gaussian row values and one of the following conditions holds:*

1. *$N_m$ is Poisson with mean $\nu_m$, and $\nu_m/m \to \gamma > 1$;*
2. *$N_m$ is binomial $\mathrm{Bin}(n_m^{\mathrm{pool}}, p_m)$, with $n_m^{\mathrm{pool}} p_m/m \to \gamma > 1$;*
3. *$N_m = \sum_{i=1}^{n_m^{\mathrm{pool}}} \xi_{im}$, where the $\xi_{im}$ are independent Bernoulli retention indicators, independent of the row values, and $\nu_m = \sum_i \mathbb{E}\xi_{im}$ satisfies $\nu_m/m \to \gamma > 1$.*

*In the third case the retained rows, conditional on the retained count, are iid Gaussian. Value-dependent thinning is excluded: it can bias the retained row law and change the extremal statistic.*

*Proof.* In each displayed case $N_m/m \to \gamma$ in probability. For Poisson and binomial variables this is Chebyshev's inequality. In the final item, $\operatorname{Var} N_m \leq \nu_m = O(m)$, so Chebyshev again gives convergence in probability. Theorem 3.14 applies. For independent thinning, conditional on the retained count, the retained row vectors are iid Gaussian and exchangeable; their order is irrelevant because the statistic minimizes over all $m$-row subsets. □

Apart from the Gaussian random-count squeeze, the distribution-free estimates above are lower-side statements only. Matching non-Gaussian upper bounds, and hence exact extremal bases, still require the pair-correlation input below.

The matching upper side is subtler. For non-Gaussian entries it is not a formal consequence of one-matrix estimates, because the events indexed by different row sets are strongly dependent.

**Definition 3.17** (Pair-correlation condition for row submatrices)**.** For $s < h(\gamma)/d_{\mathbb{F}}$, put

$$Z_m(s) = \sum_{|T|=m} \mathbf{1}_{\{\sigma_{\min}(A_{m,T}) \leq e^{-sm}\}}.$$

The ensemble satisfies the extremal hard-edge pair-correlation condition if

$$\mathbb{E}Z_m(s) \to \infty, \qquad \frac{\operatorname{Var} Z_m(s)}{(\mathbb{E}Z_m(s))^2} \to 0$$

for every fixed $s < h(\gamma)/d_{\mathbb{F}}$.

**Theorem 3.18** (Conditional non-Gaussian universality)**.** *Let $A_m$ be a continuous entry ensemble over $\mathbb{F}$. Assume:*

1. *the square row submatrices have one-matrix exponent $d_{\mathbb{F}}$;*
2. *the corresponding lower small-ball first moment diverges for every $s < h(\gamma)/d_{\mathbb{F}}$;*
3. *the pair-correlation condition above holds.*

*Then*

$$\frac{1}{m} \log \min_{|T|=m} \sigma_{\min}(A_{m,T}) \xrightarrow{\mathbb{P}} -\frac{h(\gamma)}{d_{\mathbb{F}}}.$$

*Proof.* The lower side is Theorem 3.2. The upper side is the second-moment half of Theorem 2.2. □

*Remark* 3.19. The conditionality of Theorem 3.18 is removed in Appendix B. There the pair-correlation condition is reduced (Theorem B.4) to a single decorrelation estimate for random normals — proved exactly for disjoint row sets and established for overlapping sets (Proposition B.5, Corollary B.8) through an elementary off-support resolvent estimate — proved self-contained by a leave-one-out argument (Remark B.6), with the anisotropic local law [10, 5] an optional citation. The route replaces the Gaussian normal-vector decorrelation of [15] by unconditional ingredients — a two-projection local central limit theorem (Lemma 3.20 below), a high-overlap negligibility estimate (Proposition 3.21), a resolvent-monotonicity bound, and the self-contained off-support estimate — so the universality base holds for all iid bounded-density sub-Gaussian entries, with full error control.

The pair-correlation condition is built from one estimate per pair of row sets. Conditioning on every row except the distinguished one, the two events $\{|\langle x, u_I\rangle| \le t\}$ and $\{|\langle x, u_J\rangle| \le t\}$ are slabs (resp. disks) against two fixed normals whose overlap is $\varrho = \langle u_I, u_J\rangle$. The Gaussian argument controls their joint probability through the exact Gaussian density at the origin. The following lemma reproduces that bound, with the same constants, for arbitrary bounded-density entries.

**Lemma 3.20** (Two-projection anti-concentration without Gaussianity)**.** *Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and let $x \in \mathbb{F}^m$ have iid entries with mean zero, unit variance (proper when $\mathbb{F} = \mathbb{C}$), and Lebesgue density bounded by $K$ on $\mathbb{F} \simeq \mathbb{R}^{d_\mathbb{F}}$. Let $u, v \in \mathbb{F}^m$ be deterministic unit vectors with $\|u\|_\infty, \|v\|_\infty \le m^{-1/2+\delta}$ for a fixed $\delta \in (0, 1/2)$, and put $\varrho = \langle u, v\rangle$. Fix $c_0 \in (0, 1)$. Then for every sequence $t_m \to 0$ there is a sequence $\theta_m \to 0$, depending only on $K, \delta, c_0$ and the rate of $t_m$, such that uniformly over all such $u, v$ with $|\varrho| \le 1 - c_0$,*

$$\mathbb{P}\{|\langle x, u\rangle| \le t_m,\ |\langle x, v\rangle| \le t_m\} \le (1 + \theta_m)\, \frac{\mathbb{P}\{|\langle x, u\rangle| \le t_m\}^2}{(1 - |\varrho|^2)^{d_\mathbb{F}/2}}.$$

*The regime $t_m \to 0$ is the only one used in the second moment, where $t_m = e^{-sm}$; at a fixed scale $t$ the constant degrades but the bound still holds with the Gaussian two-slab/two-disk constant of* [15].

*Proof sketch.* Let $w = (\langle x, u\rangle, \langle x, v\rangle) \in \mathbb{F}^2$. Its characteristic function factors over the coordinates of $x$, and a single mean-zero, unit-variance scalar with density at most $K$ satisfies $|\mathbb{E}e^{i\Re(\theta\bar\xi)}| \le e^{-\eta \min(|\theta|^2, 1)}$, where $\eta > 0$ depends only on $K$: the quadratic regime comes from the variance, and the bounded regime from the Riemann–Lebesgue decay of an absolutely continuous law, with $\|\widehat{f}\,\|_{L^2}^2 \le 2\pi K$ available for the far field. Delocalization forces the coefficients $\theta u_j, \theta v_j$ to be uniformly small, so on a fixed Fourier ball a Lindeberg expansion shows that the density of $w$ at the origin converges to that of the centered Gaussian on $\mathbb{F}^2$ with covariance $\left(\begin{smallmatrix} 1 & \varrho \\ \bar\varrho & 1 \end{smallmatrix}\right)$, namely $a_{d_\mathbb{F}}^2 (1 - |\varrho|^2)^{-d_\mathbb{F}/2}$; the moderate range is controlled by the Gaussian-type decay and the far range by a transverse-pair Cauchy–Schwarz bound that uses two coefficient pairs with $|\det| \gtrsim m^{-1}$ (Cauchy–Binet). Integrating the density bound over the two slabs (resp. disks) of radius $t$ and comparing with $\mathbb{P}\{|\langle x, u\rangle| \le t\} = (1 + o(1)) a_{d_\mathbb{F}} t^{d_\mathbb{F}}$ gives the stated inequality, with constants matching the two-slab and two-disk estimate of [15]. The complete proof — the one-variable characteristic-function bound, the transverse-pair integrability, the three-region Fourier inversion, and the conversion to the small-ball ratio — is given in Appendix A (cf. [12]). □

**Proposition 3.21** (Small-overlap-codimension pairs are negligible)**.** *Let $I, J$ be independent uniform $(m-1)$-subsets of $[N_m - 1]$, with $N_m/m \to \gamma > 1$, and put $D(I, J) = m - |I \cap J|$. Fix $s < h(\gamma)/d_\mathbb{F}$. If $p_m \ge ce^{-d_\mathbb{F} sm}$ for some $c > 0$, then there are $\varepsilon, \delta > 0$, depending only on $\gamma$ and $s$, such that*

$$\frac{\mathbb{P}\{2 \le D(I, J) \le \varepsilon m\}}{p_m} \le C_{\gamma, c} m^2 e^{-\delta m} \to 0.$$

*Consequently, if $q_D \le p_m$ is any joint small-ball probability indexed by the overlap codimension, then*

$$\mathbb{E}\left[\frac{q_D}{p_m^2};\, 2 \le D \le \varepsilon m\right] \to 0.$$

*Proof.* Conditioning on $I$, the variable $D-1$ has the hypergeometric law

$$\mathbb{P}\{D = d\} = \frac{\binom{m-1}{d-1}\binom{N_m-m}{d-1}}{\binom{N_m-1}{m-1}}.$$

Let $H_2(x) = -x\log x - (1-x)\log(1-x)$. Standard entropy bounds for binomial coefficients give, uniformly for $d = \alpha m + O(1)$ below the mode,

$$\frac{1}{m}\log \mathbb{P}\{D = d\} \le H_2(\alpha) + (\gamma-1)H_2\left(\frac{\alpha}{\gamma-1}\right) - h(\gamma) + o(1).$$

The right side tends to $-h(\gamma)$ as $\alpha \downarrow 0$. Since $d_{\mathbb{F}} s < h(\gamma)$, choose $\varepsilon > 0$ so small that the displayed exponent is at most $-d_{\mathbb{F}} s - 2\delta$ for $0 \le \alpha \le \varepsilon$. The hypergeometric lower tail is increasing up to its mode, so summing $O(m)$ terms gives

$$\mathbb{P}\{2 \le D \le \varepsilon m\} \le C_\gamma m^2 e^{-(d_{\mathbb{F}} s + \delta)m}.$$

Dividing by $p_m \ge c e^{-d_{\mathbb{F}} s m}$ proves the first claim. The second follows from $q_D \le p_m$. □

*Remark* 3.22 (What remains of the pair-correlation condition). Lemma 3.20 is the per-pair analytic input to the second moment, while Proposition 3.21 removes the small/intermediate-codimension range by combinatorial rarity and the trivial bound $q_D \le p_m$. Thus the remaining non-Gaussian upper-side gap is only a bulk overlap estimate: for some fixed $\varepsilon > 0$,

$$\mathbb{E}\Big[(1 - |\varrho_{I,J}|^2)^{-d_{\mathbb{F}}/2}\ ;\ D(I,J) \ge \varepsilon m\Big] \le 1 + o(1), \qquad \varrho_{I,J} = \langle u_I, u_J\rangle.$$

For Gaussian entries the normals are Haar in the orthocomplement of the shared span and this is the overlap computation of [15]. For general isotropic non-Gaussian rows the elementary bounded-density bounds recover the correct tail exponent but only with constants growing in $m$. Closing the gap is a quantitative central limit theorem for overlaps of normals in the bulk, recorded as Problem 10.1.

## 4. Atomic laws and exact singularity

The bounded-density hypothesis in Proposition 3.3 is not a cosmetic assumption. Atomic entries create exact singularity events that are invisible in the continuous hard-edge picture.

**Proposition 4.1** (First-moment atomic obstruction)**.** *Let $A \in \{\pm 1\}^{N\times m}$ have iid Rademacher entries and let*

$$X_m = \#\{T \subset [N] : |T| = m,\ A_T \text{ is singular}\}.$$

*If $N/m \to \gamma > 1$, then*

$$\mathbb{E}X_m \ge \exp\{(h(\gamma) - \log 2 + o(1))m\}.$$

*In particular, at the real phase-retrieval threshold $N = 2m-1$,*

$$\mathbb{E}X_m \ge \exp\{(\log 2 + o(1))m\}.$$

*Proof.* For a fixed $T$ of size $m$, the first two rows of $A_T$ are equal with probability $2^{-m}$. On that event $A_T$ is singular. Therefore

$$\mathbb{P}\{A_T \text{ is singular}\} \ge 2^{-m}.$$

Summing over all $T$ and using Stirling's formula gives

$$\mathbb{E}X_m \ge \binom{N}{m}2^{-m} = \exp\{(h(\gamma) - \log 2 + o(1))m\}.$$

□

*Remark* 4.2 (Why this is not yet a high-probability theorem). The proof above is intentionally only a first-moment obstruction. It cannot by itself prove that $X_m > 0$ with high probability. The equal-row event is very clustered: a single equal pair of rows in the full $N \times m$ matrix belongs to many $m$-row submatrices. For $N = O(m)$, the probability that the full Rademacher matrix contains an equal pair of rows is at most $N^2 2^{-m}$, which tends to zero. Thus the equal-row mechanism explains why continuity is necessary for the clean small-ball estimate, but it does not settle full-spark failure for random sign frames.

**Proposition 4.3** (Deterministic finite-alphabet projective-pair obstruction). *Let $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, let $\varnothing \neq S \subset \mathbb{K}$ be finite, and put*

$$\Pi(S) = \{[a : b] \in \mathbb{P}^1(\mathbb{K}) : a, b \in S,\ (a, b) \neq (0, 0)\}.$$

*If $A \in S^{N \times m}$, $m \geq 2$, and either $S = \{0\}$ with $N \geq m$, or $S \neq \{0\}$ with*

$$N \geq |\Pi(S)|(m - 1) + 1,$$

*then some $m \times m$ row submatrix of $A$ is singular over $\mathbb{K}$. When $0 \notin S$, $|\Pi(S)| = |S/S|$, so this recovers the quotient obstruction.*

*Proof.* If $S = \{0\}$, then every $m$-row submatrix is the zero matrix, provided $N \geq m$. Assume henceforth that $S \neq \{0\}$, and write $L = |\Pi(S)| \geq 1$.

For each row $i$, inspect the pair

$$v_i = (A_{i1}, A_{i2}) \in S^2.$$

Let $z$ be the number of rows with $v_i = (0, 0)$. For each projective class $\pi = [u : v] \in \Pi(S)$, let $n_\pi$ be the number of rows with $v_i \neq (0, 0)$ and $[A_{i1} : A_{i2}] = \pi$.

If $z + n_\pi \geq m$ for some $\pi = [u : v]$, select $m$ rows from the union of the zero-zero rows and the rows in that class. On every selected row,

$$vA_{i1} - uA_{i2} = 0,$$

where zero-zero rows satisfy this relation for every $[u : v]$. Thus the first two columns $C_1, C_2$ of the selected square submatrix obey the nontrivial linear relation $vC_1 - uC_2 = 0$, and the determinant is zero.

It remains to show that such a class must exist. If $z + n_\pi \leq m - 1$ for every $\pi$, then

$$N = z + \sum_{\pi \in \Pi(S)} n_\pi \leq z + L(m - 1 - z) = L(m - 1) - (L - 1)z \leq L(m - 1),$$

contradicting $N \geq L(m - 1) + 1$. Hence some $z + n_\pi$ is at least $m$, and the preceding paragraph applies. □

**Proposition 4.4** (Deterministic sign obstruction at the critical threshold). *Let $A \in \{\pm 1\}^{N \times m}$ with $m \geq 2$ and $N \geq 2m - 1$. Then some $m \times m$ row submatrix of $A$ is singular. In particular every real sign frame with $N = 2m - 1$ fails the full-spark condition required for critical real phase retrieval.*

*Proof.* Apply Proposition 4.3 with $S = \{\pm 1\}$. Then $\Pi(S) = \{[1 : 1], [1 : -1]\}$, so the threshold is $2m - 1$. □

**Corollary 4.5** (Critical Rademacher phase retrieval fails surely). *Let $A_m \in \{\pm 1\}^{(2m-1) \times m}$ be random or deterministic, $m \geq 2$. Then*

$$\min_{|T|=m} \sigma_{\min}(A_{m,T}) = 0.$$

*At $N = 2m - 1$, failure of full spark is equivalent to failure of the complement property, hence to non-injectivity, by [1, 4]. Consequently iid Rademacher critical frames fail real phase-retrieval injectivity with probability one.*

The projective-pair obstruction is deterministic finite-alphabet bookkeeping; in the sign case it lands exactly at the critical count $N = 2m - 1$. It should not be confused with a general atomic dichotomy or with an optimal threshold for all finite alphabets. Allowing zero only changes the two-column bookkeeping; it does not turn the result into a high-probability threshold below the deterministic row count. For a general discrete law, the singularity exponent of one square matrix is law-dependent and is not simply $\log(1/p_*)$, where $p_*$ is the largest atom.

**Problem 4.6** (Discrete extremal hard edge below the exact-singularity threshold)**.** For iid entries with a nontrivial atom but with

$$\mathbb{P}\{\exists\, T,\ |T| = m,\ A_T \text{ singular}\} \to 0,$$

determine the exponential scale of

$$\min_{|T|=m} \sigma_{\min}(A_T).$$

The answer should depend both on the continuous small-ball exponent away from atoms and on the arithmetic structure of the atom set.

## 5. Geometric generalizations: rectangular, higher, and two-sided submatrices

The entropy/small-ball principle of Section 2 separates the combinatorial entropy of a submatrix family from the one-matrix small-ball exponent of the singular value being tracked. Changing the shape of the submatrices, or the index of the singular value, moves one of these two inputs in a controlled way. We treat three such changes: tall submatrices with fixed excess, the $k$-th smallest singular value, and submatrices chosen on both sides. In each case the union-bound lower side is unconditional once the relevant one-matrix tail is available, while the matching exact value needs the same second-moment/pair-correlation input as the square row-submatrix theorem of Section 3. We label each statement accordingly.

5.1. **Rectangular submatrices with fixed excess.** Fix an integer $r \geq 0$ and let $A_m \in \mathbb{F}^{N_m \times m}$ have independent standard Gaussian entries with $N_m/m \to \gamma > 1$. For a row set $T \subset [N_m]$ with $|T| = m + r$, write $A_{m,T} \in \mathbb{F}^{(m+r)\times m}$ for the tall submatrix on those rows and $\sigma_{\min}(A_{m,T})$ for its smallest singular value. Put

$$M_m^{(r)} = \min_{\substack{T\subset[N_m]\\ |T|=m+r}} \sigma_{\min}(A_{m,T}).$$

For $r = 0$ this is the square quantity $M_m$ of the source theorem.

The relevant one-matrix input is the tall hard edge. For a single $(m + r) \times m$ matrix $G$ with iid $\mathbb{F}$-Gaussian entries, $\sigma_{\min}(G)^2$ is the smallest eigenvalue of the Laguerre matrix $G^*G$, whose density near the origin carries the factor $\lambda^{d_{\mathbb{F}}(r+1)/2-1}$; equivalently

$$\mathbb{P}\{\sigma_{\min}(G) \leq t\} \asymp t^{\,d_{\mathbb{F}}(r+1)}, \qquad t \to 0,$$

see [8, 9]. In the real case ($d_{\mathbb{F}} = 1$) the smallest Wishart eigenvalue has density of order $\lambda^{(r-1)/2}$ at zero, so $\mathbb{P}\{\sigma_{\min} \leq t\} \asymp t^{r+1}$; the complex case ($d_{\mathbb{F}} = 2$) replaces the exponent by $2(r + 1)$. The exponent $d_{\mathbb{F}}(r + 1)$ reduces to $d_{\mathbb{F}}$ at $r = 0$.

For the lower side we need a non-asymptotic upper form of this tail, uniform in $t$ and with only polynomial loss in $m$. The square case is Proposition 3.3; the tall case follows from the same negative-second-moment identity, applied to the columns rather than the rows.

**Lemma 5.1** (Tall one-matrix tail)**.** *Let $G \in \mathbb{F}^{\ell\times m}$ have independent standard Gaussian entries with $\ell \geq m$, and set $p = \ell - m + 1$. Then for all $m \geq 1$ and all $t > 0$,*

$$\mathbb{P}\{\sigma_{\min}(G) \leq t\} \leq m^{\,1+d_{\mathbb{F}}p/2}\, t^{\,d_{\mathbb{F}}p}.$$

*Proof.* Let $C_1, \dots, C_m \in \mathbb{F}^\ell$ be the columns of $G$, let $H_i = \operatorname{span}_{\mathbb{F}}\{C_j : j \neq i\}$, and set $d_i = \operatorname{dist}(C_i, H_i)$. Almost surely $G$ has full column rank, and the negative second-moment identity for the columns reads

$$\sum_{j=1}^{m} \sigma_j(G)^{-2} = \sum_{i=1}^{m} d_i^{-2},$$

because the $i$-th diagonal entry of $(G^*G)^{-1}$ equals $d_i^{-2}$. If $\sigma_{\min}(G) \le t$, the left side is at least $t^{-2}$, so some $i$ satisfies $d_i \le \sqrt{m}\, t$.

Condition on $\{C_j : j \neq i\}$. Then $\dim_{\mathbb{F}} H_i \le m-1$, so its orthogonal complement in $\mathbb{F}^\ell$ has $\mathbb{F}$-dimension at least $\ell-(m-1) = p$, that is real dimension at least $d_{\mathbb{F}}p$. The distance $d_i = \|P_{H_i^\perp} C_i\|$ is the norm of the projection of a standard $\mathbb{F}$-Gaussian vector onto a subspace of real dimension at least $d_{\mathbb{F}}p$; it therefore stochastically dominates the norm of a standard Gaussian vector $g$ in real dimension $d_{\mathbb{F}}p$. Comparing the volume of a ball with the maximal value of the density gives

$$\mathbb{P}\{\|g\| \le u\} \le u^{d_{\mathbb{F}}p}, \qquad u > 0.$$

Taking $u = \sqrt{m}\, t$ and a union bound over the $m$ columns yields $\mathbb{P}\{\sigma_{\min}(G) \le t\} \le m\,(\sqrt{m}\, t)^{d_{\mathbb{F}}p} = m^{1+d_{\mathbb{F}}p/2} t^{d_{\mathbb{F}}p}$. □

**Lemma 5.2** (Fixed-excess bounded-density tail)**.** *Fix $r \ge 0$ and set $p = r+1$. Let $G \in \mathbb{F}^{(m+r)\times m}$, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, have independent entries whose $d_{\mathbb{F}}$-dimensional scalar densities are bounded by $K_m \ge 1$. Then, for all $t > 0$,*

$$\mathbb{P}\{\sigma_{\min}(G) \le t\} \le C_{\mathbb{F},r} K_m^p m^{\,1+d_{\mathbb{F}}p} t^{\,d_{\mathbb{F}}p}.$$

*Proof.* Let $\ell = m + r$, and let $C_1, \dots, C_m \in \mathbb{F}^\ell$ be the columns. As in Lemma 5.1, $\sigma_{\min}(G) \le t$ implies that for some $i$,

$$\operatorname{dist}(C_i, \operatorname{span}\{C_j : j \neq i\}) \le \sqrt{m}\, t.$$

Condition on all columns except $C_i$, write $H = \operatorname{span}\{C_j : j \neq i\}$, and choose a $p$-dimensional $\mathbb{F}$-subspace $L \subset H^\perp$. Let $Q : \mathbb{F}^\ell \to \mathbb{F}^p$ have orthonormal rows spanning $L$. By Cauchy–Binet, some $p$-coordinate set $S \subset [\ell]$ has

$$|\det_{\mathbb{F}} Q_S| \ge \binom{\ell}{p}^{-1/2}.$$

Condition on the coordinates of $C_i$ outside $S$. Then

$$QC_i = Q_S (C_i)_S + c$$

for a fixed vector $c$. The joint density of $(C_i)_S$ is at most $K_m^p$, so the density of $QC_i$ on $\mathbb{F}^p \simeq \mathbb{R}^{d_{\mathbb{F}}p}$ is at most

$$K_m^p |\det_{\mathbb{F}} Q_S|^{-d_{\mathbb{F}}} \le K_m^p \binom{\ell}{p}^{d_{\mathbb{F}}/2}.$$

Consequently, with $\tau = \sqrt{m}\, t$,

$$\mathbb{P}\{\|QC_i\| \le \tau \mid H\} \le C_{\mathbb{F},p} K_m^p \binom{\ell}{p}^{d_{\mathbb{F}}/2} \tau^{d_{\mathbb{F}}p}.$$

Since $L \subset H^\perp$, the event $\operatorname{dist}(C_i, H) \le \tau$ is contained in $\{\|QC_i\| \le \tau\}$. A union bound over $i = 1, \dots, m$, together with fixed $p$, gives the stated estimate. □

**Theorem 5.3** (Fixed-excess lower side)**.** *With the notation above, for every fixed $r \ge 0$ and every $\varepsilon > 0$,*

$$\mathbb{P}\left\{ M_m^{(r)} \le \exp\left[ -\left( \frac{h(\gamma)}{d_{\mathbb{F}}(r+1)} + \varepsilon \right) m \right] \right\} \longrightarrow 0.$$

*Equivalently,* $\liminf_m \frac{1}{m} \log M_m^{(r)} \ge -\dfrac{h(\gamma)}{d_{\mathbb{F}}(r+1)}$ *in probability.*

*Proof.* Apply Corollary 2.3 to the family $\mathcal{T}_m = \{T : |T| = m + r\}$ with $Y_{m,T} = \sigma_{\min}(A_{m,T})$. Since $r$ is fixed, $\binom{N_m}{m+r} = \binom{N_m}{m}(\gamma - 1 + o(1))^r$, so

$$\frac{1}{m} \log |\mathcal{T}_m| = \frac{1}{m} \log \binom{N_m}{m+r} \to h(\gamma),$$

and the entropy is $H = h(\gamma)$. Lemma 5.1 with $p = r + 1$ supplies the uniform small-ball bound with exponent $a = d_{\mathbb{F}}(r + 1)$. The corollary gives the conclusion with base $H/a = h(\gamma)/(d_{\mathbb{F}}(r + 1))$. □

**Theorem 5.4** (Fixed-excess bounded-density lower side)**.** *Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, let $A_m \in \mathbb{F}^{N_m \times m}$ have independent entries whose $d_{\mathbb{F}}$-dimensional scalar densities are bounded by $K_m \geq 1$, uniformly in their position, and assume $N_m/m \to \gamma > 1$. Put*

$$\kappa = \limsup_{m \to \infty} \frac{1}{m} \log K_m < \infty.$$

*For every fixed $r \geq 0$, with*

$$M_m^{(r)} = \min_{\substack{T \subset [N_m] \\ |T| = m+r}} \sigma_{\min}(A_{m,T}),$$

*and every $\varepsilon > 0$,*

$$\mathbb{P}\left\{ M_m^{(r)} \leq \exp\left[ -\left( \frac{h(\gamma)}{d_{\mathbb{F}}(r + 1)} + \frac{\kappa}{d_{\mathbb{F}}} + \varepsilon \right) m \right] \right\} \longrightarrow 0.$$

*In particular, if $K_m = e^{o(m)}$, the Gaussian fixed-excess lower-side base $h(\gamma)/(d_{\mathbb{F}}(r + 1))$ remains valid.*

*Proof.* Apply Lemma 5.2 to every selected $(m + r) \times m$ submatrix, with $p = r + 1$. For $t = e^{-sm}$,

$$\mathbb{P}\{M_m^{(r)} \leq t\} \leq \binom{N_m}{m+r} C_{\mathbb{F},r} K_m^{r+1} m^{1+d_{\mathbb{F}}(r+1)} t^{d_{\mathbb{F}}(r+1)}.$$

The logarithmic limsup divided by $m$ is at most

$$h(\gamma) + (r + 1)\kappa - d_{\mathbb{F}}(r + 1)s.$$

This is negative when

$$s > \frac{h(\gamma)}{d_{\mathbb{F}}(r + 1)} + \frac{\kappa}{d_{\mathbb{F}}},$$

which is exactly the claimed lower side. □

**Corollary 5.5** (Finite-entropy fixed-excess lower side)**.** *Let $r \geq 0$ be fixed and set $p = r + 1$. Let $N_m \geq m + r$, and let $A_m \in \mathbb{F}^{N_m \times m}$, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, have independent entries whose $d_{\mathbb{F}}$-dimensional scalar densities are bounded by $K_m \geq 1$, uniformly in their position. Put*

$$L_m^{(r)} = \log \binom{N_m}{m+r}, \qquad M_m^{(r)} = \min_{|T| = m+r} \sigma_{\min}(A_{m,T}).$$

*Then, for every $u_m > 0$,*

$$\mathbb{P}\left\{ M_m^{(r)} \leq \exp\left[ -\frac{L_m^{(r)} + p \log K_m + (1 + d_{\mathbb{F}} p) \log m + O_{\mathbb{F},r}(1) + u_m}{d_{\mathbb{F}} p} \right] \right\} \leq e^{-u_m}.$$

*In particular, taking $u_m = d_{\mathbb{F}} p\, \varepsilon m$ gives the corresponding convergence-in-probability lower side at the finite entropy budget $(L_m^{(r)} + p \log K_m)/(d_{\mathbb{F}} p)$, with any fixed exponential slack. If the matrices are coupled for all $m$ and $\sum_m e^{-u_m} < \infty$, then the displayed bad event occurs only finitely often almost surely, with the $O_{\mathbb{F},r}(1)$ interpreted as one fixed constant.*

*Proof.* Apply Lemma 5.2 and Corollary 2.4 with

$$|\mathcal{T}_m| = \binom{N_m}{m+r}, \qquad C_m = C_{\mathbb{F},r} K_m^p m^{1+d_{\mathbb{F}}p}, \qquad a_m = d_{\mathbb{F}}p.$$

□

**Lemma 5.6** (A fixed-excess row-block certificate)**.** *Let $B \in \mathbb{F}^{(m+r)\times m}$, put $p = r+1$, and assume $B$ has full column rank. Then there is a row set $U \subset [m+r]$ with $|U| = m-1$ and a unit normal $u_U \in \mathbb{F}^m$ to the span of the rows indexed by $U$ such that*

$$\left(\sum_{i\notin U} |\langle R_i, u_U\rangle|^2\right)^{1/2} \le \sqrt{m}\,\sigma_{\min}(B),$$

*where $R_i$ are the rows of $B$. Equivalently, if $\sigma_{\min}(B) \le t$, then for some $U$ with $|U| = m-1$,*

$$\left(\sum_{i\notin U} |\langle R_i, u_U\rangle|^2\right)^{1/2} \le \sqrt{m}\,t.$$

*Proof.* For a row set $U$ with $|U| = m-1$, let $\Delta_U = \det(B_U B_U^*)$, and ignore sets with $\Delta_U = 0$. If $\Delta_U > 0$, let $u_U$ be a unit normal to the span of the rows of $B_U$. By exterior algebra, or equivalently by the cofactor formula, for each $i \notin U$,

$$\Delta_U |\langle R_i, u_U\rangle|^2 = \det(B_{U\cup\{i\}} B_{U\cup\{i\}}^*).$$

Summing over $U$ and $i \notin U$, every $m$-row minor is counted $m$ times. Hence Cauchy–Binet gives

$$\sum_{|U|=m-1} \Delta_U \sum_{i\notin U} |\langle R_i, u_U\rangle|^2 = m \det(B^*B).$$

Another Cauchy–Binet identity gives

$$\sum_{|U|=m-1} \Delta_U = e_{m-1}(\sigma_1(B)^2, \dots, \sigma_m(B)^2) = \det(B^*B) \sum_{j=1}^m \sigma_j(B)^{-2}.$$

Therefore the weighted average of $\sum_{i\notin U} |\langle R_i, u_U\rangle|^2$ is

$$\frac{m}{\sum_j \sigma_j(B)^{-2}} \le m\,\sigma_{\min}(B)^2.$$

At least one $U$ is no larger than this average. □

**Corollary 5.7** (Fixed-excess directional row lower side)**.** *Let $r \ge 0$ be fixed and set $p = r+1$. Let $N_m \ge m+r$, let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and let $A_m \in \mathbb{F}^{N_m\times m}$ have independent rows $R_{m,i}$. Assume that there are deterministic numbers $L_{m,i} \ge 1$ such that, for every unit vector $u \in \mathbb{F}^m$, every $z \in \mathbb{F}$, and every $0 < \rho \le 1$,*

$$\mathbb{P}\{|\langle R_{m,i}, u\rangle - z| \le \rho\} \le L_{m,i}\rho^{d_{\mathbb{F}}}.$$

*Put*

$$\mathcal{B}_{m,r}^{\mathrm{dir}} = \sum_{\substack{T\subset[N_m]\\|T|=m+r}} \sum_{\substack{S\subset T\\|S|=p}} \prod_{i\in S} L_{m,i} = \binom{N_m - p}{m-1} \sum_{\substack{S\subset[N_m]\\|S|=p}} \prod_{i\in S} L_{m,i}.$$

*Then, for every $u_m > 0$,*

$$\mathbb{P}\left\{M_m^{(r)} \le \exp\left[-\frac{\log \mathcal{B}_{m,r}^{\mathrm{dir}} + \frac{d_{\mathbb{F}}p}{2}\log m + u_m}{d_{\mathbb{F}}p}\right]\right\} \le e^{-u_m}.$$

*Moreover, on any common coupling, if* $\sum_m e^{-u_m} < \infty$, *then the displayed bad event occurs only finitely often almost surely. Consequently, if* $N_m/m \to \gamma > 1$ *and*

$$\beta_r^{\mathrm{dir}} = \limsup_{m\to\infty} \frac{1}{m} \log \left( \sum_{\substack{S\subset[N_m]\\ |S|=p}} \prod_{i\in S} L_{m,i} \right) < \infty,$$

*then for every* $\varepsilon > 0$,

$$\mathbb{P}\left\{ M_m^{(r)} \le \exp\left[ -\left( \frac{h(\gamma) + \beta_r^{\mathrm{dir}}}{d_{\mathbb{F}}(r+1)} + \varepsilon \right) m \right] \right\} \to 0.$$

*Proof.* Fix a selected row set $T$ with $|T| = m + r$. The small-ball assumption implies that every $m-1$ selected rows have rank $m-1$ almost surely, by conditioning sequentially and using the absence of atoms on affine hyperplanes. On this full-probability event, Lemma 5.6 shows that $\sigma_{\min}(A_{m,T}) \le t$ implies the existence of a set $U \subset T$, $|U| = m-1$, with $S = T \setminus U$, $|S| = p$, and

$$\left( \sum_{i\in S} |\langle R_{m,i}, u_U \rangle|^2 \right)^{1/2} \le \sqrt{m}\, t,$$

where $u_U$ is a measurable unit normal to the rows indexed by $U$. Condition on the rows in $U$. The remaining rows are independent of $u_U$, and the displayed event implies $|\langle R_{m,i}, u_U\rangle| \le \sqrt{m}\, t$ for every $i \in S$. Since the bound is trivial when $\sqrt{m}\, t > 1$, for all $t > 0$,

$$\mathbb{P}\{\sigma_{\min}(A_{m,T}) \le t\} \le m^{d_{\mathbb{F}}p/2} t^{d_{\mathbb{F}}p} \sum_{\substack{S\subset T\\ |S|=p}} \prod_{i\in S} L_{m,i}.$$

Summing this nonuniform budget over $T$ and applying Corollary 2.5 gives the finite-budget estimate, and Corollary 2.6 gives the almost-sure upgrade. The fixed-aspect statement follows by taking $u_m = d_{\mathbb{F}} p\, \varepsilon m$, since

$$\frac{1}{m} \log \binom{N_m - p}{m-1} \to h(\gamma)$$

and the polynomial term is $o(m)$. □

**Corollary 5.8** (Fixed-excess heterogeneous row-density budget)**.** *Let* $r \ge 0$ *be fixed and set* $p = r+1$. *Let* $N_m \ge m + r$, *and let* $A_m \in \mathbb{F}^{N_m \times m}$, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, *have independent entries. Assume that for each row* $i \in [N_m]$ *there is a deterministic number* $K_{m,i} \ge 1$ *such that every entry in row* $i$ *has* $d_{\mathbb{F}}$*-dimensional Lebesgue density bounded by* $K_{m,i}$. *For each selected row set* $T$, *put*

$$\kappa_T^{(r)} = \max_{\substack{S\subset T\\ |S|=p}} \prod_{i\in S} K_{m,i}, \qquad \Theta_{m,r} = \sum_{\substack{T\subset[N_m]\\ |T|=m+r}} \kappa_T^{(r)}.$$

*Then, for every* $u_m > 0$,

$$\mathbb{P}\left\{ M_m^{(r)} \le \exp\left[ -\frac{\log \Theta_{m,r} + (1 + d_{\mathbb{F}}p)\log m + O_{\mathbb{F},r}(1) + u_m}{d_{\mathbb{F}}p} \right] \right\} \le e^{-u_m}.$$

*Moreover, on any common coupling, if* $\sum_m e^{-u_m} < \infty$, *then the displayed bad event occurs only finitely often almost surely, with the* $O_{\mathbb{F},r}(1)$ *interpreted as one fixed constant. Consequently, if*

$$\theta_r = \limsup_{m\to\infty} \frac{1}{m} \log \Theta_{m,r} < \infty,$$

*then for every* $\varepsilon > 0$,

$$\mathbb{P}\left\{ M_m^{(r)} \le \exp\left[ -\left( \frac{\theta_r}{d_{\mathbb{F}}(r+1)} + \varepsilon \right) m \right] \right\} \to 0.$$

*Proof.* Repeat the proof of Lemma 5.2 for one selected row set $T$. After conditioning on all columns except $C_q$, choose a $p$-row coordinate block $S \subset T$ supplied by Cauchy–Binet. Conditioning on all other coordinates of $C_q$, the density of the projected $p$-coordinate vector is bounded by

$$\left(\prod_{a\in S} K_{m,a}\right)\binom{m+r}{p}^{d_{\mathbb{F}}/2}.$$

Taking the worst such $S$ gives, for $0 < t \le 1$,

$$\mathbb{P}\{\sigma_{\min}(A_{m,T}) \le t\} \le C_{\mathbb{F},r}\kappa_T^{(r)} m^{1+d_{\mathbb{F}}p} t^{d_{\mathbb{F}}p}.$$

Summing this bound over $T$ and applying Corollary 2.5 gives the finite-budget estimate. Corollary 2.6 gives the almost-sure upgrade. The exponential-scale statement follows by taking $u_m = d_{\mathbb{F}}p\,\varepsilon m$; the logarithmic polynomial and constant terms are $o(m)$. □

**Corollary 5.9** (Fixed-excess entrywise heterogeneous finite budget)**.** *Let $r \ge 0$ be fixed and set $p = r+1$. Let $N_m \ge m+r$, and let $A_m \in \mathbb{F}^{N_m\times m}$, $\mathbb{F} \in \{\mathbb{R},\mathbb{C}\}$, have independent entries. Assume that entry $(a,q)$ has $d_{\mathbb{F}}$-dimensional Lebesgue density bounded by a deterministic number $K_{m,a,q} \ge 1$. For $T \subset [N_m]$, $|T| = m+r$, and $q \in [m]$, put*

$$\kappa_{T,q}^{(r)} = \max_{\substack{S\subset T\\ |S|=p}} \prod_{a\in S} K_{m,a,q}, \qquad \Xi_{m,r} = \sum_{\substack{T\subset[N_m]\\ |T|=m+r}} \sum_{q=1}^{m} \kappa_{T,q}^{(r)}.$$

*Then, for every $u_m > 0$,*

$$\mathbb{P}\left\{M_m^{(r)} \le \exp\left[-\frac{\log \Xi_{m,r} + d_{\mathbb{F}}p\log m + O_{\mathbb{F},r}(1) + u_m}{d_{\mathbb{F}}p}\right]\right\} \le e^{-u_m}.$$

*Moreover, on any common coupling, if $\sum_m e^{-u_m} < \infty$, then the displayed bad event occurs only finitely often almost surely, with the $O_{\mathbb{F},r}(1)$ interpreted as one fixed constant. Consequently, if*

$$\xi_r = \limsup_{m\to\infty} \frac{1}{m}\log \Xi_{m,r} < \infty,$$

*then for every $\varepsilon > 0$,*

$$\mathbb{P}\left\{M_m^{(r)} \le \exp\left[-\left(\frac{\xi_r}{d_{\mathbb{F}}(r+1)} + \varepsilon\right) m\right]\right\} \to 0.$$

*If $K_{m,a,q} = K_{m,a}$ is row-wise and column-uniform, this recovers Corollary 5.8; if all $K_{m,a,q} = K_m$, it recovers Corollary 5.5.*

*Proof.* Fix a row set $T$ and a column $q$. The column negative second-moment identity implies that $\sigma_{\min}(A_{m,T}) \le t$ forces some column distance to be at most $\sqrt{m}\,t$. Conditioning on all columns except column $q$, choose a $p$-dimensional $\mathbb{F}$-subspace of the orthogonal complement and then a $p$-row coordinate block $S \subset T$ by Cauchy–Binet, exactly as in Lemma 5.2. Conditioning on the coordinates of column $q$ outside $S$, the projected $p$-coordinate density is bounded by

$$\left(\prod_{a\in S} K_{m,a,q}\right)\binom{m+r}{p}^{d_{\mathbb{F}}/2}.$$

Taking the worst such $S$, and then summing over the possible bad column $q$, gives

$$\mathbb{P}\{\sigma_{\min}(A_{m,T}) \le t\} \le C_{\mathbb{F},r} m^{d_{\mathbb{F}}p} t^{d_{\mathbb{F}}p} \sum_{q=1}^{m} \kappa_{T,q}^{(r)}, \qquad 0 < t \le 1.$$

Summing these constants over $T$ and applying Corollary 2.5 gives the finite-budget estimate. Corollary 2.6 gives the almost-sure upgrade. The exponential-scale statement follows by taking

$u_m = d_{\mathbb{F}} p\, \varepsilon m$. The two specializations follow from $\Xi_{m,r} = m\Theta_{m,r}$ in the row-wise case and $\Xi_{m,r} = mK_m^p \binom{N_m}{m+r}$ in the uniform case. □

**Conjecture 5.10** (Fixed-excess base)**.** For each fixed $r \geq 0$,

$$\frac{1}{m} \log M_m^{(r)} \xrightarrow{\mathbb{P}} -\frac{h(\gamma)}{d_{\mathbb{F}}(r+1)}.$$

For Gaussian entries the lower bound is Theorem 5.3; for bounded-density entries with subexponential density growth it is Theorem 5.4. The matching upper bound would follow from Theorem 2.2 once the pair-correlation condition of Section 3 is established for the tall family: one must show that

$$Z_m(s) = \sum_{|T|=m+r} \mathbf{1}_{\{\sigma_{\min}(A_{m,T}) \leq e^{-sm}\}}$$

has $\operatorname{Var} Z_m(s)/(\mathbb{E} Z_m(s))^2 \to 0$ for every $s < h(\gamma)/(d_{\mathbb{F}}(r+1))$. For Gaussian entries this is the $(r+1)$-codimensional analogue of the hyperplane second moment of [15], with one fixed row replaced by $r+1$ fixed rows defining a codimension-$(r+1)$ small-ball event; it is not carried out here. The lower side is thus proven for every fixed $r$, while the exact base remains conditional on this second-moment input.

5.2. **Proportional excess: the exponential regime is exactly fixed excess.** If the excess grows with $m$, the exponential scale disappears.

**Proposition 5.11** (Proportional excess)**.** *Let $A_m \in \mathbb{F}^{N_m \times m}$ have iid standard Gaussian entries, $N_m/m \to \gamma > 1$, and let $\ell_m$ be integers with $\ell_m/m \to \zeta \in (1, \gamma)$. Then*

$$\frac{1}{m} \log \min_{|T|=\ell_m} \sigma_{\min}(A_{m,T}) \xrightarrow{\mathbb{P}} 0.$$

*Proof.* Write $M = \min_{|T|=\ell_m} \sigma_{\min}(A_{m,T})$ and $p_m = \ell_m - m + 1$, so $p_m/m \to \zeta - 1 > 0$. For the upper bound, fix one admissible $T_0$; then $\sigma_{\min}(A_{m,T_0})^2 \leq \frac{1}{m} \operatorname{tr}(A_{m,T_0}^* A_{m,T_0})$, whose mean is $\ell_m$, so $M \leq \sigma_{\min}(A_{m,T_0}) = O_{\mathbb{P}}(\sqrt{m})$ and $\limsup_m \frac{1}{m} \log M \leq 0$.

For the lower bound, fix $\varepsilon > 0$. By Lemma 5.1 with $p = p_m$,

$$\sup_{|T|=\ell_m} \mathbb{P}\{\sigma_{\min}(A_{m,T}) \leq e^{-\varepsilon m}\} \leq m^{1+d_{\mathbb{F}} p_m/2} e^{-d_{\mathbb{F}} p_m \varepsilon m} = \exp\{-d_{\mathbb{F}}(\zeta - 1)\varepsilon m^2 (1 + o(1))\}.$$

A union bound over the at most $2^{N_m} = e^{O(m)}$ row sets gives $\mathbb{P}\{M \leq e^{-\varepsilon m}\} \to 0$, hence $\liminf_m \frac{1}{m} \log M \geq -\varepsilon$. Let $\varepsilon \downarrow 0$. □

Thus the exponentially small regime is precisely that of fixed excess $r$: there the one-matrix exponent $d_{\mathbb{F}}(r+1)$ stays bounded, the entropy survives at $h(\gamma)$, and the competition produces an exponential base. Once the excess is proportional, the one-matrix exponent grows linearly in $m$ and overwhelms the entropy.

**Proposition 5.12** (Proportional-excess density-rate lower side)**.** *Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and let $A_m \in \mathbb{F}^{N_m \times m}$ have independent entries whose $d_{\mathbb{F}}$-dimensional scalar densities are bounded by deterministic $K_m \geq 1$, uniformly in their position. Assume $N_m/m \to \gamma > 1$, $\ell_m/m \to \zeta \in (1, \gamma)$, and*

$$\kappa = \limsup_{m \to \infty} \frac{1}{m} \log K_m < \infty.$$

*Put*

$$M_m^{(\ell)} = \min_{\substack{T \subset [N_m] \\ |T|=\ell_m}} \sigma_{\min}(A_{m,T}).$$

*Then for every $\varepsilon > 0$,*

$$\mathbb{P}\left\{ M_m^{(\ell)} \le \exp\left[ -\left( \frac{\kappa}{d_{\mathbb{F}}} + \varepsilon \right) m \right] \right\} \to 0.$$

*In particular, if $K_m = e^{o(m)}$, then $\liminf_m m^{-1} \log M_m^{(\ell)} \ge 0$ in probability.*

*Proof.* It suffices to keep the dependence on $p_m = \ell_m - m + 1$, which now satisfies $p_m/m \to \zeta - 1$. For one selected $\ell_m \times m$ submatrix $G$, the proof of Lemma 5.2 gives, for $t > 0$,

$$\mathbb{P}\{\sigma_{\min}(G) \le t\} \le m K_m^{p_m} \binom{\ell_m}{p_m}^{d_{\mathbb{F}}/2} v_{d_{\mathbb{F}} p_m} (\sqrt{m}\, t)^{d_{\mathbb{F}} p_m},$$

where $v_q$ is the Euclidean unit-ball volume in $\mathbb{R}^q$. Stirling's formula gives

$$\log v_{d_{\mathbb{F}} p_m} + \frac{d_{\mathbb{F}} p_m}{2} \log m = O(m), \qquad \log \binom{\ell_m}{p_m} = O(m).$$

With $t = e^{-sm}$, the logarithm of the one-submatrix probability is therefore at most

$$p_m \log K_m - d_{\mathbb{F}} p_m s m + o(m^2) \le \big( (\zeta - 1)\kappa - d_{\mathbb{F}}(\zeta - 1)s + o(1) \big) m^2.$$

The union over all row sets contributes only $\log \binom{N_m}{\ell_m} = O(m)$. Hence the probability tends to zero whenever $s > \kappa / d_{\mathbb{F}}$, proving the claim. □

**Corollary 5.13** (Moving-excess rectangular finite budget)**.** *Let $m \le \ell_m \le N_m$, set $p_m = \ell_m - m + 1$, and let $A_m \in \mathbb{F}^{N_m \times m}$, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, have independent entries whose $d_{\mathbb{F}}$-dimensional scalar densities are bounded by a deterministic $K_m \ge 1$, uniformly in their position. Put*

$$M_m^{(\ell)} = \min_{\substack{T \subset [N_m] \\ |T| = \ell_m}} \sigma_{\min}(A_{m,T}), \qquad q_m = d_{\mathbb{F}} p_m,$$

*and let $v_q$ denote the Euclidean unit-ball volume in $\mathbb{R}^q$. Define*

$$B_m = 1 \vee \left[ \binom{N_m}{\ell_m} m K_m^{p_m} \binom{\ell_m}{p_m}^{d_{\mathbb{F}}/2} v_{q_m} m^{q_m/2} \right].$$

*Then, for every $u_m > 0$,*

$$\mathbb{P}\left\{ M_m^{(\ell)} \le \exp\left[ -\frac{\log B_m + u_m}{d_{\mathbb{F}} p_m} \right] \right\} \le e^{-u_m}.$$

*On any common coupling, if $\sum_m e^{-u_m} < \infty$, then the displayed bad event occurs only finitely often almost surely.*

*Proof.* For one fixed row set $T$, repeat the proof of Lemma 5.2 without treating $p_m$ as fixed. The same Cauchy–Binet coordinate-minor argument gives, for $t > 0$,

$$\mathbb{P}\{\sigma_{\min}(A_{m,T}) \le t\} \le m K_m^{p_m} \binom{\ell_m}{p_m}^{d_{\mathbb{F}}/2} v_{d_{\mathbb{F}} p_m} (\sqrt{m}\, t)^{d_{\mathbb{F}} p_m}.$$

Summing over the $\binom{N_m}{\ell_m}$ possible row sets and applying Corollary 2.5 with exponent $d_{\mathbb{F}} p_m$ and the deterministic budget $B_m$ gives the finite-budget bound. The almost-sure statement is Corollary 2.6. □

**Corollary 5.14** (Random-prefix proportional-excess lower sides)**.** *Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. For each $m$, let $(R_{im})_{i \ge 1}$ be independent $\mathbb{F}^m$-valued rows with independent scalar entries whose $d_{\mathbb{F}}$-dimensional Lebesgue densities are bounded by a deterministic $K_m \ge 1$, uniformly in the row and column positions. Assume*

$$\kappa = \limsup_{m \to \infty} \frac{1}{m} \log K_m < \infty.$$

*Let $\ell_m/m \to \zeta \in (1,\gamma)$, and let $\mathcal{N}_m$ be integer-valued, possibly dependent on the row array, with $\mathcal{N}_m/m \to \gamma > 1$ in probability. For finite $T \subset \mathbb{N}$, let $A_{m,T}$ be the matrix with rows $(R_{im})_{i\in T}$, and put*

$$M_m^{(\ell)}(\mathcal{N}_m) = \min_{\substack{T\subset[\mathcal{N}_m]\\ |T|=\ell_m}} \sigma_{\min}(A_{m,T}), \qquad \Lambda_m^{(\ell)}(\mathcal{N}_m) = \min_{\substack{T\subset[\mathcal{N}_m]\\ |T|=\ell_m}} \lambda_{\min}\left(\ell_m^{-1} A_{m,T}^* A_{m,T}\right),$$

*with the convention that both minima are $+\infty$ when $\mathcal{N}_m < \ell_m$. Then for every $\varepsilon > 0$,*

$$\mathbb{P}\left\{ M_m^{(\ell)}(\mathcal{N}_m) \le \exp\left[ -\left( \frac{\kappa}{d_{\mathbb{F}}} + \varepsilon \right) m \right] \right\} \to 0$$

*and*

$$\mathbb{P}\left\{ \Lambda_m^{(\ell)}(\mathcal{N}_m) \le \exp\left[ -\left( \frac{2\kappa}{d_{\mathbb{F}}} + \varepsilon \right) m \right] \right\} \to 0.$$

*If the row arrays are realized on a common probability space and $\mathcal{N}_m/m \to \gamma$ almost surely, then*

$$\liminf_{m\to\infty} \frac{1}{m} \log M_m^{(\ell)}(\mathcal{N}_m) \ge -\frac{\kappa}{d_{\mathbb{F}}}, \qquad \liminf_{m\to\infty} \frac{1}{m} \log \Lambda_m^{(\ell)}(\mathcal{N}_m) \ge -\frac{2\kappa}{d_{\mathbb{F}}} \quad \text{almost surely.}$$

*In particular, for $K_m = e^{o(m)}$, the random-prefix proportional-excess singular and normalized column-Gram lower envelopes have exponential-scale base 0. For iid standard Gaussian rows, the fixed-block upper bound also gives $m^{-1}\log M_m^{(\ell)}(\mathcal{N}_m) \to 0$ in probability. This is an exponential-scale lower-side transfer only; it does not identify the $\sqrt{m}$ Wishart edge, the left-tail large-deviation constant, or any upper/exact proportional-edge law.*

*Proof.* Fix $\gamma_+ > \gamma$ and set $n_m^+ = \lceil \gamma_+ m \rceil$. Since $\zeta < \gamma < \gamma_+$, the deterministic-count Proposition 5.12 applies to the first $n_m^+$ rows. Write $M_m^{(\ell)}(n)$ for the same minimum with $n$ in place of $\mathcal{N}_m$. On the event $\{\mathcal{N}_m \le n_m^+\}$, monotonicity in the available prefix gives

$$M_m^{(\ell)}(\mathcal{N}_m) \ge M_m^{(\ell)}(n_m^+),$$

where the $+\infty$ convention covers the infeasible case. Hence

$$\mathbb{P}\left\{ M_m^{(\ell)}(\mathcal{N}_m) \le e^{-sm} \right\} \le \mathbb{P}\{\mathcal{N}_m > n_m^+\} + \mathbb{P}\left\{ M_m^{(\ell)}(n_m^+) \le e^{-sm} \right\}.$$

Taking $s = \kappa/d_{\mathbb{F}} + \varepsilon$ proves the singular-value convergence in probability. The same deterministic upper bracket is eventually valid almost surely under a.s. convergence of $\mathcal{N}_m/m$, and Corollary 5.13, applied to the deterministic prefix $n_m^+$ with a summable error budget, gives the corresponding a.s. lower envelope.

Finally,

$$\Lambda_m^{(\ell)}(\mathcal{N}_m) = \ell_m^{-1}\left(M_m^{(\ell)}(\mathcal{N}_m)\right)^2$$

pathwise. Since $\log \ell_m = o(m)$, the normalized column-Gram bounds follow from the singular-value bounds with the exponent doubled. In the Gaussian case, on $\{\mathcal{N}_m \ge \ell_m\}$, the fixed set $\{1,\dots,\ell_m\}$ is admissible and $\sigma_{\min}(A_{m,\{1,\dots,\ell_m\}}) = O_{\mathbb{P}}(\sqrt{m})$, giving the displayed upper exponential scale. □

**Corollary 5.15** (Random-prefix proportional-excess conditioning transfer)**.** *In the setting of Corollary 5.14, let $d_{im} > 0$, $i \ge 1$, be finite real row weights, and let $S_m \in GL_m(\mathbb{F})$ be one global all-column right factor. The weights and $S_m$ may be random and dependent on $\mathcal{N}_m$ and on the row array. Assume*

$$\Delta_m^{\mathrm{propcond}} = \max_{1\le i\le \mathcal{N}_m} |\log d_{im}| + \left|\log \sigma_{\min}(S_m)\right| + \left|\log \|S_m\|\right| = o_{\mathbb{P}}(m),$$

*omitting the row-weight term when row weights are absent. For $\mathcal{N}_m \ge \ell_m$ and $T \subset [\mathcal{N}_m]$, $|T| = \ell_m$, set*

$$B_{m,T} = \mathrm{diag}(d_{im} : i \in T)\, A_{m,T} S_m$$

*and define*

$$M_{m,B}^{(\ell)} = \min_{\substack{T\subset[\mathcal{N}_m]\\|T|=\ell_m}} \sigma_{\min}(B_{m,T}), \qquad \Lambda_{m,B}^{(\ell)} = \min_{\substack{T\subset[\mathcal{N}_m]\\|T|=\ell_m}} \lambda_{\min}\big(\ell_m^{-1}B_{m,T}^*B_{m,T}\big)\,,$$

*with the convention that both minima are $+\infty$ on $\{\mathcal{N}_m < \ell_m\}$. Then for every $\varepsilon > 0$,*

$$\mathbb{P}\left\{M_{m,B}^{(\ell)} \le \exp\left[-\left(\frac{\kappa}{d_{\mathbb{F}}}+\varepsilon\right)m\right]\right\} \to 0$$

*and*

$$\mathbb{P}\left\{\Lambda_{m,B}^{(\ell)} \le \exp\left[-\left(\frac{2\kappa}{d_{\mathbb{F}}}+\varepsilon\right)m\right]\right\} \to 0.$$

*On a common coupling, if $\mathcal{N}_m/m \to \gamma$ almost surely and $\Delta_m^{\mathrm{propcond}} = o(m)$ almost surely, then*

$$\liminf_{m\to\infty}\frac{1}{m}\log M_{m,B}^{(\ell)} \ge -\frac{\kappa}{d_{\mathbb{F}}}, \qquad \liminf_{m\to\infty}\frac{1}{m}\log \Lambda_{m,B}^{(\ell)} \ge -\frac{2\kappa}{d_{\mathbb{F}}} \quad \text{almost surely.}$$

*Positive column diagonals are included by taking $S_m$ diagonal. Full-column covariance and whitening factors are included only through the absolute singular-scale hypothesis above. For Gaussian row arrays, row normalization over the realized prefix and full-prefix column normalization are included when their logarithmic norm extrema are absorbed into $\Delta_m^{\mathrm{propcond}}$.*

*Proof.* Write $A_m(n)$ for the first $n$ unweighted rows and let

$$M_{m,A}^{(\ell)}(n) = \min_{\substack{T\subset[n]\\|T|=\ell_m}} \sigma_{\min}(A_{m,T}), \qquad \Lambda_{m,A}^{(\ell)}(n) = \min_{\substack{T\subset[n]\\|T|=\ell_m}} \lambda_{\min}\big(\ell_m^{-1}A_{m,T}^*A_{m,T}\big)\,,$$

again with the $+\infty$ convention on infeasible prefixes. Fix $\gamma_+ > \gamma$ and set $n_m^+ = \lceil \gamma_+ m\rceil$. On the event $\mathcal{N}_m \le n_m^+$, the min–max singular-value sandwich gives, for every feasible $T \subset [\mathcal{N}_m]$,

$$\sigma_{\min}(B_{m,T}) \ge e^{-\Delta_m^{\mathrm{propcond}}}\sigma_{\min}(A_{m,T}).$$

Therefore

$$M_{m,B}^{(\ell)} \ge e^{-\Delta_m^{\mathrm{propcond}}} M_{m,A}^{(\ell)}(n_m^+).$$

With $c = \kappa/d_{\mathbb{F}}$,

$$\begin{aligned}\mathbb{P}\{M_{m,B}^{(\ell)} \le e^{-(c+\varepsilon)m}\} \le{}& \mathbb{P}\{\Delta_m^{\mathrm{propcond}} > \varepsilon m/2\} + \mathbb{P}\{\mathcal{N}_m > n_m^+\}\\ &+ \mathbb{P}\{M_{m,A}^{(\ell)}(n_m^+) \le e^{-(c+\varepsilon/2)m}\}.\end{aligned}$$

The first term tends to zero by hypothesis, the second by $\mathcal{N}_m/m \to \gamma$ in probability, and the third by Proposition 5.12 applied to the deterministic prefix $n_m^+$. This proves the singular lower side.

The normalized column-Gram comparison is the squared sandwich:

$$\Lambda_{m,B}^{(\ell)} \ge e^{-2\Delta_m^{\mathrm{propcond}}}\Lambda_{m,A}^{(\ell)}(n_m^+)$$

on $\{\mathcal{N}_m \le n_m^+\}$. Hence the same argument, with $\varepsilon m/4$ in the $\Delta_m^{\mathrm{propcond}}$ event and the doubled threshold from Corollary 5.14, gives the displayed Gram lower side. The almost-sure lower envelopes follow from the eventual deterministic prefix bracket, the a.s. lower envelope in Corollary 5.14, and $\Delta_m^{\mathrm{propcond}} = o(m)$.

This is only a pathwise conditioning transfer from the independent-entry base array. It does not assert that the weighted or right-multiplied matrices satisfy the density hypothesis. A condition-number bound alone is not enough, because a scalar factor can shift the exponential scale. The statement does not assert a quaternion extension, value-dependent row-selection theorem, selected-column covariance transfer, selected-submatrix column normalization, row-Gram result, proportional Wishart-edge or LDP constant, upper side, or exact proportional-edge law. □

*Remark* 5.16 (Finer scale at proportional excess). At proportional excess the right normalization is $\sqrt{m}$, not exponential. The least singular value of one $\ell_m \times m$ Gaussian matrix concentrates at the lower Marchenko–Pastur edge on the unnormalized singular-value scale, $\sqrt{\ell_m} - \sqrt{m} \sim (\sqrt{\zeta} - 1)\sqrt{m}$ [11, 9]. Minimizing over the $\exp\{H_{\mathrm{row}}(\gamma, \zeta)m + o(m)\}$ submatrices, where

$$H_{\mathrm{row}}(\gamma, \zeta) = \gamma \log \gamma - \zeta \log \zeta - (\gamma - \zeta) \log(\gamma - \zeta), \qquad 1 < \zeta < \gamma,$$

pulls the minimum below the edge to a value $x_* \sqrt{m}$ with $x_* \in (0, \sqrt{\zeta} - 1)$ determined by balancing $H_{\mathrm{row}}(\gamma, \zeta)$ against the left-tail large-deviation rate for $\sigma_{\min}(A_{m,T})/\sqrt{m}$. Identifying $x_*$ is a different problem from the hard-edge theorem and is not pursued here.

5.3. **The $k$-th smallest singular value.** Keep the submatrices square but track a higher singular value. For $A_m \in \mathbb{F}^{N_m \times m}$ with iid standard Gaussian entries and $N_m/m \to \gamma > 1$, order the singular values of each square submatrix as $\sigma_1(A_{m,T}) \geq \cdots \geq \sigma_m(A_{m,T})$, and for a fixed integer $k \geq 1$ put

$$M_m[k] = \min_{\substack{T \subset [N_m] \\ |T| = m}} \sigma_{m-k+1}(A_{m,T}).$$

Here $\sigma_{m-k+1}$ is the $k$-th smallest singular value, and $M_m[1] = M_m$.

The one-matrix input is the hard-edge small-ball exponent of the $k$-th smallest singular value of a single $m \times m$ $\mathbb{F}$-Gaussian matrix $G$. Let $\lambda_{(1)} \leq \cdots \leq \lambda_{(m)}$ be the eigenvalues of $G^*G$, the square Laguerre ensemble at rectangular excess $r = 0$; near the origin the joint density carries the Vandermonde repulsion $\prod_{i<j} |\lambda_i - \lambda_j|^{d_{\mathbb{F}}}$ and the weight $\prod_i \lambda_i^{d_{\mathbb{F}}/2-1}$. Rescaling the $k$ smallest eigenvalues by $\delta$ collects a power $d_{\mathbb{F}}\binom{k}{2}$ from the repulsion among them, $k(d_{\mathbb{F}}/2 - 1)$ from the weight, and $k$ from the measure, for a total exponent $d_{\mathbb{F}}k^2/2$. Hence $\mathbb{P}\{\lambda_{(k)} \leq \delta\} \asymp \delta^{\,d_{\mathbb{F}}k^2/2}$, and since $\sigma_{m-k+1} = \sqrt{\lambda_{(k)}}$,

$$\mathbb{P}\{\sigma_{m-k+1}(G) \leq t\} \asymp t^{\,e_k}, \qquad e_k = d_{\mathbb{F}}k^2,$$

as $t \to 0$; see [9]. At $k = 1$ this is $e_1 = d_{\mathbb{F}}$, recovering the least-singular-value exponent.

The exponent computation alone is not the finite-budget input. The lower side below uses the following uniform finite-$m$, polynomial-loss version, obtained from the Laguerre–Selberg normalization formula.

**Lemma 5.17** (Fixed-$k$ Gaussian Laguerre tail)**.** *Fix $k \geq 1$ and $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. There are constants $C_{\mathbb{F},k} \geq 1$ and $B_{\mathbb{F},k} \geq 0$ such that, for every $m \geq k$, every $m \times m$ matrix $G$ with iid standard $\mathbb{F}$-Gaussian entries, and every $0 < t \leq 1$,*

$$\mathbb{P}\{\sigma_{m-k+1}(G) \leq t\} \leq C_{\mathbb{F},k} m^{B_{\mathbb{F},k}} t^{\,d_{\mathbb{F}}k^2}.$$

*Proof.* Put $b = d_{\mathbb{F}} \in \{1, 2\}$, $q = b/2$, and $\delta = t^2$. In ordered coordinates $0 \leq \lambda_1 \leq \cdots \leq \lambda_m$, the eigenvalues of $G^*G$ have Laguerre density

$$\frac{m!}{Z_m(q)} e^{-\eta \sum_i \lambda_i} \prod_{i=1}^{m} \lambda_i^{q-1} \prod_{i<j} (\lambda_j - \lambda_i)^b,$$

where $\eta = 1/2$ in the real normalization and $\eta = 1$ in the complex normalization. This is the standard Laguerre ensemble density; we use the Laguerre–Selberg product formula for $Z_m(q)$, see [9, Chs. 3–4].

On the event $\lambda_k \leq \delta$, write $x_i = \lambda_i$ for $1 \leq i \leq k$ and $y_j = \lambda_{k+j}$ for $1 \leq j \leq m - k$. Since $0 \leq x_i \leq y_j$,

$$\prod_{i \leq k < j} (\lambda_j - \lambda_i)^b \leq \prod_{j=1}^{m-k} y_j^{bk}.$$

The integral over the first $k$ variables contributes

$$\int_{[0,\delta]^k} \prod_i x_i^{q-1} \prod_{i<j} |x_i - x_j|^b \, dx = C_{b,k} \delta^{kq + b\binom{k}{2}} = C_{b,k} \delta^{bk^2/2}.$$

The remaining variables are bounded by a Laguerre–Selberg integral with parameter $q + bk$. Thus

$$\mathbb{P}\{\lambda_k \le \delta\} \le C_{b,k} \delta^{bk^2/2} \frac{m!}{(m-k)!} \frac{Z_{m-k}(q+bk)}{Z_m(q)}.$$

For fixed $k$, the product formula for $Z_n(a)$ and Stirling's formula give

$$\frac{m!}{(m-k)!} \frac{Z_{m-k}(q+bk)}{Z_m(q)} \le C_{b,k} m^{B_{b,k}},$$

after absorbing the finitely many cases $m < 2k$ into the constant. Therefore

$$\mathbb{P}\{\lambda_k \le \delta\} \le C_{\mathbb{F},k} m^{B_{\mathbb{F},k}} \delta^{d_{\mathbb{F}} k^2/2}.$$

Since $\sigma_{m-k+1}(G) = \sqrt{\lambda_k}$, taking $\delta = t^2$ gives the claimed bound. □

**Lemma 5.18** (Fixed-excess fixed-$k$ Gaussian Laguerre tail)**.** *Fix $r \ge 0$, $k \ge 1$, and $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. There are constants $C_{\mathbb{F},r,k} \ge 1$ and $B_{\mathbb{F},r,k} \ge 0$ such that, for every $m \ge k$, every $G \in \mathbb{F}^{(m+r)\times m}$ with iid standard $\mathbb{F}$-Gaussian entries, and every $0 < t \le 1$,*

$$\mathbb{P}\{\sigma_{m-k+1}(G) \le t\} \le C_{\mathbb{F},r,k} m^{B_{\mathbb{F},r,k}} t^{\, d_{\mathbb{F}} k(k+r)}.$$

*Proof.* Put $b = d_{\mathbb{F}} \in \{1, 2\}$, $q = b(r+1)/2$, and $\delta = t^2$. In ordered coordinates $0 \le \lambda_1 \le \cdots \le \lambda_m$, the eigenvalues of $G^*G$ have Laguerre density

$$\frac{m!}{Z_m(q)} e^{-\eta \sum_i \lambda_i} \prod_{i=1}^m \lambda_i^{q-1} \prod_{i<j} (\lambda_j - \lambda_i)^b,$$

with the same real/complex value of $\eta$ as above. On $\lambda_k \le \delta$, write $x_i = \lambda_i$ for $1 \le i \le k$ and $y_j = \lambda_{k+j}$ for $1 \le j \le m-k$. Since $0 \le x_i \le y_j$,

$$\prod_{i \le k < j} (\lambda_j - \lambda_i)^b \le \prod_{j=1}^{m-k} y_j^{bk}.$$

The first $k$ variables contribute

$$\int_{[0,\delta]^k} \prod_i x_i^{q-1} \prod_{i<j} |x_i - x_j|^b \, dx = C_{b,r,k} \delta^{kq + b\binom{k}{2}} = C_{b,r,k} \delta^{bk(k+r)/2}.$$

The remaining variables are bounded by a Laguerre–Selberg integral with parameter $q + bk$. Hence, with $Z_0 = 1$,

$$\mathbb{P}\{\lambda_k \le \delta\} \le C_{b,r,k} \delta^{bk(k+r)/2} \frac{m!}{(m-k)!} \frac{Z_{m-k}(q+bk)}{Z_m(q)}.$$

For fixed $r, k$, the Laguerre–Selberg product and Stirling's formula give

$$\frac{m!}{(m-k)!} \frac{Z_{m-k}(q+bk)}{Z_m(q)} \le C_{b,r,k} m^{B_{b,r,k}},$$

after absorbing the finitely many small $m$ into the constant. Taking $\delta = t^2$ proves the displayed singular-value tail. □

**Theorem 5.19** (Fixed-excess fixed-$k$ Gaussian finite-budget lower side)**.** *Fix $r \geq 0$, $k \geq 1$, let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, let $N_m \geq m + r$, and let $A_m \in \mathbb{F}^{N_m \times m}$ have iid standard $\mathbb{F}$-Gaussian entries. Put*

$$M_m^{(r)}[k] = \min_{\substack{T \subset [N_m] \\ |T| = m+r}} \sigma_{m-k+1}(A_{m,T}).$$

*Then $M_m^{(r)}[1] = M_m^{(r)}$ and $M_m^{(0)}[k] = M_m[k]$. There are constants $C_{\mathbb{F},r,k} \geq 1$ and $B_{\mathbb{F},r,k} \geq 0$ such that, for every $u_m > 0$,*

$$\mathbb{P}\left\{ M_m^{(r)}[k] \leq \exp\left[ -\frac{\log \binom{N_m}{m+r} + B_{\mathbb{F},r,k} \log m + \log C_{\mathbb{F},r,k} + u_m}{d_{\mathbb{F}} k(k+r)} \right] \right\} \leq e^{-u_m}.$$

*Proof.* Apply Lemma 5.18 to every selected $(m+r)$-row submatrix. For $0 < t \leq 1$,

$$\mathbb{P}\{M_m^{(r)}[k] \leq t\} \leq \binom{N_m}{m+r} C_{\mathbb{F},r,k} m^{B_{\mathbb{F},r,k}} t^{d_{\mathbb{F}} k(k+r)}.$$

Substituting the displayed finite-budget threshold gives the result. □

**Corollary 5.20** (Fixed-excess fixed-$k$ Gaussian lower side)**.** *Under the hypotheses of Theorem 5.19, if $N_m/m \to \gamma > 1$, then for every $\varepsilon > 0$,*

$$\mathbb{P}\left\{ M_m^{(r)}[k] \leq \exp\left[ -\left( \frac{h(\gamma)}{d_{\mathbb{F}} k(k+r)} + \varepsilon \right) m \right] \right\} \longrightarrow 0.$$

*Equivalently,*

$$\liminf_{m \to \infty} \frac{1}{m} \log M_m^{(r)}[k] \geq -\frac{h(\gamma)}{d_{\mathbb{F}} k(k+r)} \qquad \text{in probability.}$$

*Proof.* Use $m^{-1} \log \binom{N_m}{m+r} \to h(\gamma)$ and take $u_m = d_{\mathbb{F}} k(k+r)\varepsilon m$, absorbing the logarithmic terms into the slack. □

**Corollary 5.21** (Fixed-excess fixed-$k$ Gram eigenvalue lower side)**.** *Under the hypotheses of Theorem 5.19, write*

$$0 \leq \lambda_{(1)}(C) \leq \cdots \leq \lambda_{(m)}(C)$$

*for the eigenvalues of a positive semidefinite $m \times m$ matrix, and put*

$$\Lambda_m^{(r)}[k] = \min_{\substack{T \subset [N_m] \\ |T| = m+r}} \lambda_{(k)}\left( (m+r)^{-1} A_{m,T}^* A_{m,T} \right).$$

*With the constants $C_{\mathbb{F},r,k}, B_{\mathbb{F},r,k}$ of Theorem 5.19, for every $u_m > 0$,*

$$\mathbb{P}\left\{ \Lambda_m^{(r)}[k] \leq (m+r)^{-1} \exp\left[ -\frac{2\left( \log \binom{N_m}{m+r} + B_{\mathbb{F},r,k} \log m + \log C_{\mathbb{F},r,k} + u_m \right)}{d_{\mathbb{F}} k(k+r)} \right] \right\} \leq e^{-u_m}.$$

*Consequently, if $N_m/m \to \gamma > 1$, then for every $\varepsilon > 0$,*

$$\mathbb{P}\left\{ \Lambda_m^{(r)}[k] \leq \exp\left[ -\left( \frac{2h(\gamma)}{d_{\mathbb{F}} k(k+r)} + \varepsilon \right) m \right] \right\} \longrightarrow 0,$$

*or equivalently*

$$\liminf_{m \to \infty} \frac{1}{m} \log \Lambda_m^{(r)}[k] \geq -\frac{2h(\gamma)}{d_{\mathbb{F}} k(k+r)} \qquad \text{in probability.}$$

*Proof.* For each row set $T$, the eigenvalues of $A_{m,T}^* A_{m,T}$ are the squared singular values in increasing order:

$$\lambda_{(k)}(A_{m,T}^* A_{m,T}) = \sigma_{m-k+1}(A_{m,T})^2.$$

Therefore, pathwise,

$$\Lambda_m^{(r)}[k] = (m+r)^{-1} \left( M_m^{(r)}[k] \right)^2.$$

The finite-budget estimate follows from Theorem 5.19 by squaring the singular-value threshold and multiplying by $(m+r)^{-1}$. The fixed-aspect statement follows because $m^{-1}\log(m+r)\to 0$. □

*Remark* 5.22. These fixed-excess fixed-$k$ singular-value and Gram eigenvalue statements are only lower-side statements. The matching convergence

$$\frac{1}{m}\log M_m^{(r)}[k] \xrightarrow{\mathbb{P}} -\frac{h(\gamma)}{d_{\mathbb{F}}k(k+r)}$$

would require a fixed-$r$, fixed-$k$ pair-correlation or second-moment theorem for overlapping $(m+r)$-row submatrices. That input is not proved here, so no exact law or almost-sure exact law is asserted.

**Theorem 5.23** (Fixed-$k$ Gaussian finite-budget lower side)**.** *Fix $k\ge 1$, let $N_m\ge m$, and let $A_m\in\mathbb{F}^{N_m\times m}$ have iid standard $\mathbb{F}$-Gaussian entries. There are constants $C_{\mathbb{F},k}\ge 1$ and $B_{\mathbb{F},k}\ge 0$ such that, for every $u_m>0$,*

$$\mathbb{P}\left\{M_m[k]\le \exp\left[-\frac{\log\binom{N_m}{m}+B_{\mathbb{F},k}\log m+\log C_{\mathbb{F},k}+u_m}{d_{\mathbb{F}}k^2}\right]\right\}\le e^{-u_m}.$$

*Consequently, if $N_m/m\to\gamma>1$, then for every $\varepsilon>0$,*

$$\mathbb{P}\left\{M_m[k]\le \exp\left[-\left(\frac{h(\gamma)}{d_{\mathbb{F}}k^2}+\varepsilon\right)m\right]\right\}\longrightarrow 0.$$

*Equivalently,*

$$\liminf_{m\to\infty}\frac{1}{m}\log M_m[k]\ge -\frac{h(\gamma)}{d_{\mathbb{F}}k^2}\qquad \text{in probability.}$$

*Proof.* Apply Lemma 5.17 to each selected square row submatrix. For $0<t\le 1$,

$$\mathbb{P}\{M_m[k]\le t\}\le\binom{N_m}{m}C_{\mathbb{F},k}m^{B_{\mathbb{F},k}}t^{d_{\mathbb{F}}k^2}.$$

Substituting the displayed finite-budget threshold gives the $e^{-u_m}$ bound. If $N_m/m\to\gamma>1$, then $m^{-1}\log\binom{N_m}{m}\to h(\gamma)$, and taking $u_m=d_{\mathbb{F}}k^2\varepsilon m$ gives the fixed-aspect lower side after absorbing the logarithmic terms into the slack. □

**Corollary 5.24** (Fixed-$k$ Gram eigenvalue lower side)**.** *Fix $k\ge 1$, let $\mathbb{F}\in\{\mathbb{R},\mathbb{C}\}$, let $N_m\ge m$, and let $A_m\in\mathbb{F}^{N_m\times m}$ have iid standard $\mathbb{F}$-Gaussian entries. For a positive semidefinite $m\times m$ matrix $C$, write its eigenvalues as*

$$0\le\lambda_{(1)}(C)\le\cdots\le\lambda_{(m)}(C).$$

*Put*

$$\Lambda_m[k]=\min_{\substack{T\subset[N_m]\\|T|=m}}\lambda_{(k)}\left(m^{-1}A_{m,T}^*A_{m,T}\right).$$

*With the constants $C_{\mathbb{F},k},B_{\mathbb{F},k}$ of Theorem 5.23, for every $u_m>0$,*

$$\mathbb{P}\left\{\Lambda_m[k]\le m^{-1}\exp\left[-\frac{2\left(\log\binom{N_m}{m}+B_{\mathbb{F},k}\log m+\log C_{\mathbb{F},k}+u_m\right)}{d_{\mathbb{F}}k^2}\right]\right\}\le e^{-u_m}.$$

*Consequently, if $N_m/m\to\gamma>1$, then for every $\varepsilon>0$,*

$$\mathbb{P}\left\{\Lambda_m[k]\le \exp\left[-\left(\frac{2h(\gamma)}{d_{\mathbb{F}}k^2}+\varepsilon\right)m\right]\right\}\longrightarrow 0,$$

*or equivalently*

$$\liminf_{m\to\infty}\frac{1}{m}\log\Lambda_m[k]\ge -\frac{2h(\gamma)}{d_{\mathbb{F}}k^2}\qquad \text{in probability.}$$

*Proof.* For each row set $T$, the ordered eigenvalues of $A_{m,T}^* A_{m,T}$ are the squared singular values in increasing order:

$$\lambda_{(k)}(A_{m,T}^* A_{m,T}) = \sigma_{m-k+1}(A_{m,T})^2.$$

Hence, pathwise,

$$\Lambda_m[k] = m^{-1} M_m[k]^2.$$

Substitute the finite-budget threshold of Theorem 5.23 and square it. The fixed-aspect lower side follows because $m^{-1} \log m \to 0$. □

**Conjecture 5.25** (Extremal $k$-th hard edge)**.** For each fixed $k \geq 1$,

$$\frac{1}{m} \log M_m[k] \xrightarrow{\mathbb{P}} -\frac{h(\gamma)}{d_{\mathbb{F}} k^2}.$$

The lower side is Theorem 5.23. The upper side needs a $k$-dimensional analogue of the hyperplane second moment of Section 3: one fixes $k$ rows and compares the resulting codimension-$k$ small-ball events across overlapping row sets. This pair-correlation step is not carried out here, so the exact extremal base remains conjectural.

5.4. **Two-sided submatrices.** Finally, choose submatrices on both sides. Let $A \in \mathbb{F}^{N \times n}$ have iid standard Gaussian entries with $N/k \to \gamma_R > 1$ and $n/k \to \gamma_C > 1$, and for $R \subset [N]$, $C \subset [n]$ with $|R| = |C| = k$ write $A_{R,C} \in \mathbb{F}^{k \times k}$ for the square submatrix on those rows and columns. Put

$$M = \min_{\substack{R \subset [N],\, |R|=k \\ C \subset [n],\, |C|=k}} \sigma_{\min}(A_{R,C}).$$

The scaling parameter is the submatrix size $k$.

Because the choices on the two sides are independent, the entropy adds:

$$\frac{1}{k} \log \left( \binom{N}{k} \binom{n}{k} \right) = \frac{1}{k} \log \binom{N}{k} + \frac{1}{k} \log \binom{n}{k} \longrightarrow h(\gamma_R) + h(\gamma_C),$$

by Stirling's formula applied to each factor. Each $A_{R,C}$ is a square $k \times k$ $\mathbb{F}$-Gaussian matrix, so the one-matrix exponent is $d_{\mathbb{F}}$, as in Proposition 3.3.

**Theorem 5.26** (Two-sided lower side)**.** *With $H = h(\gamma_R) + h(\gamma_C)$, for every $\varepsilon > 0$,*

$$\mathbb{P}\left\{ M \leq \exp\left[ -\left( \frac{H}{d_{\mathbb{F}}} + \varepsilon \right) k \right] \right\} \longrightarrow 0,$$

*so* $\liminf_k \frac{1}{k} \log M \geq -\dfrac{h(\gamma_R) + h(\gamma_C)}{d_{\mathbb{F}}}$ *in probability.*

*Proof.* Apply Corollary 2.3, with the scaling index $m$ there taken to be $k$, to the family $\mathcal{T} = \{(R, C) : |R| = |C| = k\}$ and $Y_{R,C} = \sigma_{\min}(A_{R,C})$. The entropy bound is the display above. The uniform small-ball bound $\mathbb{P}\{\sigma_{\min}(A_{R,C}) \leq t\} \leq C_{\mathbb{F}} k^{1+d_{\mathbb{F}}} t^{d_{\mathbb{F}}}$ is Proposition 3.3. Hence the base is $H/d_{\mathbb{F}}$. □

**Theorem 5.27** (Two-sided density-rate lower side)**.** *In the two-sided setup above, replace the Gaussian entries by independent entries whose $d_{\mathbb{F}}$-dimensional scalar densities are bounded by deterministic $K_k \geq 1$, uniformly in their position, and put*

$$\kappa = \limsup_{k \to \infty} \frac{1}{k} \log K_k < \infty.$$

*With $H = h(\gamma_R) + h(\gamma_C)$, for every $\varepsilon > 0$,*

$$\mathbb{P}\left\{ M \leq \exp\left[ -\left( \frac{H + \kappa}{d_{\mathbb{F}}} + \varepsilon \right) k \right] \right\} \longrightarrow 0.$$

*In particular, $K_k = e^{o(k)}$ preserves the Gaussian two-sided lower-side base $H/d_{\mathbb{F}}$.*

*Proof.* For one selected $k \times k$ minor, the proof of Proposition 3.3 gives

$$\mathbb{P}\{\sigma_{\min}(A_{R,C}) \le t\} \le C_{\mathbb{F}} K_k k^{d_{\mathbb{F}}+1} t^{d_{\mathbb{F}}}, \qquad 0 < t \le 1.$$

Thus, at $t = e^{-sk}$,

$$\mathbb{P}\{M \le e^{-sk}\} \le \binom{N}{k}\binom{n}{k} C_{\mathbb{F}} K_k k^{d_{\mathbb{F}}+1} e^{-d_{\mathbb{F}} sk}.$$

The logarithmic limsup divided by $k$ is at most $H+\kappa-d_{\mathbb{F}}s$, which is negative for $s > (H+\kappa)/d_{\mathbb{F}}$. □

**Corollary 5.28** (Finite-entropy two-sided lower side)**.** *Let $N_k, n_k \ge k$, and let $A_k \in \mathbb{F}^{N_k \times n_k}$, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, have independent entries whose $d_{\mathbb{F}}$-dimensional scalar densities are bounded by $K_k \ge 1$, uniformly in their position. Put*

$$L_k^{\mathrm{two}} = \log \binom{N_k}{k} + \log \binom{n_k}{k}, \qquad M_k^{\mathrm{two}} = \min_{\substack{R \subset [N_k],\, |R|=k \\ C \subset [n_k],\, |C|=k}} \sigma_{\min}((A_k)_{R,C}).$$

*Then, for every $u_k > 0$,*

$$\mathbb{P}\left\{ M_k^{\mathrm{two}} \le \exp\left[ -\frac{L_k^{\mathrm{two}} + \log K_k + (d_{\mathbb{F}}+1)\log k + O_{\mathbb{F}}(1) + u_k}{d_{\mathbb{F}}} \right] \right\} \le e^{-u_k}.$$

*In particular, taking $u_k = d_{\mathbb{F}} \varepsilon k$ gives the convergence-in-probability lower side at the finite two-sided entropy budget $(L_k^{\mathrm{two}} + \log K_k)/d_{\mathbb{F}}$, with any fixed exponential slack. If the matrices are coupled for all $k$ and $\sum_k e^{-u_k} < \infty$, then the displayed bad event occurs only finitely often almost surely, with the $O_{\mathbb{F}}(1)$ interpreted as one fixed constant. This almost-sure finite-budget upgrade still uses only the union bound; overlap between row-column minors is irrelevant for this lower-side statement.*

*Proof.* For one selected $k \times k$ minor, the proof of Proposition 3.3 gives

$$\mathbb{P}\{\sigma_{\min}((A_k)_{R,C}) \le t\} \le C_{\mathbb{F}} K_k k^{d_{\mathbb{F}}+1} t^{d_{\mathbb{F}}}, \qquad 0 < t \le 1.$$

Apply Corollary 2.4 to the family of row-column pairs. Only a union bound is used; overlap between two tested minors is irrelevant for this lower-side estimate. □

**Corollary 5.29** (Heterogeneous-density two-sided finite budget)**.** *Let $N_k, n_k \ge k$, and let $A_k \in \mathbb{F}^{N_k \times n_k}$, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, have independent entries. Assume that entry $(i,j)$ has a $d_{\mathbb{F}}$-dimensional Lebesgue density bounded by a deterministic number $K_{k,i,j} \ge 1$. For $C \subset [n_k]$, $|C| = k$, put*

$$K_{k,i,C} = \max_{j \in C} K_{k,i,j},$$

*and define*

$$\Lambda_k^{\mathrm{two,het}} = \log \binom{N_k - 1}{k-1} + \log \left( \sum_{i=1}^{N_k} \sum_{\substack{C \subset [n_k] \\ |C|=k}} K_{k,i,C} \right).$$

*For*

$$M_k^{\mathrm{two}} = \min_{\substack{R \subset [N_k],\, |R|=k \\ C \subset [n_k],\, |C|=k}} \sigma_{\min}((A_k)_{R,C}),$$

*every $u_k > 0$ satisfies*

$$\mathbb{P}\left\{ M_k^{\mathrm{two}} \le \exp\left[ -\frac{\Lambda_k^{\mathrm{two,het}} + d_{\mathbb{F}} \log k + O_{\mathbb{F}}(1) + u_k}{d_{\mathbb{F}}} \right] \right\} \le e^{-u_k}.$$

*If the matrices are coupled for all $k$ and $\sum_k e^{-u_k} < \infty$, then the displayed bad event occurs only finitely often almost surely, with the $O_{\mathbb{F}}(1)$ interpreted as one fixed constant. In the uniform case $K_{k,i,j} = K_k$, this recovers Corollary 5.28.*

*Proof.* Fix $R, C$ with $|R| = |C| = k$. As in the proof of Proposition 3.3, the negative second-moment identity implies that $\sigma_{\min}((A_k)_{R,C}) \le t$ forces, for some $i \in R$, the $i$-th row distance to be at most $\sqrt{k}\, t$. Conditioning on the other rows of the minor and on all but one coordinate of the chosen row, select a coordinate $j \in C$ of the unit normal with modulus at least $k^{-1/2}$. The remaining scalar density is bounded by $K_{k,i,j} k^{d_{\mathbb{F}}/2} \le K_{k,i,C} k^{d_{\mathbb{F}}/2}$, so for $0 < t \le 1$,

$$\mathbb{P}\{\sigma_{\min}((A_k)_{R,C}) \le t\} \le C_{\mathbb{F}} k^{d_{\mathbb{F}}} t^{d_{\mathbb{F}}} \sum_{i \in R} K_{k,i,C}.$$

Summing these one-minor constants over all row-column choices gives

$$C_{\mathbb{F}} k^{d_{\mathbb{F}}} \sum_{\substack{|R|=k \\ |C|=k}} \sum_{i \in R} K_{k,i,C} = C_{\mathbb{F}} k^{d_{\mathbb{F}}} \binom{N_k - 1}{k-1} \sum_{i=1}^{N_k} \sum_{\substack{C \subset [n_k] \\ |C|=k}} K_{k,i,C}.$$

Apply Corollary 2.5 with exponent $d_{\mathbb{F}}$. Corollary 2.6 gives the stated almost-sure upgrade when $\sum_k e^{-u_k} < \infty$. When $K_{k,i,j} = K_k$, the last budget equals $C_{\mathbb{F}} K_k k^{d_{\mathbb{F}}+1} \binom{N_k}{k} \binom{n_k}{k}$, which is precisely the uniform finite-entropy budget up to the $O_{\mathbb{F}}(1)$ constant. □

**Corollary 5.30** (Two-sided smoothed deterministic lower side)**.** *Let $F_k \in \mathbb{F}^{N_k \times n_k}$ be arbitrary deterministic matrices, let $G_k$ have iid standard $\mathbb{F}$-Gaussian entries, and set $A_k = F_k + \sigma_k G_k$. Assume $N_k/k \to \gamma_R > 1$, $n_k/k \to \gamma_C > 1$, and*

$$\eta = \limsup_{k \to \infty} \frac{1}{k} \log \sigma_k^{-1} < \infty.$$

*For*

$$M_{\sigma,k}^{\mathrm{two}} = \min_{\substack{R \subset [N_k],\, |R|=k \\ C \subset [n_k],\, |C|=k}} \sigma_{\min}((A_k)_{R,C}),$$

*and $H = h(\gamma_R) + h(\gamma_C)$, every $\varepsilon > 0$ satisfies*

$$\mathbb{P}\left\{ M_{\sigma,k}^{\mathrm{two}} \le \exp\left[ -\left( \frac{H}{d_{\mathbb{F}}} + \eta + \varepsilon \right) k \right] \right\} \longrightarrow 0.$$

*Here $F_k$ is the deterministic background and $\sigma_k G_k$ the Gaussian perturbation of the observed matrix $A_k = F_k + \sigma_k G_k$.*

*Proof.* Translation by $F_k$ does not change density suprema. Scaling a standard $\mathbb{F}$-Gaussian scalar by $\sigma_k$ gives a $d_{\mathbb{F}}$-dimensional density bounded by $C_{\mathbb{F}} \sigma_k^{-d_{\mathbb{F}}}$. Apply Theorem 5.27. □

**Corollary 5.31** (Random-prefix two-sided density-rate lower side)**.** *Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. For each $k$, let $(a_{ij}^{(k)})_{i,j \ge 1}$ be an infinite array whose scalar entries are independent and whose $d_{\mathbb{F}}$-dimensional densities are bounded by a deterministic $K_k \ge 1$, uniformly in $i, j$. Put*

$$\kappa = \limsup_{k \to \infty} \frac{1}{k} \log K_k < \infty.$$

*Let $N_k^R$ and $N_k^C$ be integer-valued prefix counts, possibly dependent on the entry array and on each other, with*

$$\frac{N_k^R}{k} \xrightarrow{\mathbb{P}} \gamma_R > 1, \qquad \frac{N_k^C}{k} \xrightarrow{\mathbb{P}} \gamma_C > 1.$$

*For row sets $R$ and column sets $C$, write $A_{R,C}^{(k)}$ for the corresponding finite submatrix of the array, and define*

$$M_k^{\mathrm{rpref,two}} = \min_{\substack{R \subset [N_k^R],\, |R|=k \\ C \subset [N_k^C],\, |C|=k}} \sigma_{\min}(A_{R,C}^{(k)}),$$

*with the convention* $M_k^{\mathrm{rpref,two}} = +\infty$ *if either prefix has size less than* $k$*. Then, for every* $\varepsilon > 0$*,*

$$\mathbb{P}\left\{ M_k^{\mathrm{rpref,two}} \le \exp\left[ -\left( \frac{h(\gamma_R) + h(\gamma_C) + \kappa}{d_{\mathbb{F}}} + \varepsilon \right) k \right] \right\} \to 0.$$

*If the arrays and counts are realized on a common probability space and both prefix counts converge almost surely, then*

$$\liminf_{k\to\infty} \frac{1}{k} \log M_k^{\mathrm{rpref,two}} \ge -\frac{h(\gamma_R) + h(\gamma_C) + \kappa}{d_{\mathbb{F}}} \qquad \textit{almost surely.}$$

*Proof.* Fix $\gamma_R^+ > \gamma_R$, $\gamma_C^+ > \gamma_C$, and set

$$N_k^+ = \lceil \gamma_R^+ k \rceil, \qquad n_k^+ = \lceil \gamma_C^+ k \rceil.$$

Let $M_k^{\mathrm{two}}(N, n)$ denote the two-sided minimum formed from the deterministic prefixes $[N]$ and $[n]$, again with the $+\infty$ convention. On the event $\{N_k^R \le N_k^+,\, N_k^C \le n_k^+\}$, the random-prefix tested family is a subfamily of the deterministic-prefix family, so

$$M_k^{\mathrm{rpref,two}} \ge M_k^{\mathrm{two}}(N_k^+, n_k^+).$$

The probability of the complementary count event tends to zero. Applying Theorem 5.27 to the deterministic prefixes gives, for every fixed $\gamma_R^+, \gamma_C^+$,

$$\mathbb{P}\left\{ M_k^{\mathrm{two}}(N_k^+, n_k^+) \le \exp\left[ -\left( \frac{h(\gamma_R^+) + h(\gamma_C^+) + \kappa}{d_{\mathbb{F}}} + \varepsilon \right) k \right] \right\} \to 0.$$

Letting $\gamma_R^+ \downarrow \gamma_R$ and $\gamma_C^+ \downarrow \gamma_C$, and using continuity of $h$, proves the convergence-in-probability statement.

For the almost-sure lower envelope, take countable decreasing bracket sequences $\gamma_{R,j}^+ \downarrow \gamma_R$ and $\gamma_{C,j}^+ \downarrow \gamma_C$. For each fixed pair of brackets, Corollary 5.28 with a summable error budget gives the deterministic-prefix lower envelope almost surely. On the event of a.s. count convergence, the prefix counts are eventually bounded by those deterministic brackets. Intersecting over the countable bracket pairs and then sending $j \to \infty$ gives the displayed liminf.

This is a lower-side transfer for ordered prefixes only. It does not assert an upper side, exact two-sided law, pair-correlation estimate, quaternion extension, value-dependent thinning or reindexing theorem, selected-column covariance transfer, non-diagonal right-factor transfer, or upper/exact Gram/sample-covariance law. Diagonal row/column weighting is recorded separately in Corollary 5.32; the deterministic selected-minor normalized Gram lower-side transfer is recorded separately in Corollary 5.34. □

**Corollary 5.32** (Random-prefix two-sided diagonal conditioning transfer)**.** *In the setting of Corollary 5.31, let* $d_{i,k} > 0$*,* $i \ge 1$*, and* $e_{j,k} > 0$*,* $j \ge 1$*, be finite real row and column weights. The weights may be random and dependent on the entry array and on both prefix counts. Assume*

$$\Delta_k^{\mathrm{twodiag}} = \max_{1 \le i \le N_k^R} |\log d_{i,k}| + \max_{1 \le j \le N_k^C} |\log e_{j,k}| = o_{\mathbb{P}}(k),$$

*omitting a maximum when the corresponding weights are absent. For feasible* $R \subset [N_k^R]$*,* $C \subset [N_k^C]$*,* $|R| = |C| = k$*, set*

$$B_{R,C}^{(k)} = \mathrm{diag}(d_{i,k} : i \in R)\, A_{R,C}^{(k)}\, \mathrm{diag}(e_{j,k} : j \in C)$$

*and define*

$$M_{k,B}^{\mathrm{rpref,twodiag}} = \min_{\substack{R \subset [N_k^R],\, |R| = k \\ C \subset [N_k^C],\, |C| = k}} \sigma_{\min}(B_{R,C}^{(k)}),$$

*with value $+\infty$ if either prefix is infeasible. Then, for every $\varepsilon > 0$,*

$$\mathbb{P}\left\{ M_{k,B}^{\mathrm{rpref,twodiag}} \le \exp\left[ -\left( \frac{h(\gamma_R) + h(\gamma_C) + \kappa}{d_{\mathbb{F}}} + \varepsilon \right) k \right] \right\} \to 0.$$

*On a common coupling, if both prefix counts converge almost surely and $\Delta_k^{\mathrm{twodiag}} = o(k)$ almost surely, then*

$$\liminf_{k\to\infty} \frac{1}{k} \log M_{k,B}^{\mathrm{rpref,twodiag}} \ge -\frac{h(\gamma_R) + h(\gamma_C) + \kappa}{d_{\mathbb{F}}} \qquad \textit{almost surely.}$$

*Proof.* For every feasible pair $(R, C)$, the diagonal min–max sandwich gives

$$\sigma_{\min}(B_{R,C}^{(k)}) \ge \exp(-\Delta_k^{\mathrm{twodiag}})\, \sigma_{\min}(A_{R,C}^{(k)}).$$

Consequently

$$M_{k,B}^{\mathrm{rpref,twodiag}} \ge \exp(-\Delta_k^{\mathrm{twodiag}})\, M_k^{\mathrm{rpref,two}}.$$

With

$$c = \frac{h(\gamma_R) + h(\gamma_C) + \kappa}{d_{\mathbb{F}}},$$

we have

$$\begin{aligned}\mathbb{P}\{M_{k,B}^{\mathrm{rpref,twodiag}} \le e^{-(c+\varepsilon)k}\} \le\ & \mathbb{P}\{\Delta_k^{\mathrm{twodiag}} > \varepsilon k/2\} \\ & + \mathbb{P}\{M_k^{\mathrm{rpref,two}} \le e^{-(c+\varepsilon/2)k}\}.\end{aligned}$$

The first term tends to zero by hypothesis and the second by Corollary 5.31. The almost-sure lower envelope follows from the same pathwise inequality, the a.s. lower envelope in Corollary 5.31, and $\Delta_k^{\mathrm{twodiag}} = o(k)$.

This is only a pathwise transfer from the unweighted independent-entry base array. It does not assert that the diagonally weighted matrix satisfies the density hypothesis. Non-diagonal column covariance or right-factor conditioning does not commute with column selection and is not included. A condition-number bound alone is insufficient because a scalar row or column factor can shift the exponential scale. The statement does not assert an upper side, exact two-sided law, pair-correlation theorem, quaternion extension, value-dependent thinning or reindexing theorem, selected-submatrix renormalization theorem, row/column weighting beyond fixed global prefix diagonals, upper/exact Gram/sample-covariance law, or Wishart-edge/LDP statement. The deterministic selected-minor normalized Gram lower-side transfer is recorded separately in Corollary 5.34. □

**Corollary 5.33** (Gaussian random-prefix two-sided full-prefix normalization)**.** *Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. For each $k$, let $(g_{ij}^{(k)})_{i,j\ge 1}$ be an infinite iid standard $\mathbb{F}$-Gaussian array, and let $N_k^R, N_k^C$ be integer-valued prefix counts, possibly dependent on the array, such that*

$$\frac{N_k^R}{k} \xrightarrow{\mathbb{P}} \gamma_R > 1, \qquad \frac{N_k^C}{k} \xrightarrow{\mathbb{P}} \gamma_C > 1.$$

*On feasible prefixes define the full-prefix row and column norms*

$$\rho_{i,k} = \left( \sum_{j=1}^{N_k^C} |g_{ij}^{(k)}|^2 \right)^{1/2}, \qquad 1 \le i \le N_k^R,$$

*and*

$$\chi_{j,k} = \left( \sum_{i=1}^{N_k^R} |g_{ij}^{(k)}|^2 \right)^{1/2}, \qquad 1 \le j \le N_k^C.$$

*Choose $\alpha_R, \alpha_C \in \{0,1\}$. For $R \subset [N_k^R]$, $C \subset [N_k^C]$, $|R| = |C| = k$, set*

$$B_{R,C}^{(k)} = \operatorname{diag}(\rho_{i,k}^{-\alpha_R} : i \in R)\, G_{R,C}^{(k)}\, \operatorname{diag}(\chi_{j,k}^{-\alpha_C} : j \in C),$$

*where $G_{R,C}^{(k)} = (g_{ij}^{(k)})_{i\in R, j\in C}$, and define*

$$M_{k,B}^{\mathrm{rpref,norm}} = \min_{\substack{R\subset[N_k^R],\,|R|=k\\ C\subset[N_k^C],\,|C|=k}} \sigma_{\min}(B_{R,C}^{(k)}),$$

*with value $+\infty$ if either prefix is infeasible. Then, for every $\varepsilon > 0$,*

$$\mathbb{P}\left\{ M_{k,B}^{\mathrm{rpref,norm}} \le \exp\left[ -\left( \frac{h(\gamma_R) + h(\gamma_C)}{d_{\mathbb{F}}} + \varepsilon \right) k \right] \right\} \to 0.$$

*On a common coupling, if both prefix counts converge almost surely, then*

$$\liminf_{k\to\infty} \frac{1}{k} \log M_{k,B}^{\mathrm{rpref,norm}} \ge -\frac{h(\gamma_R) + h(\gamma_C)}{d_{\mathbb{F}}} \qquad \textit{almost surely.}$$

*Proof.* The row and column norm factors are computed once from the raw realized prefix before any row or column subsets are selected, and are then frozen. On feasible prefixes Gaussian rows and columns are nonzero almost surely. Thus this is a special case of Corollary 5.32 with $\kappa = 0$, once we prove

$$\Delta_k^{\mathrm{norm}} = \alpha_R \max_{i\le N_k^R} |\log \rho_{i,k}| + \alpha_C \max_{j\le N_k^C} |\log \chi_{j,k}| = o_{\mathbb{P}}(k),$$

and, under a.s. count convergence, $\Delta_k^{\mathrm{norm}} = o(k)$ almost surely.

We verify the norm extrema by deterministic prefix brackets; this avoids conditioning on the possibly entry-dependent realized counts. Fix $C_0 > \max\{\gamma_R, \gamma_C\}$. With probability tending to one, and eventually almost surely under a.s. count convergence,

$$k \le N_k^R, N_k^C \le C_0 k.$$

On this event every realized row norm is at most the norm of the first $\lceil C_0 k \rceil$ coordinates of the same row and at least the norm of the first $k$ coordinates of that row. Therefore, for every $\delta > 0$, Markov's inequality gives the summable upper-tail bound

$$\mathbb{P}\left\{ \max_{i\le C_0 k} \left( \sum_{j=1}^{\lceil C_0 k\rceil} |g_{ij}^{(k)}|^2 \right)^{1/2} > e^{\delta k} \right\} \le O(k^2 e^{-2\delta k}),$$

and the ball-volume estimate in real dimension $d_{\mathbb{F}} k$ gives the summable lower-tail bound

$$\mathbb{P}\left\{ \min_{i\le C_0 k} \left( \sum_{j=1}^{k} |g_{ij}^{(k)}|^2 \right)^{1/2} < e^{-\delta k} \right\} \le \exp\{-d_{\mathbb{F}}\delta k^2 + O(k\log k)\}.$$

The same two estimates, with rows and columns interchanged, control the full-prefix column norms $\chi_{j,k}$. Hence the displayed $\Delta_k^{\mathrm{norm}}$ is $o_{\mathbb{P}}(k)$, and Borel–Cantelli gives $o(k)$ almost surely on any common coupling with a.s. count convergence.

This is a one-shot full-prefix diagonal normalization of the raw Gaussian array. It does not include per-minor normalization, such as normalizing a selected minor using only its selected rows or columns, nor any iterative balancing. When both row and column factors are applied, the resulting matrix is only the one-shot diagonally scaled matrix; it is not asserted to have both unit row norms and unit column norms, and no Sinkhorn-type property is claimed. The normalized entries are dependent and need not satisfy an independent-entry density hypothesis. No non-Gaussian normalization theorem, non-diagonal covariance/right-factor transfer, value-dependent thinning or reindexing, selected-submatrix renormalization theorem, upper side, exact two-sided law, quaternion

extension, or upper/exact Gram/sample-covariance law is asserted. The deterministic selected-minor normalized Gram lower-side transfer is recorded separately in Corollary 5.34. □

**Corollary 5.34** (Random-prefix two-sided normalized Gram/sample-covariance transfer)**.** *In the setting of Corollary 5.32, define*

$$\Lambda_{k,B}^{\mathrm{rpref,twodiag}} = \min_{\substack{R\subset[N_k^R],\,|R|=k\\ C\subset[N_k^C],\,|C|=k}} \lambda_{\min}\left(k^{-1}(B_{R,C}^{(k)})^* B_{R,C}^{(k)}\right),$$

*with value $+\infty$ if either prefix is infeasible. Then, for every $\varepsilon>0$,*

$$\mathbb{P}\left\{\Lambda_{k,B}^{\mathrm{rpref,twodiag}} \le \exp\left[-\left(\frac{2(h(\gamma_R)+h(\gamma_C)+\kappa)}{d_{\mathbb{F}}}+\varepsilon\right)k\right]\right\}\to 0.$$

*On a common coupling, if both prefix counts converge almost surely and $\Delta_k^{\mathrm{twodiag}} = o(k)$ almost surely, then*

$$\liminf_{k\to\infty}\frac{1}{k}\log\Lambda_{k,B}^{\mathrm{rpref,twodiag}} \ge -\frac{2(h(\gamma_R)+h(\gamma_C)+\kappa)}{d_{\mathbb{F}}} \qquad \textit{almost surely.}$$

*The same conclusion applies to the Gaussian full-prefix-normalized matrices of Corollary 5.33, with $\kappa=0$ and $B_{R,C}^{(k)}$ interpreted as in that corollary.*

*Proof.* For every feasible $R, C$,

$$\lambda_{\min}\left(k^{-1}(B_{R,C}^{(k)})^* B_{R,C}^{(k)}\right) = k^{-1}\sigma_{\min}(B_{R,C}^{(k)})^2.$$

Taking minima over the same feasible family gives the pathwise identity

$$\Lambda_{k,B}^{\mathrm{rpref,twodiag}} = k^{-1}\left(M_{k,B}^{\mathrm{rpref,twodiag}}\right)^2.$$

Let $c=(h(\gamma_R)+h(\gamma_C)+\kappa)/d_{\mathbb{F}}$. If $\Lambda_{k,B}^{\mathrm{rpref,twodiag}} \le e^{-(2c+\varepsilon)k}$, then

$$M_{k,B}^{\mathrm{rpref,twodiag}} \le k^{1/2}e^{-(c+\varepsilon/2)k} \le e^{-(c+\varepsilon/4)k}$$

for all large $k$, since $(\log k)/k\to 0$. The probability statement therefore follows from Corollary 5.32. The almost-sure lower envelope follows from

$$\frac{1}{k}\log\Lambda_{k,B}^{\mathrm{rpref,twodiag}} = 2\,\frac{1}{k}\log M_{k,B}^{\mathrm{rpref,twodiag}} - \frac{\log k}{k}$$

and the a.s. lower envelope in the same corollary.

This is only deterministic squaring for already selected $k\times k$ minors with the selected-minor normalization $k^{-1}$. It is not an upper or exact two-sided Gram/sample-covariance law, a Wishart-edge or LDP theorem, a pair-correlation theorem, a non-diagonal covariance or right-factor transfer, a row-Gram result for rectangular matrices, a per-minor normalization theorem, a value-dependent selection theorem, or a quaternion statement. □

**Corollary 5.35** (Almost-sure geometric lower envelopes)**.** *Assume the corresponding matrices for all dimensions are realized on a common probability space.*

*For fixed excess $r\ge 0$, the Gaussian lower side of Theorem 5.3 strengthens to*

$$\liminf_{m\to\infty}\frac{1}{m}\log M_m^{(r)} \ge -\frac{h(\gamma)}{d_{\mathbb{F}}(r+1)} \qquad \textit{almost surely.}$$

*Under the bounded-density rate hypotheses of Theorem 5.4,*

$$\liminf_{m\to\infty}\frac{1}{m}\log M_m^{(r)} \ge -\frac{h(\gamma)}{d_{\mathbb{F}}(r+1)} - \frac{\kappa}{d_{\mathbb{F}}} \qquad \textit{almost surely.}$$

*Under the directional row hypotheses of Corollary 5.7,*

$$\liminf_{m\to\infty}\frac{1}{m}\log M_m^{(r)} \geq -\frac{h(\gamma)+\beta_r^{\mathrm{dir}}}{d_{\mathbb{F}}(r+1)} \qquad \textit{almost surely.}$$

*In the proportional-excess setting of Proposition 5.12,*

$$\liminf_{m\to\infty}\frac{1}{m}\log M_m^{(\ell)} \geq -\frac{\kappa}{d_{\mathbb{F}}} \qquad \textit{almost surely.}$$

*For the fixed-$k$ Gaussian square-row setting of Theorem 5.23,*

$$\liminf_{m\to\infty}\frac{1}{m}\log M_m[k] \geq -\frac{h(\gamma)}{d_{\mathbb{F}}k^2} \qquad \textit{almost surely.}$$

*For the fixed-excess fixed-$k$ Gaussian setting of Corollary 5.20,*

$$\liminf_{m\to\infty}\frac{1}{m}\log M_m^{(r)}[k] \geq -\frac{h(\gamma)}{d_{\mathbb{F}}k(k+r)} \qquad \textit{almost surely.}$$

*For the normalized Gram eigenvalue statistic of Corollary 5.24,*

$$\liminf_{m\to\infty}\frac{1}{m}\log \Lambda_m[k] \geq -\frac{2h(\gamma)}{d_{\mathbb{F}}k^2} \qquad \textit{almost surely.}$$

*For the normalized fixed-excess fixed-$k$ Gram statistic of Corollary 5.21,*

$$\liminf_{m\to\infty}\frac{1}{m}\log \Lambda_m^{(r)}[k] \geq -\frac{2h(\gamma)}{d_{\mathbb{F}}k(k+r)} \qquad \textit{almost surely.}$$

*For two-sided minors, with $H = h(\gamma_R)+h(\gamma_C)$, Theorem 5.26 gives*

$$\liminf_{k\to\infty}\frac{1}{k}\log M \geq -\frac{H}{d_{\mathbb{F}}} \qquad \textit{almost surely,}$$

*Theorem 5.27 gives*

$$\liminf_{k\to\infty}\frac{1}{k}\log M \geq -\frac{H+\kappa}{d_{\mathbb{F}}} \qquad \textit{almost surely,}$$

*and Corollary 5.30 gives*

$$\liminf_{k\to\infty}\frac{1}{k}\log M_{\sigma,k}^{\mathrm{two}} \geq -\frac{H}{d_{\mathbb{F}}}-\eta \qquad \textit{almost surely.}$$

*Proof.* The fixed-excess and two-sided proofs give exponentially summable bad-event probabilities at any threshold larger than the displayed constant, so Corollary 2.8 applies. The directional fixed-excess statement uses the finite-budget estimate in Corollary 5.7 with summable $e^{-u_m}$. The fixed-$k$ statement uses the finite-budget estimate in Theorem 5.23 with summable $e^{-u_m}$; the fixed-excess fixed-$k$ statement uses Theorem 5.19 in the same way; the Gram statements follow from $\Lambda_m[k] = m^{-1}M_m[k]^2$ and $\Lambda_m^{(r)}[k] = (m+r)^{-1}(M_m^{(r)}[k])^2$. In the proportional-excess case, the one-matrix exponent is proportional to $m$, so the bad-event probability is $\exp(-c_\varepsilon m^2 + O(m))$ at every threshold $\kappa/d_{\mathbb{F}}+\varepsilon$, which is also summable.

For the two-sided smoothed statement, use the scaling

$$F_k + \sigma_k G_k = \sigma_k(\sigma_k^{-1}F_k + G_k)$$

and the two-sided Gaussian lower envelope for the shifted standard Gaussian matrix inside the parentheses, exactly as in Corollary 3.13. □

**Conjecture 5.36** (Two-sided base, conditional)**.** For iid Gaussian entries under the hypotheses of Theorem 5.26,

$$\frac{1}{k}\log M \xrightarrow{\mathbb{P}} -\frac{h(\gamma_R)+h(\gamma_C)}{d_{\mathbb{F}}},$$

conditional on the two-sided pair-correlation input below.

The Gaussian lower bound is Theorem 5.26. The density-rate and smoothed statements above are lower-side estimates only. The Gaussian upper bound is again the second-moment half of Theorem 2.2: one needs

$$\frac{\operatorname{Var} Z(s)}{(\mathbb{E}Z(s))^2} \to 0, \qquad Z(s) = \sum_{|R|=|C|=k} \mathbf{1}_{\{\sigma_{\min}(A_{R,C}) \le e^{-sk}\}},$$

for every $s < (h(\gamma_R) + h(\gamma_C))/d_{\mathbb{F}}$. The new feature relative to Section 3 is that two submatrices can overlap in their rows, in their columns, or in both, so the pair-correlation analysis runs over the two-parameter family of overlaps $(|R \cap R'|, |C \cap C'|)$ and must control the joint small-ball behaviour of two $k \times k$ Gaussian minors that share a sub-block. We do not establish this estimate; the two-sided base is therefore conditional on it, exactly as the square base of Section 3 is conditional on the one-sided pair-correlation condition.

5.5. **Endpoint aspect asymptotics.** The aspect ratio enters every fixed-shape row-submatrix estimate through the Stirling entropy

$$h(\gamma) = \gamma \log \gamma - (\gamma - 1)\log(\gamma - 1), \qquad c_\gamma = e^{h(\gamma)} = \frac{\gamma^\gamma}{(\gamma-1)^{\gamma-1}}, \qquad \gamma > 1.$$

The following calculation is deterministic. It calibrates the fixed-aspect formulas above; it is not a uniform moving-aspect random-matrix theorem.

**Proposition 5.37** (Endpoint asymptotics of the entropy base)**.** *The function $h$ is increasing and concave on $(1, \infty)$. As $\gamma = 1 + \delta \downarrow 1$, with $L = \log(1/\delta)$,*

$$h(1+\delta) = \delta L + \delta + \frac{\delta^2}{2} - \frac{\delta^3}{6} + O(\delta^4),$$

*and*

$$c_{1+\delta} = 1 + \delta(L+1) + \frac{\delta^2}{2}\big((L+1)^2 + 1\big) + O(\delta^3 L^3).$$

*In particular $h(1+\delta) \sim \delta \log(1/\delta)$ and $c_{1+\delta} \to 1$. As $\gamma \to \infty$,*

$$h(\gamma) = \log \gamma + 1 - \frac{1}{2\gamma} - \frac{1}{6\gamma^2} - \frac{1}{12\gamma^3} + O(\gamma^{-4}),$$

*and*

$$c_\gamma = e\gamma\left(1 - \frac{1}{2\gamma} - \frac{1}{24\gamma^2} - \frac{1}{48\gamma^3} + O(\gamma^{-4})\right).$$

*Proof.* The derivatives are

$$h'(\gamma) = \log \frac{\gamma}{\gamma - 1}, \qquad h''(\gamma) = -\frac{1}{\gamma(\gamma-1)}.$$

Thus $h$ is increasing and concave. For $\gamma = 1 + \delta$,

$$h(1+\delta) = (1+\delta)\log(1+\delta) - \delta \log \delta,$$

and the Taylor expansion of $(1+\delta)\log(1+\delta)$ gives the first asymptotic display. Exponentiating gives the expansion of $c_{1+\delta}$.

For large $\gamma$,

$$h(\gamma) = \log \gamma - (\gamma - 1)\log(1 - \gamma^{-1}) = \log \gamma + 1 - \sum_{j \ge 1} \frac{1}{j(j+1)\gamma^j}.$$

Expanding the exponential of the first three correction terms gives the stated formula for $c_\gamma$. □

*Remark* 5.38 (Interpretation and moving aspects). For fixed selected excess $r$, the proved lower threshold and the conjectural exact fixed-excess base involve

$$B_{r,\mathbb{F}}(\gamma) = \frac{h(\gamma)}{d_{\mathbb{F}}(r+1)}.$$

Thus barely overcomplete row pools have $B_{r,\mathbb{F}}(1+\delta) \sim \delta \log(1/\delta)/(d_{\mathbb{F}}(r+1))$, while very large fixed aspect ratios increase the exponent only like $\log\gamma/(d_{\mathbb{F}}(r+1))$. The complex exponent is half the real exponent.

The preceding asymptotics are pointwise in $\gamma$. Matching exact bases are proved or conjectured above for $N_m/m \to \gamma > 1$ with fixed limiting $\gamma$. If $\gamma_m = N_m/m$ tends to 1 or to infinity with $m$, an exact moving-aspect law would need uniform small-ball and pair-correlation input. The lower side is less rigid: Corollaries 3.5, 5.5, and 5.28 keep the finite binomial entropy budget instead of passing to a fixed $h(\gamma)$. The entropy identity itself gives, for $N_m = m + r_m$ with $r_m = o(m)$,

$$mh(N_m/m) = r_m \log(m/r_m) + r_m + O(r_m^2/m),$$

but this is only an entropy-scale statement. When this entropy is comparable to $\log m$, polynomial hard-edge factors and Stirling constants can affect the actual scale.

## 6. Large deviations of the extremal hard edge

Throughout this section $A_m \in \mathbb{F}^{N_m \times m}$ has independent standard Gaussian entries, $N_m/m \to \gamma > 1$, and

$$M_m = M_m^{\mathbb{F}}(A_m) = \min_{\substack{T\subset[N_m]\\|T|=m}} \sigma_{\min}(A_{m,T}).$$

Write $d = d_{\mathbb{F}}$ and $h = h(\gamma)$, so that the source theorem gives $\frac{1}{m}\log M_m \xrightarrow{\mathbb{P}} -h/d$. Two large-deviation events surround this limit. In the *upper tail* the minimum is atypically small, $M_m \le e^{-sm}$ with $s > h/d$; in the *lower tail* it is atypically large, $M_m \ge e^{-sm}$ with $s < h/d$. The two regimes are not symmetric. The upper tail is a first-moment computation whose rate we pin between two explicit lines, and the matching value is a clean conjecture. The lower tail is a strongly dependent avoidance problem and is left open. Both are organised by the counting variable

$$Z_m(s) = \sum_{\substack{T\subset[N_m]\\|T|=m}} \mathbf{1}_{\{\sigma_{\min}(A_{m,T})\le e^{-sm}\}}, \qquad \{M_m \le e^{-sm}\} = \{Z_m(s) \ge 1\}. \tag{1}$$

### 6.1. The upper tail: an atypically small minimum.

The union-bound half of the theory gives a clean upper-tail estimate for the event that the extremal hard edge is even smaller than its typical exponential scale.

**Proposition 6.1** (Small-minimum upper tail). *Let $Y_m^* = \min_{T\in\mathcal{T}_m} Y_{m,T}$. Assume*

$$|\mathcal{T}_m| \le \exp((H+o(1))m)$$

*and, for every fixed $s > 0$,*

$$\sup_T \mathbb{P}\{Y_{m,T} \le e^{-sm}\} \le \exp\{-(as+o(1))m\}.$$

*Then for every $s > H/a$,*

$$\mathbb{P}\{Y_m^* \le e^{-sm}\} \le \exp\{-(as-H+o(1))m\}.$$

*Proof.* This is the same union bound as in Theorem 2.2:

$$\mathbb{P}\{Y_m^* \le e^{-sm}\} \le |\mathcal{T}_m| \sup_T \mathbb{P}\{Y_{m,T} \le e^{-sm}\}.$$

□

For square row submatrices with $H = h(\gamma)$ and $a = d_{\mathbb{F}}$, this gives

$$\mathbb{P}\{M_m \le e^{-sm}\} \le \exp\{-(d_{\mathbb{F}} s - h(\gamma) + o(1))m\}, \qquad s > \frac{h(\gamma)}{d_{\mathbb{F}}}, \tag{2}$$

whenever the one-matrix exponential small-ball estimate holds. For the Gaussian ensemble that estimate, and its matching lower companion, are the following two-sided bound.

**Lemma 6.2** (Two-sided least-singular-value tail)**.** *Let $G \in \mathbb{F}^{m\times m}$ have independent standard Gaussian entries. There are constants $0 < c_{\mathbb{F}} \le C_{\mathbb{F}} < \infty$, depending only on $\mathbb{F}$, such that for all $0 < t \le 1$,*

$$c_{\mathbb{F}}\, t^d \;\le\; \mathbb{P}\{\sigma_{\min}(G) \le t\} \;\le\; C_{\mathbb{F}}\, m^{1+d/2}\, t^d. \tag{3}$$

*Proof.* The upper bound is the polynomial Gaussian tail of [15] underlying Definition 3.1 with exponent $a = d_{\mathbb{F}}$.

For the lower bound, let $R_1, \dots, R_m$ be the rows of $G$ and put $d_m = \operatorname{dist}\big(R_m, \operatorname{span}_{\mathbb{F}}\{R_1, \dots, R_{m-1}\}\big)$. Almost surely $R_1, \dots, R_{m-1}$ span a hyperplane; let $w \in \mathbb{F}^m$ be a unit vector orthogonal to it, so $\langle R_j, w\rangle = 0$ for $j \le m-1$ and $d_m = |\langle R_m, w\rangle|$. Set $u = \overline{w}$ (so $u = w$ when $\mathbb{F} = \mathbb{R}$); then $\|u\| = 1$ and the vector $Gu$ has $j$-th coordinate $\sum_k (R_j)_k \overline{w_k} = \langle R_j, w\rangle$, which vanishes for $j \le m-1$. Hence $Gu = \langle R_m, w\rangle\, e_m$ and $\|Gu\| = d_m$, so

$$\sigma_{\min}(G) = \min_{\|v\|=1} \|Gv\| \le \|Gu\| = d_m.$$

Conditional on $R_1, \dots, R_{m-1}$, the unit vector $w$ is fixed and $R_m$ is a standard Gaussian vector, so $\langle R_m, w\rangle$ is a standard scalar Gaussian over $\mathbb{F}$. Hence $\mathbb{P}\{\sigma_{\min}(G) \le t\} \ge \mathbb{P}\{d_m \le t\} = p_{\mathbb{F}}(t)$, where $p_{\mathbb{R}}(t) = \mathbb{P}\{|N(0,1)| \le t\} \ge \sqrt{2/\pi}\, e^{-1/2}\, t$ and $p_{\mathbb{C}}(t) = 1 - e^{-t^2} \ge t^2/2$ for $0 < t \le 1$. In both cases $p_{\mathbb{F}}(t) \ge c_{\mathbb{F}} t^d$. □

The exponent in (2) is the natural large-deviation rate. Define

$$I_-(s) = \liminf_{m\to\infty} \Big( -\tfrac{1}{m} \log \mathbb{P}\{M_m \le e^{-sm}\} \Big). \tag{4}$$

Then (2) reads

$$I_-(s) \ge d_{\mathbb{F}}\, s - h(\gamma), \qquad s \ge \tfrac{h}{d}, \qquad \text{with } I_-(s) = 0 \text{ for } s \le \tfrac{h}{d}. \tag{5}$$

The boundary statement holds because for $s < h/d$ one has $e^{-sm} > e^{-(h/d)m}$, so $\mathbb{P}\{M_m \le e^{-sm}\} \to 1$ by the source theorem and $I_-(s) = 0$; the value $I_-(h/d) = 0$ follows by monotonicity. A single submatrix bounds the rate from above.

**Proposition 6.3** (Upper bound on the rate)**.** *For every fixed $s > 0$ and every fixed $T_0$ with $|T_0| = m$,*

$$\mathbb{P}\{M_m \le e^{-sm}\} \ge \mathbb{P}\{\sigma_{\min}(A_{m,T_0}) \le e^{-sm}\} \ge c_{\mathbb{F}}\, e^{-d_{\mathbb{F}} sm}, \tag{6}$$

*so that $I_-(s) \le d_{\mathbb{F}}\, s$. In particular*

$$d_{\mathbb{F}}\, s - h(\gamma) \;\le\; I_-(s) \;\le\; d_{\mathbb{F}}\, s, \qquad s \ge \tfrac{h}{d}. \tag{7}$$

*Proof.* The first inequality in (6) holds because $M_m \le \sigma_{\min}(A_{m,T_0})$, and the second is the lower half of Lemma 6.2 with $t = e^{-sm} \le 1$. The sandwich combines this with (5). □

The width of the window (7) is exactly the entropy $h(\gamma)$: a single submatrix sees only the small-ball exponent $d_{\mathbb{F}}$, while the union bound credits all $e^{hm}$ submatrices. Closing the window to the lower line is the natural guess.

**Conjecture 6.4** (Exact upper-tail rate)**.** For every fixed $s > h(\gamma)/d_{\mathbb{F}}$ the limit defining $I_-(s)$ exists and

$$I_-(s) = d_{\mathbb{F}}\, s - h(\gamma). \tag{8}$$

The heuristic behind (8) is a first-moment computation. By Lemma 6.2,

$$\mathbb{E}Z_m(s) = \binom{N_m}{m}\,\mathbb{P}\{\sigma_{\min}(A_{m,T}) \le e^{-sm}\} = \exp\{(h - d_{\mathbb{F}}s + o(1))m\}, \tag{9}$$

which tends to 0 for $s > h/d$. If the small-submatrix events behaved like a Poisson family one would expect $\mathbb{P}\{Z_m(s) \ge 1\} \sim \mathbb{E}Z_m(s) = e^{-(d_{\mathbb{F}}s - h + o(1))m}$, which is (8). Turning this into a proof requires a matching lower bound on $\mathbb{P}\{Z_m(s) \ge 1\}$, for instance through the Bonferroni inequality

$$\mathbb{P}\{Z_m(s) \ge 1\} \ge \mathbb{E}Z_m(s) - \tfrac{1}{2}\,\mathbb{E}\big[Z_m(s)(Z_m(s) - 1)\big],$$

which is useful only when $\mathbb{E}[Z_m(s)(Z_m(s) - 1)] = o(\mathbb{E}Z_m(s))$. This is exactly what fails at the crude level. The pairs $I, J$ with $|I \cap J| = m - 1$ number $\binom{N_m}{m}\, m(N_m - m) = \Theta(m^2)\binom{N_m}{m}$, and for such nearly coincident row sets the events $\{\sigma_{\min}(A_{m,I}) \le e^{-sm}\}$ and $\{\sigma_{\min}(A_{m,J}) \le e^{-sm}\}$ are strongly positively correlated rather than asymptotically independent: their pair probability does not factorise into $\mathbb{P}\{\sigma_{\min} \le e^{-sm}\}^2$. This is the same overlap dependence that [15] controls at the typical scale through the second-moment estimate over random hyperplanes; in the rarer regime $\mathbb{E}Z_m(s) \to 0$ that control is not available. Restricting to a subfamily of pairwise disjoint submatrices removes the dependence, but only $\lfloor \gamma \rfloor$ of them fit inside $[N_m]$, so it improves (6) by a constant factor and does not change the rate. We therefore record (8) as a conjecture, with (7) as the proven two-sided bound.

6.2. **The lower tail: an atypically large minimum.** Fix $s < h/d$. By (1) the lower-tail probability is the probability that the counting variable vanishes, up to the null boundary event $\{M_m = e^{-sm}\}$,

$$\mathbb{P}\{M_m \ge e^{-sm}\} = \mathbb{P}\{M_m > e^{-sm}\} = \mathbb{P}\{Z_m(s) = 0\} = \mathbb{P}\Big\{\,\sigma_{\min}(A_{m,T}) > e^{-sm} \text{ for all } |T| = m\,\Big\}, \tag{10}$$

that is, the probability that all $\binom{N_m}{m} = e^{hm + o(m)}$ square row submatrices simultaneously avoid being $e^{-sm}$-small. Since by (9) $\mathbb{E}Z_m(s) = e^{(h - d_{\mathbb{F}}s + o(1))m} \to \infty$ for $s < h/d$, this is the lower-tail large deviation of $Z_m(s)$ all the way to zero. It is a strongly dependent avoidance event: the indicators in (1) are far from independent, because overlapping row sets share most of their rows. This is the open problem recorded as Problem 10.3; here we collect what can and cannot be said rigorously.

The only unconditional bound available is one-sided and merely polynomial.

**Proposition 6.5** (Polynomial bound on the lower tail)**.** *Assume $N_m = \gamma m + O(1)$. For every fixed $s < h(\gamma)/d_{\mathbb{F}}$,*

$$\mathbb{P}\{M_m \ge e^{-sm}\} = O_{\gamma,s,\mathbb{F}}(m^{-1}). \tag{11}$$

*Proof.* Pick $s < s' < h/d$. Then $\{M_m \ge e^{-sm}\} \subseteq \{M_m > e^{-s'm}\}$, and the second-moment upper estimate of [15], equivalently the Chebyshev half of Theorem 2.2, gives $\mathbb{P}\{M_m > e^{-s'm}\} \le \operatorname{Var} Z_m(s')/(\mathbb{E}Z_m(s'))^2 = O(m^{-1})$. □

Estimate (11) shows the lower-tail event is at least polynomially rare, but says nothing about an exponential rate; in particular it does not even exclude polynomial decay. The independence heuristic points the opposite way: if the events in (1) were independent then $\mathbb{P}\{Z_m(s) = 0\} = \prod_T (1 - \mathbb{P}\{\sigma_{\min}(A_{m,T}) \le e^{-sm}\}) \approx e^{-\mathbb{E}Z_m(s)} = \exp\{-e^{(h - d_{\mathbb{F}}s)m}\}$, a doubly exponential decay. The conjectural picture is that the true decay is much slower than this independent-product prediction, but proving this unconditionally is part of the avoidance problem. A construction supplies a complementary bound, conditional on a deterministic geometry input.

**Proposition 6.6** (Conditional lower bound on the lower tail)**.** *Fix $s > 0$. Suppose that for all large $m$ there are deterministic matrices $A_{0,m} \in \mathbb{F}^{N_m \times m}$ with*

$$\sup_{m,i,j} |(A_{0,m})_{ij}| \le B < \infty, \qquad M_m(A_{0,m}) \ge 2e^{-sm}.$$

*Then there is $C = C(\gamma, B, \mathbb{F})$ such that, for all large $m$,*

$$\mathbb{P}\{M_m \geq e^{-sm}\} \geq \exp\{-C(s\,m^3 + m^2 \log m)\}. \tag{12}$$

*Proof.* Write $A_0 = A_{0,m}$. If $\|A_m - A_0\|_{\mathrm{op}} \leq e^{-sm}$, then for every $T$, $\sigma_{\min}(A_{m,T}) \geq \sigma_{\min}((A_0)_T) - \|A_{m,T} - (A_0)_T\|_{\mathrm{op}} \geq 2e^{-sm} - e^{-sm} = e^{-sm}$, so $M_m \geq e^{-sm}$. Hence $\mathbb{P}\{M_m \geq e^{-sm}\} \geq \mathbb{P}\{\|A_m - A_0\|_F \leq e^{-sm}\}$, the standard Gaussian measure of the Frobenius ball of radius $r = e^{-sm}$ about $A_0$ in real dimension $n = d_{\mathbb{F}} N_m m$. Bounding the integrand below by its value on the ball,

$$\mathbb{P}\{\|A_m - A_0\|_F \leq r\} \geq c_{\mathbb{F}}^n e^{-C_{\mathbb{F}}(\|A_0\|_F + r)^2} \operatorname{vol}(B_r^n) \geq \exp\Big\{ - C_{\mathbb{F}} n \log \tfrac{n}{r^2} - C_{\mathbb{F}}\big(n + \|A_0\|_F^2\big)\Big\}.$$

Since $\|A_0\|_F^2 \leq d_{\mathbb{F}} N_m m B^2 = \Theta(m^2)$, $n = \Theta(m^2)$ and $\log(1/r) = sm$, the exponent is bounded below by $-C(s\,m^3 + m^2 \log m)$, giving (12). □

Under this deterministic geometry hypothesis, Proposition 6.6 would rule out the doubly exponential independence prediction, but only under the hypothesis that a uniformly well-conditioned bounded frame $A_0$ with $M_m(A_0) \geq 2e^{-sm}$ exists, and only with the non-sharp speed $m^3$ coming from the use of a Frobenius ball. For $s < h/d$ the hypothesis is the bounded-amplitude quantitative achievability condition that, for some fixed $B$,

$$\mathfrak{m}_m(B) = \sup_{\|A_0\|_\infty \leq B} M_m(A_0) \geq 2e^{-sm}$$

eventually. Whether this holds for some $s < h/d$ is itself open; it is a quantitative full-spark or Kruskal-rank conditioning problem at linear oversampling, not a consequence of the usual generic full-spark statements. A typical Gaussian frame, or a random unit-norm frame, has $M_m = e^{-(h/d)m + o(m)}$, so it does not realise the lower-tail event; the construction must use an atypical configuration.

We summarise the expected picture, with the speed flagged as the main unknown.

**Conjecture 6.7** (Lower-tail large deviation)**.** For each fixed $s < h(\gamma)/d_{\mathbb{F}}$ the hole probability (10) decays at exponential speed $m^2$: there is a rate $K(s) \in (0, \infty)$ with

$$\mathbb{P}\{M_m \geq e^{-sm}\} = \exp\{-(K(s) + o(1))\, m^2\}. \tag{13}$$

The rate $K$ is nonincreasing in $s$, with $K(s) \downarrow 0$ as $s \uparrow h(\gamma)/d_{\mathbb{F}}$ and $K(s) \uparrow$ as $s$ decreases.

The speed $m^2$ is suggested by the Coulomb-gas large deviations of Laguerre (Wishart) ensembles, where forcing a global change of the spectrum costs an exponential of order $m^2$ [9]. Forcing every $m$-row submatrix to be simultaneously well conditioned is a global constraint of this type. What is known is far from this: the lower bound on the rate (an exponential, rather than polynomial, upper bound on the probability) is open, so even the positivity $K(s) > 0$ is unproven, while the matching upper bound on the rate is open beyond the conditional, non-sharp estimate (12). A positive-association (FKG-type) heuristic for the avoidance events $\{\sigma_{\min}(A_{m,T}) > e^{-sm}\}$ would make the hole probability exceed the independent product, consistent with (13) lying far above the doubly exponential prediction; but $\sigma_{\min}(A_{m,T})$ is not monotone in the Gaussian entries, so no FKG inequality applies directly.

*Remark* 6.8 (Relation to the fluctuation problem). The lower tail is the hard side of the deviation problem left open by the source paper, whose quantitative rate is only the polynomial (11). A description of $K(s)$, together with the matching upper-tail Conjecture 6.4, would determine the full large-deviation profile of $\frac{1}{m}\log M_m$ about its limit $-h/d$, and is tied to the conjectured Gumbel fluctuation law of $\log M_m$ in Conjecture 10.2. The first-moment intensity at the centered scale is recorded there; the full Poisson law requires critical-scale factorial moments and overlap-cluster control, beyond the bulk pair-correlation estimate used for the exponential-scale second moment.

## 7. Other ensembles: division algebras, variance profiles, sample covariance, general rows

This section records ensemble extensions that are immediate from the principle, and separates them from extensions that require new input.

7.1. **Division algebras.** The real and complex Gaussian cases correspond to real dimensions $d = 1$ and $d = 2$ of one scalar coordinate. The same hard-edge bookkeeping applies to quaternionic Gaussian matrices, where one scalar has real dimension $d = 4$.

For quaternionic matrices we use the following conventions; see [17] for quaternion matrix background. Row vectors in $\mathbb{H}^m$ carry the inner product

$$\langle x, y\rangle_{\mathbb{H}} = \sum_{j=1}^{m} x_j \overline{y_j},$$

linear in the first variable, and $\operatorname{span}_{\mathbb{H}}$ denotes left $\mathbb{H}$-span. For $A \in \mathbb{H}^{n\times m}$,

$$\sigma_{\min}(A) = \inf_{\|x\|_2=1} \|Ax\|_2.$$

Equivalently, $\sigma_{\min}(A)^2$ is the least eigenvalue of $A^*A$, and it is also the least singular value of the realification; the realification only repeats quaternionic singular values with multiplicity four. A standard quaternion Gaussian scalar is

$$q = q_0 + q_1\mathbf{i} + q_2\mathbf{j} + q_3\mathbf{k}, \qquad q_\ell \sim N(0, 1/4) \quad \text{iid},$$

so $\mathbb{E}|q|^2 = 1$, $|q|^2 = (1/4)\chi^2_4$, and

$$p_{\mathbb{H}}(t) := \mathbb{P}\{|q| \le t\} = 2t^4 + O(t^6) \qquad (t \downarrow 0).$$

The compact symplectic group is

$$\mathrm{Sp}(m) = \{U \in \mathbb{H}^{m\times m} : U^*U = UU^* = I_m\},$$

acting on row vectors by right multiplication.

**Theorem 7.1** (Quaternion Gaussian extremal hard edge)**.** *Let $A_m \in \mathbb{H}^{N_m\times m}$ have iid standard quaternion Gaussian entries and $N_m/m \to \gamma > 1$. Put*

$$M_m^{\mathbb{H}} = \min_{\substack{T\subset[N_m]\\ |T|=m}} \sigma_{\min}(A_{m,T}).$$

*Then*

$$\frac{1}{m}\log M_m^{\mathbb{H}} \xrightarrow{\mathbb{P}} -\frac{h(\gamma)}{4}.$$

*Proof.* The lower side is a quaternionic one-matrix hard-edge bound. Let $G \in \mathbb{H}^{m\times m}$ have iid standard quaternion Gaussian rows $a_i$. The quaternionic negative second-moment identity gives

$$\sum_{k=1}^{m} \sigma_k(G)^{-2} = \sum_{i=1}^{m} \operatorname{dist}\left(a_i, \operatorname{span}_{\mathbb{H}}\{a_j : j \ne i\}\right)^{-2}.$$

Thus $\sigma_{\min}(G) \le t$ forces, for some $i$,

$$\operatorname{dist}\left(a_i, \operatorname{span}_{\mathbb{H}}\{a_j : j \ne i\}\right) \le \sqrt{m}\, t.$$

Conditioning on the other rows, this distance is $|\langle a_i, u\rangle_{\mathbb{H}}|$ for an $\mathbb{H}$-unit normal $u$, and this scalar is a standard quaternion Gaussian. Hence, for $0 < t \le 1$,

$$\mathbb{P}\{\sigma_{\min}(G) \le t\} \le m\, p_{\mathbb{H}}(\sqrt{m}\, t) \le Cm^3t^4.$$

The union bound over $\binom{N_m}{m} = e^{(h(\gamma)+o(1))m}$ row sets gives

$$\mathbb{P}\{M_m^{\mathbb{H}} \le e^{-sm}\} \to 0 \qquad (s > h(\gamma)/4).$$

For the upper side, fix $s < h(\gamma)/4$, set $t = e^{-sm}$, and write $N = N_m$. Let $x = a_N$. For each $(m-1)$-set $S \subset [N-1]$, let $L_S$ be the quaternionic normal line to $\operatorname{span}_{\mathbb{H}}\{a_i : i \in S\}$, and define

$$E_S = \{\operatorname{dist}(x, \operatorname{span}_{\mathbb{H}}\{a_i : i \in S\}) \le t\}.$$

Equivalently, for any unit representative $u_S \in L_S$, $E_S = \{|\langle x, u_S\rangle_{\mathbb{H}}| \le t\}$; this condition is independent of the unit-quaternion phase of $u_S$. Let

$$Z_t = \sum_{\substack{S\subset[N-1]\\|S|=m-1}} \mathbf{1}_{E_S}, \qquad B_m = \binom{N-1}{m-1}.$$

If $Z_t > 0$, then $M_m^{\mathbb{H}} \le t$. Since $x$ is independent of the normal lines,

$$\mathbb{E}Z_t = B_m p_{\mathbb{H}}(t) = \exp\{(h(\gamma) - 4s + o(1))m\} \to \infty.$$

It remains to show $\mathbb{E}Z_t^2/(\mathbb{E}Z_t)^2 \to 1$. Fix two sets $S, T$ and put

$$D = m - |S \cap T|.$$

The case $D = 1$ is the diagonal $S = T$. If $D \ge 2$, condition on the rows indexed by $S \cap T$. Their orthogonal complement is isomorphic to $\mathbb{H}^D$. The remaining rows in $S \setminus T$ and $T \setminus S$ are independent $\mathrm{Sp}(D)$-invariant quaternion Gaussian vectors, so the normal lines $L_S, L_T$ are independent Haar points of quaternionic projective space $\mathbb{H}P^{D-1}$.

Choose unit representatives $u, v$ and set $\rho = \langle u, v\rangle_{\mathbb{H}}$. The quantities below depend only on $|\rho|$. If $D \ge 3$, then

$$|\rho|^2 \sim \mathrm{Beta}(2, 2(D-1))$$

and

$$\mathbb{E}\big[(1-|\rho|^2)^{-2} \mid D\big] = \frac{(2D-1)(2D-2)}{(2D-3)(2D-4)} = 1 + \frac{2}{D} + O(D^{-2}).$$

This negative moment is finite exactly in the range $D \ge 3$, which is why the case $D = 2$ is split off below.

For fixed $u, v$ with $|\rho| < 1$, a quaternionic unitary rotation and a unit-quaternion phase change put

$$u = e_1, \qquad v = e_1\rho + e_2(1-|\rho|^2)^{1/2}.$$

In the two quaternionic coordinates $(x_1, x_2)$, the real Jacobian of

$$(x_1, x_2) \mapsto \big(\langle x, u\rangle_{\mathbb{H}}, \langle x, v\rangle_{\mathbb{H}}\big)$$

is $(1-|\rho|^2)^2$. Since the joint quaternion Gaussian density is maximal at the origin,

$$\mathbb{P}\{|\langle x, u\rangle_{\mathbb{H}}| \le t,\ |\langle x, v\rangle_{\mathbb{H}}| \le t \mid u, v\} \le (1 + O(t^2))p_{\mathbb{H}}(t)^2(1-|\rho|^2)^{-2}.$$

For $D = 2$ we only use the trivial bound by $p_{\mathbb{H}}(t)$.

For a uniformly chosen ordered pair $(S, T)$ of $(m-1)$-sets in $[N-1]$,

$$\mathbb{P}\{D = d\} = \frac{\binom{m-1}{d-1}\binom{N-m}{d-1}}{\binom{N-1}{m-1}}, \qquad 1 \le d \le m,$$

and

$$\mathbb{E}\frac{1}{D} = \frac{N}{m(N-m+1)} = O(m^{-1}).$$

Also

$$\mathbb{P}\{D = 2\} = \frac{(m-1)(N-m)}{\binom{N-1}{m-1}}.$$

Therefore

$$\frac{\mathbb{P}\{D=2\}}{p_{\mathbb{H}}(t)} \le m^{O(1)} \exp\{-(h(\gamma)-4s+o(1))m\} \to 0,$$

and the same bound with $\mathbb{P}\{D=1\} = 1/B_m$ handles the diagonal contribution. Combining the estimates,

$$\frac{\mathbb{E}Z_t^2}{(\mathbb{E}Z_t)^2} \le o(1) + (1+O(t^2)) \sum_{d=3}^{m} \mathbb{P}\{D=d\}\big(1+2/d+O(d^{-2})\big) = 1+o(1).$$

The reverse inequality is automatic from $\operatorname{Var} Z_t \ge 0$. Chebyshev's inequality gives $\mathbb{P}\{Z_t=0\} \to 0$, hence $\mathbb{P}\{M_m^{\mathbb{H}} \le e^{-sm}\} \to 1$ for every $s < h(\gamma)/4$. Together with the lower side, this proves the claimed convergence. □

**Theorem 7.2** (Quaternion random number of Gaussian rows)**.** *For each $m$, let $(R_{im})_{i\ge 1}$ be an infinite sequence of independent standard quaternion Gaussian rows in $\mathbb{H}^m$. For an integer $n \ge m$, put*

$$M_m^{\mathbb{H}}(n) = \min_{\substack{T\subset[n]\\|T|=m}} \sigma_{\min}((R_{im})_{i\in T}).$$

*Let $N_m$ be integer-valued random variables, possibly dependent on the row array, such that*

$$\frac{N_m}{m} \xrightarrow{\mathbb{P}} \gamma > 1.$$

*Extend $M_m^{\mathbb{H}}(N_m)$ by any fixed positive value on the event $N_m < m$. This event has probability tending to zero, and*

$$\frac{1}{m}\log M_m^{\mathbb{H}}(N_m) \xrightarrow{\mathbb{P}} -\frac{h(\gamma)}{4}.$$

*The normalized square Gram statistic*

$$\Gamma_m^{\mathbb{H}}(N_m) = \min_{\substack{T\subset[N_m]\\|T|=m}} \lambda_{\min}\big(m^{-1}R_T^*R_T\big)$$

*has logarithmic limit $-h(\gamma)/2$, with the same convention on $\{N_m < m\}$. If the row arrays are realized on a common probability space and $N_m/m \to \gamma$ almost surely, then the corresponding lower envelopes also hold:*

$$\liminf_{m\to\infty} \frac{1}{m}\log M_m^{\mathbb{H}}(N_m) \ge -\frac{h(\gamma)}{4}, \qquad \liminf_{m\to\infty} \frac{1}{m}\log \Gamma_m^{\mathbb{H}}(N_m) \ge -\frac{h(\gamma)}{2} \qquad \text{a.s.}$$

*Proof.* The map $n \mapsto M_m^{\mathbb{H}}(n)$ is nonincreasing, since increasing the number of available prefix rows only enlarges the family of tested square submatrices. Fix

$$1 < \gamma_- < \gamma < \gamma_+.$$

With probability tending to one,

$$\lfloor \gamma_- m \rfloor \le N_m \le \lceil \gamma_+ m \rceil.$$

On this event,

$$M_m^{\mathbb{H}}(\lceil \gamma_+ m \rceil) \le M_m^{\mathbb{H}}(N_m) \le M_m^{\mathbb{H}}(\lfloor \gamma_- m \rfloor).$$

Theorem 7.1, applied to the two deterministic row counts, gives limits $-h(\gamma_+)/4$ and $-h(\gamma_-)/4$. Letting $\gamma_- \uparrow \gamma$ and $\gamma_+ \downarrow \gamma$ proves the convergence in probability. No independence between $N_m$ and the row array is used for this prefix-count statement.

The Gram convergence follows from Corollary 7.31. For the almost-sure lower envelope, fix $\gamma_+ > \gamma$. Eventually almost surely, $N_m \le \lceil \gamma_+ m \rceil$, hence

$$M_m^{\mathbb{H}}(N_m) \ge M_m^{\mathbb{H}}(\lceil \gamma_+ m \rceil).$$

The summable lower-side estimate in the proof of Theorem 7.1 gives the deterministic-row-count lower envelope at $\gamma_+$; then let $\gamma_+ \downarrow \gamma$. The Gram lower envelope follows by the same pathwise squaring identity. □

**Corollary 7.3** (Quaternion Poissonization and independent thinning)**.** *The conclusion of Theorem 7.2 applies when $N_m$ is independent of the quaternion Gaussian row values and one of the following conditions holds:*

(1) *$N_m$ is Poisson with mean $\nu_m$, and $\nu_m/m \to \gamma > 1$;*
(2) *$N_m$ is binomial $\mathrm{Bin}(n_m^{\mathrm{pool}}, p_m)$, with $n_m^{\mathrm{pool}} p_m/m \to \gamma > 1$;*
(3) *$N_m = \sum_{i=1}^{n_m^{\mathrm{pool}}} \xi_{im}$, where the $\xi_{im}$ are independent Bernoulli retention indicators, independent of the row values, and $\nu_m = \sum_i \mathbb{E}\xi_{im}$ satisfies $\nu_m/m \to \gamma > 1$.*

*In the thinning case, conditional on the retained count, the retained rows can be reindexed as iid quaternion Gaussian rows; the statistic is invariant under row order. Value-dependent thinning or selection is not covered.*

**Corollary 7.4** (Quaternion random-prefix conditioning transfer)**.** *For each $m$, let $(R_{im})_{i\ge1}$ be an infinite sequence of independent standard quaternion Gaussian rows in $\mathbb{H}^m$. Let $N_m$ be integer-valued, possibly dependent on the row array, with*

$$\frac{N_m}{m} \xrightarrow{\mathbb{P}} \gamma > 1.$$

*Let $d_{im} > 0$, $i \ge 1$, be finite real random row weights and let $S_m \in \mathbb{H}^{m\times m}$ be invertible. The weights and $S_m$ may be random and dependent on $N_m$ and on the row array. For $T \subset [N_m]$, $|T| = m$, set*

$$B_{m,T} = \mathrm{diag}(d_{im} : i \in T)\,(R_{im})_{i\in T}\,S_m.$$

*Assume*

$$\Delta_m^{\mathrm{rpref}} := \max_{1\le i\le N_m} |\log d_{im}| + \big|\log \sigma_{\min}(S_m)\big| + \big|\log \|S_m\|\big| = o_{\mathbb{P}}(m).$$

*Absent row weights or the absent right factor are omitted. On the event $N_m < m$, give the statistics below any fixed positive value. Then*

$$\frac{1}{m}\log \min_{\substack{T\subset[N_m]\\|T|=m}} \sigma_{\min}(B_{m,T}) \xrightarrow{\mathbb{P}} -\frac{h(\gamma)}{4}$$

*and*

$$\frac{1}{m}\log \min_{\substack{T\subset[N_m]\\|T|=m}} \lambda_{\min}\big(m^{-1}B_{m,T}^* B_{m,T}\big) \xrightarrow{\mathbb{P}} -\frac{h(\gamma)}{2}.$$

*If the row arrays are realized on a common probability space, $N_m/m \to \gamma$ almost surely, and $\Delta_m^{\mathrm{rpref}} = o(m)$ almost surely, then the corresponding lower envelopes hold:*

$$\liminf_{m\to\infty} \frac{1}{m}\log \min_{\substack{T\subset[N_m]\\|T|=m}} \sigma_{\min}(B_{m,T}) \ge -\frac{h(\gamma)}{4},$$

*and the normalized square Gram lower envelope has base $-h(\gamma)/2$.*

*Positive real column diagonal factors are included by taking $S_m$ diagonal. Consequently, row-normalized random-prefix quaternion Gaussian matrices and full-prefix column-normalized quaternion Gaussian matrices have the same square-row bases, provided the prefix count hypotheses above hold. For independent Bernoulli thinning, this is read after the retained iid rows are reindexed and after the same conditioning-extrema hypothesis is imposed.*

*Proof.* On the event $N_m \geq m$, for every $T \subset [N_m]$, $|T| = m$,

$$e^{-\Delta_m^{\mathrm{rpref}}} \sigma_{\min}((R_{im})_{i\in T}) \leq \sigma_{\min}(B_{m,T}) \leq e^{\Delta_m^{\mathrm{rpref}}} \sigma_{\min}((R_{im})_{i\in T}).$$

The inequalities are the usual singular-value bounds for a positive real left diagonal factor and a global right factor. They are pathwise and uniform in $T$, so after taking minima,

$$\left| \frac{1}{m} \log \min_{\substack{T\subset[N_m]\\|T|=m}} \sigma_{\min}(B_{m,T}) - \frac{1}{m} \log M_m^{\mathbb{H}}(N_m) \right| \leq \frac{\Delta_m^{\mathrm{rpref}}}{m}$$

outside the event $N_m < m$, whose probability tends to zero. The singular convergence follows from Theorem 7.2. The Gram convergence follows from the pathwise identity in Corollary 7.31. The almost-sure lower envelopes use the same lower sandwich together with the almost-sure lower envelopes in Theorem 7.2.

For row normalization, on the high-probability bracket $N_m \leq \lceil \gamma_+ m \rceil$, the row-norm extrema over the realized prefix are bounded by the extrema over the deterministic prefix $\lceil \gamma_+ m \rceil$. For column normalization, on $\lfloor \gamma_- m \rfloor \leq N_m \leq \lceil \gamma_+ m \rceil$, each column norm is bracketed between the corresponding deterministic-prefix column norms. The same $\chi^2$ upper-tail and ball-volume lower-tail estimates used in Corollary 7.6 give subexponential extrema in probability and, on common couplings with eventual deterministic brackets, almost surely. □

**Corollary 7.5** (All-field square random-prefix row and right-factor exact transfer)**.** *Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$, with $d_{\mathbb{F}} = 1, 2, 4$, respectively. For each $m$, let $(R_{im})_{i\geq 1}$ be an infinite sequence of iid standard $\mathbb{F}^m$-Gaussian rows. Let $N_m$ be integer-valued, possibly dependent on the row array, with*

$$\frac{N_m}{m} \xrightarrow{\mathbb{P}} \gamma > 1.$$

*Let $d_{im} > 0$, $i \geq 1$, be finite real random row weights and let $S_m \in GL_m(\mathbb{F})$. The weights and $S_m$ may be random and dependent on the rows and on $N_m$. Assume*

$$\Delta_m^{\mathrm{sqrf}} = \max_{1\leq i\leq N_m} |\log d_{im}| + |\log \sigma_{\min}(S_m)| + |\log \|S_m\|| = o_{\mathbb{P}}(m),$$

*with an absent row factor replaced by $d_{im} \equiv 1$ and an absent right factor replaced by $S_m = I_m$. For $N_m \geq m$ and $T \subset [N_m]$, $|T| = m$, set*

$$B_{m,T} = \mathrm{diag}(d_{im} : i \in T)\, (R_{im})_{i\in T}\, S_m.$$

*On the event $N_m < m$, give the following two statistics any fixed positive finite value. Then*

$$\frac{1}{m} \log \min_{\substack{T\subset[N_m]\\|T|=m}} \sigma_{\min}(B_{m,T}) \xrightarrow{\mathbb{P}} -\frac{h(\gamma)}{d_{\mathbb{F}}},$$

*and*

$$\frac{1}{m} \log \min_{\substack{T\subset[N_m]\\|T|=m}} \lambda_{\min}\big(m^{-1} B_{m,T}^* B_{m,T}\big) \xrightarrow{\mathbb{P}} -\frac{2h(\gamma)}{d_{\mathbb{F}}}.$$

*If the row arrays are realized on a common probability space, $N_m/m \to \gamma$ almost surely, and $\Delta_m^{\mathrm{sqrf}} = o(m)$ almost surely, then the corresponding lower envelopes hold:*

$$\liminf_{m\to\infty} \frac{1}{m} \log \min_{\substack{T\subset[N_m]\\|T|=m}} \sigma_{\min}(B_{m,T}) \geq -\frac{h(\gamma)}{d_{\mathbb{F}}},$$

*and the normalized square column-Gram lower envelope has base $-2h(\gamma)/d_{\mathbb{F}}$.*

*Positive real column diagonal factors are included by taking $S_m$ diagonal. Consequently, the same conclusions apply to row-normalized random-prefix Gaussian rows and to full-prefix column-normalized Gaussian columns, where the column normalization is computed from all rows in the realized prefix $[N_m]$.*

*Proof.* On the event $N_m \geq m$, for every $T \subset [N_m]$, $|T| = m$,

$$e^{-\Delta_m^{\mathrm{sqrf}}} \sigma_{\min}((R_{im})_{i\in T}) \leq \sigma_{\min}(B_{m,T}) \leq e^{\Delta_m^{\mathrm{sqrf}}} \sigma_{\min}((R_{im})_{i\in T}).$$

The selected rows commute with the single global all-column right factor: $(RS_m)_T = R_T S_m$. The displayed estimate is the min–max singular-value inequality for the positive real row diagonal factor and $S_m$. Taking minima over $T$ gives

$$\left| \frac{1}{m} \log \min_{\substack{T\subset[N_m]\\|T|=m}} \sigma_{\min}(B_{m,T}) - \frac{1}{m} \log M_{\mathbb{F},m}(N_m) \right| \leq \frac{\Delta_m^{\mathrm{sqrf}}}{m}$$

outside the event $N_m < m$, where $M_{\mathbb{F},m}(N_m)$ denotes the corresponding unweighted square-prefix statistic. The event $N_m < m$ has probability tending to zero. For $\mathbb{F} = \mathbb{R}, \mathbb{C}$, use Theorem 3.14; for $\mathbb{F} = \mathbb{H}$, use Theorem 7.2. The singular-value convergence follows because $\Delta_m^{\mathrm{sqrf}} = o_{\mathbb{P}}(m)$.

The Gram statistic is transferred by the eigenvalue identity

$$\lambda_{\min}\big(m^{-1} B_{m,T}^* B_{m,T}\big) = m^{-1} \sigma_{\min}(B_{m,T})^2.$$

For the almost-sure lower envelopes, keep only the lower side of the pathwise sandwich. In the real and complex cases, the source lower envelope is Corollary 3.15 with bounded Gaussian density and $\kappa = 0$; in the quaternion case it is the a.s. part of Theorem 7.2. The Gram lower envelope again follows by squaring.

For row normalization, on the high-probability bracket $N_m \leq \lceil \gamma_+ m \rceil$, the realized-prefix row-norm extrema are bounded by deterministic-prefix extrema. For full-prefix column normalization, on $\lfloor \gamma_- m \rfloor \leq N_m \leq \lceil \gamma_+ m \rceil$, each column norm is bracketed between the corresponding deterministic-prefix column norms. Standard Gaussian norm upper tails and ball-volume lower tails give $\Delta_m^{\mathrm{sqrf}} = o_{\mathbb{P}}(m)$, and give the a.s. version under common a.s. prefix-count convergence.

This is only a square random-prefix row/right-factor exact transfer. It subsumes Corollary 7.4 in the quaternion case and contains the positive column-diagonal case. It is the exact square strengthening of Corollary 7.12 at $r = 0, k = 1$, but it does not extend to fixed-excess exact laws. Column selection does not commute with a non-diagonal right factor, so no selected-column or two-sided non-diagonal covariance statement is implied. Nor is there a value-dependent row-selection theorem, selected-submatrix column normalization, row-Gram statement for $r > 0$, quaternionic non-real row weighting, condition-number-only covariance claim, or almost-sure exact law. □

**Corollary 7.6** (Quaternion square diagonal and normalized exact base)**.** *Let $G_m \in \mathbb{H}^{N_m \times m}$ have iid standard quaternion Gaussian entries with $N_m/m \to \gamma > 1$. Let*

$$\widetilde{G}_m = D_m^{\mathrm{row}} G_m E_m,$$

*where $D_m^{\mathrm{row}}$ and $E_m$ are positive real diagonal matrices, random and possibly dependent on $G_m$, with strictly positive finite entries, and where either diagonal factor may be absent. Assume*

$$\Delta_m^{\mathrm{diag}} := \max_i |\log(D_m^{\mathrm{row}})_{ii}| + \max_j |\log(E_m)_{jj}| = o_{\mathbb{P}}(m),$$

*omitting absent terms. Put*

$$M_{\widetilde{G},\mathbb{H}} = \min_{\substack{T\subset[N_m]\\|T|=m}} \sigma_{\min}((\widetilde{G}_m)_T)$$

*and*

$$\Gamma_{\widetilde{G},\mathbb{H}} = \min_{\substack{T\subset[N_m]\\|T|=m}} \lambda_{\min}\Big(m^{-1}(\widetilde{G}_m)_T^*(\widetilde{G}_m)_T\Big).$$

*Then*

$$\frac{1}{m}\log M_{\widetilde{G},\mathbb{H}} \xrightarrow{\mathbb{P}} -\frac{h(\gamma)}{4}, \qquad \frac{1}{m}\log \Gamma_{\widetilde{G},\mathbb{H}} \xrightarrow{\mathbb{P}} -\frac{h(\gamma)}{2}.$$

*If $\Delta_m^{\mathrm{diag}} = o(m)$ almost surely on a common coupling, then the corresponding lower envelopes hold:*

$$\liminf_{m\to\infty} \frac{1}{m}\log M_{\widetilde{G},\mathbb{H}} \geq -\frac{h(\gamma)}{4}, \qquad \liminf_{m\to\infty} \frac{1}{m}\log \Gamma_{\widetilde{G},\mathbb{H}} \geq -\frac{h(\gamma)}{2} \qquad \text{a.s.}$$

*In particular, the same square-row convergence in probability and lower envelopes hold for row-normalized quaternion Gaussian rows and for full-matrix column-normalized quaternion Gaussian columns.*

*Proof.* For every row set $T$ with $|T| = m$,

$$(\widetilde{G}_m)_T = (D_m^{\mathrm{row}})_T (G_m)_T E_m.$$

The entries of the diagonal factors are positive real scalars, hence central in $\mathbb{H}$. The min–max singular-value inequalities therefore give, uniformly in $T$,

$$e^{-\Delta_m^{\mathrm{diag}}}\sigma_{\min}((G_m)_T) \leq \sigma_{\min}((\widetilde{G}_m)_T) \leq e^{\Delta_m^{\mathrm{diag}}}\sigma_{\min}((G_m)_T).$$

Taking the minimum over $T$, logs, and division by $m$ gives

$$\left|\frac{1}{m}\log M_{\widetilde{G},\mathbb{H}} - \frac{1}{m}\log M_{G,\mathbb{H}}\right| \leq \frac{\Delta_m^{\mathrm{diag}}}{m}$$

on the almost-sure event that all square row submatrices of $G_m$ are nonsingular. Theorem 7.1 and $\Delta_m^{\mathrm{diag}} = o_{\mathbb{P}}(m)$ give the singular-value convergence. The Gram convergence follows from Corollary 7.31.

For the almost-sure lower envelope, use the same pathwise comparison together with the $r = 0, k = 1$ lower-envelope case of Theorem 7.9 and Corollary 7.10. For row normalization, one row norm squared is $(1/4)\chi^2_{4m}$; for column normalization, one column norm squared is $(1/4)\chi^2_{4N_m}$. The Markov upper-tail and ball-volume lower-tail estimates used below for quaternion normalization show that the corresponding logarithmic extrema are $o_{\mathbb{P}}(m)$, and are $o(m)$ almost surely after the $O(m)$ row or column unions. The rows and columns are nonzero almost surely. □

**Corollary 7.7** (Quaternion square row and right-factor conditioning)**.** *Let $G_m \in \mathbb{H}^{N_m\times m}$ have iid standard quaternion Gaussian entries with $N_m/m \to \gamma > 1$. Let*

$$B_m = D_m^{\mathrm{row}} G_m S_m,$$

*where $D_m^{\mathrm{row}}$ is a positive real diagonal row factor with strictly positive finite entries, possibly absent, and $S_m \in \mathbb{H}^{m\times m}$ is invertible. The factors may be random and dependent on $G_m$. Assume*

$$\Delta_m^{\mathrm{rf}} := \max_i |\log(D_m^{\mathrm{row}})_{ii}| + \big|\log\sigma_{\min}(S_m)\big| + \big|\log\|S_m\|\big| = o_{\mathbb{P}}(m),$$

*omitting the row term if $D_m^{\mathrm{row}}$ is absent. Put*

$$M_{B,\mathbb{H}} = \min_{\substack{T\subset[N_m]\\|T|=m}} \sigma_{\min}((B_m)_T)$$

*and*

$$\Gamma_{B,\mathbb{H}} = \min_{\substack{T\subset[N_m]\\|T|=m}} \lambda_{\min}\big(m^{-1}(B_m)_T^*(B_m)_T\big).$$

*Then*

$$\frac{1}{m}\log M_{B,\mathbb{H}} \xrightarrow{\mathbb{P}} -\frac{h(\gamma)}{4}, \qquad \frac{1}{m}\log \Gamma_{B,\mathbb{H}} \xrightarrow{\mathbb{P}} -\frac{h(\gamma)}{2}.$$

*If $\Delta_m^{\mathrm{rf}} = o(m)$ almost surely on a common coupling, then the corresponding lower envelopes hold with the same bases.*

*In particular, deterministic uniformly elliptic quaternion right factors, including $S_m = \Sigma_m^{1/2}$ with $0 < c \le \lambda_{\min}(\Sigma_m) \le \lambda_{\max}(\Sigma_m) \le C < \infty$, preserve the square-row quaternion bases. Absolute singular-value scale is needed: a subexponential condition number alone would not rule out an exponential scalar shift.*

*Proof.* For every row set $T$ with $|T| = m$,

$$(B_m)_T = (D_m^{\mathrm{row}})_T (G_m)_T S_m.$$

Using the quaternionic operator norm and singular-value conventions fixed above,

$$\sigma_{\min}((D_m^{\mathrm{row}})_T)\sigma_{\min}(S_m)\sigma_{\min}((G_m)_T) \le \sigma_{\min}((B_m)_T) \le \|(D_m^{\mathrm{row}})_T\|\,\|S_m\|\,\sigma_{\min}((G_m)_T).$$

The bounds are uniform over $T$. Taking minima, logs, and division by $m$ shows

$$\left|\frac{1}{m}\log M_{B,\mathbb{H}} - \frac{1}{m}\log M_{G,\mathbb{H}}\right| \le \frac{\Delta_m^{\mathrm{rf}}}{m}$$

on the almost-sure event that all square row submatrices of $G_m$ are nonsingular. The singular-value convergence follows from Theorem 7.1; the Gram convergence follows from Corollary 7.31. The lower-envelope statement uses the same pathwise comparison and the $r = 0, k = 1$ lower-envelope cases of Theorem 7.9 and Corollary 7.10. □

**Lemma 7.8** (Quaternion fixed-excess fixed-$k$ Laguerre tail)**.** *Fix $r \ge 0$ and $k \ge 1$. There are constants $C_{\mathbb{H},r,k} \ge 1$ and $B_{\mathbb{H},r,k} \ge 0$ such that, for every $m \ge k$, every iid standard quaternion Gaussian matrix $G \in \mathbb{H}^{(m+r)\times m}$, and every $0 < t \le 1$,*

$$\mathbb{P}\{\sigma_{m-k+1}(G) \le t\} \le C_{\mathbb{H},r,k} m^{B_{\mathbb{H},r,k}} t^{4k(k+r)}.$$

*Proof.* Let $0 \le \lambda_1 \le \cdots \le \lambda_m$ be the eigenvalues of the quaternion Hermitian matrix $G^*G$, each counted once as a quaternionic eigenvalue. With the standard quaternion Gaussian normalization fixed above, these eigenvalues have the $\beta = 4$ Laguerre density

$$\frac{m!}{Z_m(q)} e^{-2\sum_i \lambda_i} \prod_{i=1}^m \lambda_i^{q-1} \prod_{i<j} (\lambda_j - \lambda_i)^4, \qquad q = 2(r+1),$$

in ordered coordinates; this is the quaternion Wishart/Laguerre ensemble, see [9, Chs. 3–4] together with the quaternion matrix conventions in [17]. Put $\delta = t^2$. On $\lambda_k \le \delta$, write $x_i = \lambda_i$ for $1 \le i \le k$ and $y_j = \lambda_{k+j}$ for $1 \le j \le m-k$. Since $0 \le x_i \le y_j$,

$$\prod_{i \le k < j} (\lambda_j - \lambda_i)^4 \le \prod_{j=1}^{m-k} y_j^{4k}.$$

The integral over the $k$ bottom variables contributes

$$\int_{[0,\delta]^k} \prod_i x_i^{q-1} \prod_{i<j} |x_i - x_j|^4 \, dx = C_{r,k} \delta^{kq + 4\binom{k}{2}} = C_{r,k} \delta^{2k(k+r)}.$$

The remaining variables are bounded by a Laguerre–Selberg integral with parameter $q + 4k$, so

$$\mathbb{P}\{\lambda_k \le \delta\} \le C_{r,k} \delta^{2k(k+r)} \frac{m!}{(m-k)!} \frac{Z_{m-k}(q+4k)}{Z_m(q)}.$$

For fixed $r, k$, the Laguerre–Selberg product and Stirling's formula bound the ratio by $C_{r,k}m^{B_{r,k}}$, after absorbing finitely many small $m$ into the constant. Since $\sigma_{m-k+1}(G) = \sqrt{\lambda_k}$, this gives the claimed singular-value tail. □

**Theorem 7.9** (Quaternion fixed-excess fixed-$k$ lower side)**.** *Fix $r \geq 0$ and $k \geq 1$. Let $A_m \in \mathbb{H}^{N_m \times m}$ have iid standard quaternion Gaussian entries, with $N_m \geq m + r$, and put*

$$M^{(r)}_{\mathbb{H},m}[k] = \min_{\substack{T \subset [N_m] \\ |T| = m+r}} \sigma_{m-k+1}(A_{m,T}).$$

*There are constants $C_{\mathbb{H},r,k} \geq 1$ and $B_{\mathbb{H},r,k} \geq 0$ such that, for every $u_m > 0$,*

$$\mathbb{P}\left\{ M^{(r)}_{\mathbb{H},m}[k] \leq \exp\left[ -\frac{\log \binom{N_m}{m+r} + B_{\mathbb{H},r,k} \log m + \log C_{\mathbb{H},r,k} + u_m}{4k(k+r)} \right] \right\} \leq e^{-u_m}.$$

*If $N_m/m \to \gamma > 1$, then, for every $\varepsilon > 0$,*

$$\mathbb{P}\left\{ M^{(r)}_{\mathbb{H},m}[k] \leq \exp\left[ -\left( \frac{h(\gamma)}{4k(k+r)} + \varepsilon \right) m \right] \right\} \to 0.$$

*On any common coupling over $m$,*

$$\liminf_{m \to \infty} \frac{1}{m} \log M^{(r)}_{\mathbb{H},m}[k] \geq -\frac{h(\gamma)}{4k(k+r)} \qquad a.s.$$

*Proof.* The finite-budget estimate is the union bound over the $\binom{N_m}{m+r}$ selected row sets, using Lemma 7.8. Taking $u_m = 4k(k+r)\varepsilon m$ and absorbing the logarithmic terms gives the fixed-aspect lower side. The same choice with, say, half the displayed $\varepsilon$-slack gives summable failure probabilities, so the almost-sure lower envelope follows from the first Borel–Cantelli lemma. □

**Corollary 7.10** (Quaternion fixed-excess fixed-$k$ column-Gram lower side)**.** *Under the hypotheses of Theorem 7.9, let*

$$\Lambda^{(r)}_{\mathbb{H},m}[k] = \min_{\substack{T \subset [N_m] \\ |T| = m+r}} \lambda_{(k)}\left( (m+r)^{-1} A^*_{m,T} A_{m,T} \right),$$

*where $\lambda_{(k)}$ is the $k$-th smallest quaternionic eigenvalue of the Hermitian column Gram matrix. If $N_m/m \to \gamma > 1$, then, for every $\varepsilon > 0$,*

$$\mathbb{P}\left\{ \Lambda^{(r)}_{\mathbb{H},m}[k] \leq \exp\left[ -\left( \frac{h(\gamma)}{2k(k+r)} + \varepsilon \right) m \right] \right\} \to 0,$$

*and, on any common coupling,*

$$\liminf_{m \to \infty} \frac{1}{m} \log \Lambda^{(r)}_{\mathbb{H},m}[k] \geq -\frac{h(\gamma)}{2k(k+r)} \qquad a.s.$$

*Proof.* Pathwise,

$$\Lambda^{(r)}_{\mathbb{H},m}[k] = (m+r)^{-1} \left( M^{(r)}_{\mathbb{H},m}[k] \right)^2,$$

and $m^{-1} \log(m+r) \to 0$. Apply Theorem 7.9. □

**Corollary 7.11** (Random-prefix fixed-excess fixed-$k$ lower sides)**.** *Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$, with $d_{\mathbb{F}} = 1, 2, 4$, respectively. For each $m$, let $(R_{im})_{i \geq 1}$ be an infinite sequence of iid standard $\mathbb{F}^m$-Gaussian rows. Fix $r \geq 0$ and $k \geq 1$. For an integer $n \geq m + r$, put*

$$M^{(r)}_{\mathbb{F},m}[k](n) = \min_{\substack{T \subset [n] \\ |T| = m+r}} \sigma_{m-k+1}((R_{im})_{i \in T}),$$

*and*

$$\Lambda^{(r)}_{\mathbb{F},m}[k](n) = \min_{\substack{T\subset[n]\\|T|=m+r}} \lambda_{(k)}\big((m+r)^{-1}(R_{im})^*_{i\in T}(R_{im})_{i\in T}\big)\,.$$

*Let $N_m$ be integer-valued, possibly dependent on the row array, with*

$$\frac{N_m}{m} \xrightarrow{\mathbb{P}} \gamma > 1.$$

*Extend both statistics by $+\infty$ on the event $N_m < m+r$. This event has probability tending to zero; the convention only makes the monotonicity comparison literal. Then, for every $\varepsilon > 0$,*

$$\mathbb{P}\left\{M^{(r)}_{\mathbb{F},m}[k](N_m) \le \exp\left[-\left(\frac{h(\gamma)}{d_{\mathbb{F}}k(k+r)}+\varepsilon\right)m\right]\right\} \to 0$$

*and*

$$\mathbb{P}\left\{\Lambda^{(r)}_{\mathbb{F},m}[k](N_m) \le \exp\left[-\left(\frac{2h(\gamma)}{d_{\mathbb{F}}k(k+r)}+\varepsilon\right)m\right]\right\} \to 0.$$

*On a common coupling, if $N_m/m \to \gamma$ almost surely, then*

$$\liminf_{m\to\infty}\frac{1}{m}\log M^{(r)}_{\mathbb{F},m}[k](N_m) \ge -\frac{h(\gamma)}{d_{\mathbb{F}}k(k+r)}$$

*and*

$$\liminf_{m\to\infty}\frac{1}{m}\log \Lambda^{(r)}_{\mathbb{F},m}[k](N_m) \ge -\frac{2h(\gamma)}{d_{\mathbb{F}}k(k+r)} \qquad a.s.$$

*Proof.* The map $n \mapsto M^{(r)}_{\mathbb{F},m}[k](n)$ is nonincreasing, because increasing the prefix row count enlarges the tested family. Fix $\varepsilon > 0$, and choose $\gamma_+ > \gamma$ so close that

$$\frac{h(\gamma_+)}{d_{\mathbb{F}}k(k+r)} < \frac{h(\gamma)}{d_{\mathbb{F}}k(k+r)} + \frac{\varepsilon}{2}.$$

With probability tending to one,

$$N_m \le \lceil \gamma_+ m\rceil.$$

On this event, including the convention $M = +\infty$ when $N_m < m+r$,

$$M^{(r)}_{\mathbb{F},m}[k](N_m) \ge M^{(r)}_{\mathbb{F},m}[k](\lceil\gamma_+ m\rceil).$$

Apply the deterministic-count fixed-aspect lower side Corollary 5.20 for $\mathbb{F} = \mathbb{R}, \mathbb{C}$, and Theorem 7.9 for $\mathbb{F} = \mathbb{H}$, at the deterministic aspect $\gamma_+$. At the threshold

$$\exp\left[-\left(\frac{h(\gamma)}{d_{\mathbb{F}}k(k+r)}+\varepsilon\right)m\right],$$

this deterministic lower side applies with at least $\varepsilon/2$ slack. No independence between $N_m$ and the rows is used.

The column-Gram statement is pathwise:

$$\Lambda^{(r)}_{\mathbb{F},m}[k](N_m) = (m+r)^{-1}\big(M^{(r)}_{\mathbb{F},m}[k](N_m)\big)^2,$$

on the event $N_m \ge m+r$, while both sides are $+\infty$ under the extension on the complement; $m^{-1}\log(m+r) \to 0$. For the almost-sure lower envelopes, fix a countable decreasing sequence $\gamma_j \downarrow \gamma$. Almost surely, eventually $N_m \le \lceil\gamma_j m\rceil$ for each $j$; combine the preceding pathwise comparison with the deterministic-count almost-sure lower envelope at $\gamma_j$, then let $j \to \infty$.

This is only the lower-side monotonicity transfer. In the overlapping square case $r = 0, k = 1$, the exact random-prefix laws are the earlier square results; the present corollary adds fixed-excess fixed-$k$ lower sides and normalized column-Gram lower sides only. It does not assert an upper side, exact fixed-excess law, value-dependent row-selection theorem, row-Gram statement for $r > 0$, beta interpolation, or proportional Wishart-edge law. □

**Corollary 7.12** (Random-prefix diagonal fixed-excess fixed-$k$ lower sides)**.** *Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$, with $d_{\mathbb{F}} = 1, 2, 4$, respectively. For each $m$, let $(R_{im})_{i\ge 1}$ be an infinite sequence of iid standard $\mathbb{F}^m$-Gaussian rows. Let $N_m$ be integer-valued, possibly dependent on the row array, with*

$$\frac{N_m}{m} \xrightarrow{\mathbb{P}} \gamma > 1.$$

*Fix $r \ge 0$ and $k \ge 1$. Let $d_{im} > 0$, $i \ge 1$, and $e_{jm} > 0$, $1 \le j \le m$, be finite real random weights, possibly dependent on the rows and on $N_m$. Assume*

$$\Delta_m^{\mathrm{rpdiag}} = \max_{1\le i\le N_m} |\log d_{im}| + \max_{1\le j\le m} |\log e_{jm}| = o_{\mathbb{P}}(m),$$

*with the convention that an absent row or column factor is replaced by weights identically equal to 1. For $N_m \ge m + r$ and $T \subset [N_m]$, $|T| = m + r$, set*

$$B_{m,T} = \operatorname{diag}(d_{im} : i \in T)\, (R_{im})_{i\in T}\, \operatorname{diag}(e_{1m}, \dots, e_{mm})$$

*and define*

$$M^{(r)}_{\mathrm{rpdiag},\mathbb{F}}[k] = \min_{\substack{T\subset[N_m]\\ |T|=m+r}} \sigma_{m-k+1}(B_{m,T}),$$

$$\Lambda^{(r)}_{\mathrm{rpdiag},\mathbb{F}}[k] = \min_{\substack{T\subset[N_m]\\ |T|=m+r}} \lambda_{(k)}\big((m+r)^{-1} B_{m,T}^* B_{m,T}\big)\,.$$

*Extend both statistics by $+\infty$ on $\{N_m < m + r\}$. Then, for every $\varepsilon > 0$,*

$$\mathbb{P}\left\{ M^{(r)}_{\mathrm{rpdiag},\mathbb{F}}[k] \le \exp\left[ -\left( \frac{h(\gamma)}{d_{\mathbb{F}} k(k+r)} + \varepsilon \right) m \right] \right\} \to 0$$

*and*

$$\mathbb{P}\left\{ \Lambda^{(r)}_{\mathrm{rpdiag},\mathbb{F}}[k] \le \exp\left[ -\left( \frac{2h(\gamma)}{d_{\mathbb{F}} k(k+r)} + \varepsilon \right) m \right] \right\} \to 0.$$

*On a common coupling, if $N_m/m \to \gamma$ almost surely and $\Delta_m^{\mathrm{rpdiag}} = o(m)$ almost surely, then*

$$\liminf_{m\to\infty} \frac{1}{m} \log M^{(r)}_{\mathrm{rpdiag},\mathbb{F}}[k] \ge -\frac{h(\gamma)}{d_{\mathbb{F}} k(k+r)}$$

*and*

$$\liminf_{m\to\infty} \frac{1}{m} \log \Lambda^{(r)}_{\mathrm{rpdiag},\mathbb{F}}[k] \ge -\frac{2h(\gamma)}{d_{\mathbb{F}} k(k+r)} \qquad \text{a.s.}$$

*The same conclusions apply to row-normalized random-prefix Gaussian rows and to full-prefix column-normalized Gaussian columns, where the column normalization is computed from all rows in the realized prefix $[N_m]$. For independent thinning, this is read only after the retained iid rows are reindexed; value-dependent thinning or row selection is not covered.*

*Proof.* On the feasible event $\{N_m \ge m + r\}$, for every selected row set $T$,

$$\sigma_{m-k+1}(B_{m,T}) \ge e^{-\Delta_m^{\mathrm{rpdiag}}} \sigma_{m-k+1}((R_{im})_{i\in T})\,.$$

Taking minima over $T$ gives the same lower sandwich for the random-prefix singular-value statistics, and the normalized column-Gram statistic has the squared sandwich. Hence, with $c = h(\gamma)/(d_{\mathbb{F}} k(k+r))$,

$$\mathbb{P}\{M^{(r)}_{\mathrm{rpdiag},\mathbb{F}}[k] \le e^{-(c+\varepsilon)m}\} \le \mathbb{P}\{\Delta_m^{\mathrm{rpdiag}} > \varepsilon m/2\} + \mathbb{P}\{M^{(r)}_{\mathbb{F},m}[k](N_m) \le e^{-(c+\varepsilon/2)m}\}.$$

The first term tends to zero by the diagonal-extrema hypothesis, and the second by Corollary 7.11. The Gram claim follows in the same way, using the lower sandwich $\Lambda^{(r)}_{\mathrm{rpdiag},\mathbb{F}}[k] \ge e^{-2\Delta_m^{\mathrm{rpdiag}}} \Lambda^{(r)}_{\mathbb{F},m}[k](N_m)$ on the feasible event and splitting the slack accordingly. The almost-sure lower envelopes use the same pathwise inequalities together with the a.s. part of Corollary 7.11.

For row normalization, on the high-probability bracket $N_m \le \lceil \gamma_+ m \rceil$, the row-norm extrema over the realized prefix are bounded by the extrema over the deterministic prefix $\lceil \gamma_+ m \rceil$. For column normalization, on $\lfloor \gamma_- m \rfloor \le N_m \le \lceil \gamma_+ m \rceil$, each full-prefix column norm is bracketed between the corresponding deterministic-prefix column norms. The same Gaussian norm upper-tail and ball-volume lower-tail estimates used in the fixed-count normalization corollaries give $\Delta_m^{\mathrm{rpdiag}} = o_{\mathbb{P}}(m)$ and, under common a.s. count convergence, $\Delta_m^{\mathrm{rpdiag}} = o(m)$ almost surely.

This is only a random-prefix diagonal lower-side transfer. In the overlapping square case $r = 0, k = 1$, the exact random-prefix square laws above are stronger. No upper side, exact fixed-excess law, value-dependent row-selection theorem, row-Gram result for $r > 0$, selected-submatrix column normalization, non-diagonal covariance statement, beta interpolation, proportional Wishart edge, or supercritical phase-retrieval consequence is implied. □

**Corollary 7.13** (Random-prefix all-column fixed-excess fixed-$k$ lower sides)**.** *Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$, with $d_{\mathbb{F}} = 1, 2, 4$, respectively. For each $m$, let $(R_{im})_{i \ge 1}$ be an infinite sequence of iid standard $\mathbb{F}^m$-Gaussian rows. Let $N_m$ be integer-valued, possibly dependent on the row array, with*

$$\frac{N_m}{m} \xrightarrow{\mathbb{P}} \gamma > 1.$$

*Fix $r \ge 0$ and $k \ge 1$. Let $d_{im} > 0$, $i \ge 1$, be finite real random row weights, and let $S_m \in \mathbb{F}^{m \times m}$ be invertible. The row weights and $S_m$ may be random and dependent on $N_m$ and on the row array. Assume*

$$\Delta_m^{\mathrm{rpcond}} = \max_{1 \le i \le N_m} |\log d_{im}| + \left|\log \sigma_{\min}(S_m)\right| + \left|\log \|S_m\|\right| = o_{\mathbb{P}}(m),$$

*omitting the row-weight term when row weights are absent. For $N_m \ge m + r$ and $T \subset [N_m]$, $|T| = m + r$, set*

$$B_{m,T} = \mathrm{diag}(d_{im} : i \in T)\,(R_{im})_{i \in T} S_m$$

*and define*

$$M^{(r)}_{\mathrm{rpcond},\mathbb{F}}[k] = \min_{\substack{T \subset [N_m] \\ |T| = m+r}} \sigma_{m-k+1}(B_{m,T}),$$

$$\Lambda^{(r)}_{\mathrm{rpcond},\mathbb{F}}[k] = \min_{\substack{T \subset [N_m] \\ |T| = m+r}} \lambda_{(k)}\left((m+r)^{-1} B^*_{m,T} B_{m,T}\right).$$

*Extend both statistics by $+\infty$ on $\{N_m < m + r\}$. Then, for every $\varepsilon > 0$,*

$$\mathbb{P}\left\{ M^{(r)}_{\mathrm{rpcond},\mathbb{F}}[k] \le \exp\left[ -\left( \frac{h(\gamma)}{d_{\mathbb{F}} k(k+r)} + \varepsilon \right) m \right] \right\} \to 0$$

*and*

$$\mathbb{P}\left\{ \Lambda^{(r)}_{\mathrm{rpcond},\mathbb{F}}[k] \le \exp\left[ -\left( \frac{2h(\gamma)}{d_{\mathbb{F}} k(k+r)} + \varepsilon \right) m \right] \right\} \to 0.$$

*On a common coupling, if $N_m/m \to \gamma$ almost surely and $\Delta_m^{\mathrm{rpcond}} = o(m)$ almost surely, then*

$$\liminf_{m \to \infty} \frac{1}{m} \log M^{(r)}_{\mathrm{rpcond},\mathbb{F}}[k] \ge -\frac{h(\gamma)}{d_{\mathbb{F}} k(k+r)}$$

*and*

$$\liminf_{m \to \infty} \frac{1}{m} \log \Lambda^{(r)}_{\mathrm{rpcond},\mathbb{F}}[k] \ge -\frac{2h(\gamma)}{d_{\mathbb{F}} k(k+r)} \qquad \text{a.s.}$$

*Positive column diagonals are included by taking $S_m$ diagonal. Deterministic or random full-column covariance and whitening factors are included only under the absolute singular-scale hypothesis above. Row-normalized random-prefix Gaussian rows and full-prefix column-normalized Gaussian columns may be combined with this all-column right factor when their realized-prefix logarithmic extrema are included in $\Delta_m^{\mathrm{rpcond}}$.*

*Proof.* On the feasible event $\{N_m \ge m+r\}$, for every selected row set $T$,

$$\sigma_{m-k+1}(B_{m,T}) \ge e^{-\Delta_m^{\mathrm{rpcond}}}\sigma_{m-k+1}((R_{im})_{i\in T})\,.$$

For $\mathbb{F}=\mathbb{H}$, the positive row weights are real scalars and hence central, while $S_m$ is one global quaternionic right factor; the same min–max singular-value sandwich applies. Taking minima over $T$ gives the lower sandwich for the random-prefix singular-value statistics. With $c = h(\gamma)/(d_{\mathbb{F}}k(k+r))$,

$$\mathbb{P}\{M^{(r)}_{\mathrm{rpcond},\mathbb{F}}[k] \le e^{-(c+\varepsilon)m}\} \le \mathbb{P}\{\Delta_m^{\mathrm{rpcond}} > \varepsilon m/2\} + \mathbb{P}\{M^{(r)}_{\mathbb{F},m}[k](N_m) \le e^{-(c+\varepsilon/2)m}\},$$

and both terms tend to zero by the conditioning hypothesis and Corollary 7.11. No independence among the count, row array, row weights, and $S_m$ is used.

For the normalized column Gram, the identity

$$\lambda_{(k)}\big((m+r)^{-1}B^*_{m,T}B_{m,T}\big) = (m+r)^{-1}\sigma_{m-k+1}(B_{m,T})^2$$

gives

$$\Lambda^{(r)}_{\mathrm{rpcond},\mathbb{F}}[k] \ge e^{-2\Delta_m^{\mathrm{rpcond}}}\Lambda^{(r)}_{\mathbb{F},m}[k](N_m)$$

on the feasible event. Hence

$$\mathbb{P}\{\Lambda^{(r)}_{\mathrm{rpcond},\mathbb{F}}[k] \le e^{-(2c+\varepsilon)m}\} \le \mathbb{P}\{\Delta_m^{\mathrm{rpcond}} > \varepsilon m/4\} + \mathbb{P}\{\Lambda^{(r)}_{\mathbb{F},m}[k](N_m) \le e^{-(2c+\varepsilon/2)m}\},$$

which tends to zero by the same two inputs. The almost-sure lower envelopes follow by intersecting the a.s. lower-envelope event from Corollary 7.11 with $\Delta_m^{\mathrm{rpcond}} = o(m)$.

This is only a random-prefix row-submatrix lower-side transfer under one global all-column right factor. A subexponential condition number alone is insufficient, since a scalar factor $e^{cm}I$ shifts the exponential scale. The statement does not assert an upper side, fixed-excess exact law, value-dependent row-selection theorem, selected-column covariance transfer, selected-submatrix column normalization, row-Gram statement for $r>0$, beta interpolation, proportional Wishart-edge law, or supercritical phase-retrieval consequence. □

**Conjecture 7.14** (Quaternionic fixed-excess hard edge)**.** Let $A_m \in \mathbb{H}^{N_m\times m}$ have iid standard quaternion Gaussian entries and $N_m/m \to \gamma > 1$. For each fixed integer $r \ge 1$, one expects

$$\frac{1}{m}\log \min_{\substack{T\subset[N_m]\\ |T|=m+r}} \sigma_{\min}(A_{m,T}) \xrightarrow{\mathbb{P}} -\frac{h(\gamma)}{4(r+1)}.$$

The lower side is the $k=1$ case of Theorem 7.9; the matching upper side still requires a quaternionic fixed-excess pair-correlation theorem.

*Remark* 7.15. The quaternion results above use quaternionic singular values, not the repeated realification indices. They are lower-side extensions except for the square least-singular-value theorem and its positive-real diagonal/normalization transfer and square all-column right-factor transfer, whose upper sides are proved separately. They do not assert a quaternionic version of the spherical-radial transfer theorem, whose statement in this paper is restricted to $\mathbb{F}\in\{\mathbb{R},\mathbb{C}\}$, and they do not promote fixed-excess quaternion exact laws, selected-column non-diagonal covariance transfers, proportional-excess Wishart-edge laws, beta-interpolation statements, or row-Gram statements for $r>0$.

### 7.2. **Deterministic row and column conditioning.**

**Proposition 7.16** (Subexponential deterministic conditioning transfer)**.** *Let $G_m \in \mathbb{F}^{N_m\times m}$, $N_m/m \to \gamma > 1$, have iid standard Gaussian entries. Let $D_m = \mathrm{diag}(d_{1m},\dots,d_{N_m,m})$ be deterministic with positive diagonal entries, let $\Sigma_m$ be deterministic positive definite, and put*

$$A_m = D_m G_m \Sigma_m^{1/2}.$$

*Assume*

$$\frac{1}{m}\log\left(\min_i d_{im}\sqrt{\lambda_{\min}(\Sigma_m)}\right)\to 0, \qquad \frac{1}{m}\log\left(\max_i d_{im}\sqrt{\lambda_{\max}(\Sigma_m)}\right)\to 0.$$

*Then*

$$\frac{1}{m}\log\min_{|T|=m}\sigma_{\min}(A_{m,T})-\frac{1}{m}\log\min_{|T|=m}\sigma_{\min}(G_{m,T})\to 0$$

*deterministically. Consequently*

$$\frac{1}{m}\log\min_{|T|=m}\sigma_{\min}(A_{m,T})\xrightarrow{\mathbb{P}}-\frac{h(\gamma)}{d_{\mathbb{F}}}.$$

*Proof.* For every row set $T$, write $D_T$ for the diagonal submatrix of $D_m$ on the selected rows. Then

$$A_{m,T}=D_T G_{m,T}\Sigma_m^{1/2},$$

and therefore

$$\left(\min_i d_{im}\right)\sqrt{\lambda_{\min}(\Sigma_m)}\,\sigma_{\min}(G_{m,T})\le\sigma_{\min}(A_{m,T})\le\left(\max_i d_{im}\right)\sqrt{\lambda_{\max}(\Sigma_m)}\,\sigma_{\min}(G_{m,T}).$$

Taking minima over $T$, logs, and dividing by $m$ gives the deterministic comparison. The limit then follows from the Gaussian source theorem. □

**Corollary 7.17** (Uniformly elliptic row-scaled Gaussian frames)**.** *If, for constants $0<c<C<\infty$,*

$$c\le d_{im}\le C, \qquad c\le\lambda_{\min}(\Sigma_m)\le\lambda_{\max}(\Sigma_m)\le C$$

*for all $i,m$, then the extremal row-submatrix base is the Gaussian base $h(\gamma)/d_{\mathbb{F}}$. This includes Gaussian frames with common covariance and bounded deterministic row weights.*

*Proof.* The uniform ellipticity assumptions imply the two logarithmic conditioning hypotheses in Proposition 7.16. □

**Proposition 7.18** (Random row-weight transfer)**.** *Let $A_m\in\mathbb{F}^{N_m\times n_m}$ be arbitrary random matrices and let $\mathcal{T}_m$ be any family of row sets for which the selected least singular values are defined. Put*

$$M_m(A)=\min_{T\in\mathcal{T}_m}\sigma_{\min}((A_m)_T).$$

*Let $D_m^{\mathrm{row}}=\operatorname{diag}(w_{1m},\dots,w_{N_m,m})$ have positive random diagonal entries, possibly dependent on $A_m$, and set $B_m=D_m^{\mathrm{row}}A_m$. For any scale $a_m\to\infty$, define*

$$\Delta_m=\max_{1\le i\le N_m}|\log w_{im}|.$$

*On the event $M_m(A)>0$,*

$$\left|\frac{1}{a_m}\log M_m(B)-\frac{1}{a_m}\log M_m(A)\right|\le\frac{\Delta_m}{a_m}.$$

*Consequently, if $\mathbb{P}\{M_m(A)>0\}\to 1$, $\Delta_m=o_{\mathbb{P}}(a_m)$, and $a_m^{-1}\log M_m(A)\to L$ in probability, then*

$$\frac{1}{a_m}\log M_m(B)\xrightarrow{\mathbb{P}}L.$$

*Proof.* For every row set $T$,

$$(B_m)_T=(D_m^{\mathrm{row}})_T(A_m)_T.$$

Since $e^{-\Delta_m}\le w_{im}\le e^{\Delta_m}$ for every row,

$$e^{-\Delta_m}\sigma_{\min}((A_m)_T)\le\sigma_{\min}((B_m)_T)\le e^{\Delta_m}\sigma_{\min}((A_m)_T).$$

The constants are independent of $T$. Taking minima over $T$, logs, and division by $a_m$ gives the displayed comparison. □

**Proposition 7.19** (Random column-weight transfer)**.** *Let $A_m \in \mathbb{F}^{N_m \times n_m}$ be arbitrary random matrices. Let $\mathcal{I}_m$ be any nonempty family of row-column pairs $(R, C)$ for which the least singular values of $(A_m)_{R,C}$ are defined, and put*

$$M_m(A) = \min_{(R,C)\in\mathcal{I}_m} \sigma_{\min}((A_m)_{R,C}).$$

*Let $E_m = \operatorname{diag}(e_{1m}, \dots, e_{n_m,m})$ have strictly positive finite random diagonal entries, possibly dependent on $A_m$, and set $B_m = A_m E_m$. For any scale $a_m \to \infty$, define*

$$\Delta_m^{\mathrm{col}} = \max_{1\le j\le n_m} |\log e_{jm}|.$$

*On the event $M_m(A) > 0$,*

$$\left|\frac{1}{a_m}\log M_m(B) - \frac{1}{a_m}\log M_m(A)\right| \le \frac{\Delta_m^{\mathrm{col}}}{a_m}.$$

*Consequently, if $\mathbb{P}\{M_m(A) > 0\} \to 1$, $\Delta_m^{\mathrm{col}} = o_{\mathbb{P}}(a_m)$, and $a_m^{-1}\log M_m(A) \to L$ in probability, then*

$$\frac{1}{a_m}\log M_m(B) \xrightarrow{\mathbb{P}} L.$$

*Proof.* For every selected pair $(R, C)$,

$$(B_m)_{R,C} = (A_m)_{R,C}(E_m)_C,$$

where $(E_m)_C$ is the positive diagonal submatrix of $E_m$ on the selected columns. Since

$$e^{-\Delta_m^{\mathrm{col}}} \le e_{jm} \le e^{\Delta_m^{\mathrm{col}}} \qquad (1 \le j \le n_m),$$

the singular-value inequalities for a right diagonal factor give

$$e^{-\Delta_m^{\mathrm{col}}}\sigma_{\min}((A_m)_{R,C}) \le \sigma_{\min}((B_m)_{R,C}) \le e^{\Delta_m^{\mathrm{col}}}\sigma_{\min}((A_m)_{R,C}).$$

The constants are uniform over $(R, C)$. Taking minima, logs, and division by $a_m$ proves the comparison. □

**Corollary 7.20** (Gaussian row-normalized frames)**.** *Let $G_m \in \mathbb{F}^{N_m\times m}$ have iid standard Gaussian entries, with $N_m/m \to \gamma > 1$. Let $U_m$ be the row-normalized matrix*

$$(U_m)_i = \frac{(G_m)_i}{\|(G_m)_i\|_2}.$$

*Then*

$$\frac{1}{m}\log \min_{|T|=m} \sigma_{\min}((U_m)_T) \xrightarrow{\mathbb{P}} -\frac{h(\gamma)}{d_{\mathbb{F}}}.$$

*On any common coupling over $m$, the lower envelope also satisfies*

$$\liminf_{m\to\infty} \frac{1}{m}\log \min_{|T|=m} \sigma_{\min}((U_m)_T) \ge -\frac{h(\gamma)}{d_{\mathbb{F}}} \qquad \text{almost surely.}$$

*Proof.* Apply Proposition 7.18 with

$$w_{im} = \|(G_m)_i\|_2^{-1}, \qquad a_m = m.$$

The Gaussian rows are nonzero almost surely, and all square Gaussian row submatrices are nonsingular almost surely, so the positive-weight and $M_m(G) > 0$ hypotheses of the transfer proposition hold with probability one. It remains to check

$$\max_i |\log \|(G_m)_i\|_2| = o_{\mathbb{P}}(m).$$

For every fixed $\delta > 0$, standard Gaussian norm tails give

$$\mathbb{P}\left\{\max_{i\le N_m} \|(G_m)_i\|_2 > e^{\delta m}\right\} \to 0.$$

For the lower tail, a ball-volume estimate in real dimension $d_{\mathbb{F}}m$ gives

$$\mathbb{P}\left\{\min_{i\le N_m}\|(G_m)_i\|_2 < e^{-\delta m}\right\} \le \exp\{-d_{\mathbb{F}}\delta m^2 + O(m\log m)\} \to 0.$$

Thus the random row weights have subexponential extrema, and the Gaussian source theorem transfers to $U_m$.

For the almost-sure lower envelope, only the upper row-norm extreme is needed:

$$\min_{|T|=m}\sigma_{\min}((U_m)_T) \ge \frac{\min_{|T|=m}\sigma_{\min}((G_m)_T)}{\max_{i\le N_m}\|(G_m)_i\|_2}.$$

Since $N_m = O(m)$, Markov's inequality gives

$$\sum_m \mathbb{P}\left\{\max_{i\le N_m}\|(G_m)_i\|_2 > e^{\delta m}\right\} \le \sum_m N_m\,\mathbb{E}\|(G_m)_1\|_2^2 e^{-2\delta m} < \infty.$$

Corollary 3.13, Borel–Cantelli for this row-norm bound, and then $\delta \downarrow 0$ give the almost-sure lower envelope. □

*Remark* 7.21 (Normalized non-Gaussian rows). For lower-side estimates, one only needs the upper row-norm extreme. If $B_i = A_i/\|A_i\|_2$, the rows are nonzero with high probability, and

$$\max_{i\le N_m}\|A_i\|_2 = e^{o_{\mathbb{P}}(m)},$$

then $M_m(B) \ge e^{-o_{\mathbb{P}}(m)}M_m(A)$. Hence any lower-side exponential bound already proved for $A_m$ transfers to the row-normalized ensemble. Bounded scalar density alone does not imply this row-tail condition for pathologically heavy-tailed laws; it should be assumed, or replaced by a sufficient tail hypothesis such as

$$N_m m\,\mathbb{P}\{|X_m| > e^{\delta m}/\sqrt{m}\} \to 0 \qquad (\delta > 0).$$

The normalized rows are sphere-supported and their coordinates are dependent, so the bounded-density entry theorem cannot be applied directly to $B_m$.

**Corollary 7.22** (Gaussian column-normalized matrices). *Let $G_m \in \mathbb{F}^{N_m\times m}$ have iid standard Gaussian entries, with $N_m/m \to \gamma > 1$. Let $V_m$ be the column-normalized matrix*

$$(V_m)^{(j)} = \frac{(G_m)^{(j)}}{\|(G_m)^{(j)}\|_2}, \qquad 1 \le j \le m,$$

*where $(G_m)^{(j)}$ denotes the $j$-th column. Then*

$$\frac{1}{m}\log\min_{|T|=m}\sigma_{\min}((V_m)_T) \xrightarrow{\mathbb{P}} -\frac{h(\gamma)}{d_{\mathbb{F}}}.$$

*On any common coupling over $m$, the lower envelope also satisfies*

$$\liminf_{m\to\infty}\frac{1}{m}\log\min_{|T|=m}\sigma_{\min}((V_m)_T) \ge -\frac{h(\gamma)}{d_{\mathbb{F}}} \qquad \textit{almost surely.}$$

*Proof.* Write $V_m = G_mE_m$, where

$$E_m = \operatorname{diag}\big(\|(G_m)^{(1)}\|_2^{-1},\dots,\|(G_m)^{(m)}\|_2^{-1}\big).$$

The Gaussian columns are nonzero almost surely. We apply Proposition 7.19 to the row-column family $(T,[m])$, $|T| = m$; Gaussian square row submatrices are nonsingular almost surely, so the required positivity event has probability one. By Proposition 7.19, it is enough for convergence in probability to prove

$$\max_{1\le j\le m}\left|\log\|(G_m)^{(j)}\|_2\right| = o_{\mathbb{P}}(m).$$

For the almost-sure lower envelope, the same estimates will be summable. For the upper tail, Markov's inequality gives, for every fixed $\delta > 0$,

$$\mathbb{P}\{\|(G_m)^{(j)}\|_2 > e^{\delta m}\} \le N_m e^{-2\delta m},$$

and the union over $j \le m$ is summable in $m$. For the lower tail, the column lives in real dimension $d_{\mathbb{F}} N_m$, so a ball-volume estimate gives

$$\mathbb{P}\{\|(G_m)^{(j)}\|_2 < e^{-\delta m}\} \le \exp\{-d_{\mathbb{F}}\delta N_m m + O(N_m \log N_m)\}.$$

Since $N_m/m \to \gamma > 1$, the union over $j \le m$ is again summable. Thus the column weights have subexponential extrema in probability and almost surely. The Gaussian source theorem gives the convergence in probability, and Corollary 3.13 plus the pathwise sandwich gives the almost-sure lower envelope. □

*Remark* 7.23 (Product variance profiles). For real Gaussian entries $A_{ij} = d_i g_{ij} c_j$, this is a bounded product variance profile. It is not a theorem for arbitrary variance profiles $\sigma_{ij} g_{ij}$: row types may have different entropy and small-ball prefactors, so the sharp base should be profile-dependent.

**Corollary 7.24** (Fixed-excess transfer)**.** *Under the hypotheses of Proposition 7.16, fix $r \ge 0$ and define*

$$M_A^{(r)} = \min_{|T|=m+r} \sigma_{\min}(A_{m,T}), \qquad M_G^{(r)} = \min_{|T|=m+r} \sigma_{\min}(G_{m,T}).$$

*Then*

$$\frac{1}{m}\log M_A^{(r)} - \frac{1}{m}\log M_G^{(r)} \to 0$$

*deterministically. Consequently the fixed-excess lower side of Theorem 5.3 transfers to $A_m$: for every $\varepsilon > 0$,*

$$\mathbb{P}\left\{ M_A^{(r)} \le \exp\left[ -\left( \frac{h(\gamma)}{d_{\mathbb{F}}(r+1)} + \varepsilon \right) m \right] \right\} \to 0.$$

*Proof.* The singular-value sandwich in the proof of Proposition 7.16 does not use $|T| = m$. It applies uniformly to every $(m+r)$-row set $T$. Taking minima, logs, and dividing by $m$ gives the deterministic comparison, and Theorem 5.3 gives the lower side. □

**Corollary 7.25** (Fixed-excess fixed-$k$ transfer)**.** *Under the hypotheses of Proposition 7.16, with $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, fix integers $r \ge 0$ and $k \ge 1$. For $m \ge k$, put*

$$M_A^{(r)}[k] = \min_{\substack{T\subset[N_m]\\|T|=m+r}} \sigma_{m-k+1}(A_{m,T}), \qquad M_G^{(r)}[k] = \min_{\substack{T\subset[N_m]\\|T|=m+r}} \sigma_{m-k+1}(G_{m,T}).$$

*Then*

$$\frac{1}{m}\log M_A^{(r)}[k] - \frac{1}{m}\log M_G^{(r)}[k] \to 0$$

*deterministically. Consequently the fixed-excess fixed-k lower side of Corollary 5.20 transfers to $A_m$: for every $\varepsilon > 0$,*

$$\mathbb{P}\left\{ M_A^{(r)}[k] \le \exp\left[ -\left( \frac{h(\gamma)}{d_{\mathbb{F}}k(k+r)} + \varepsilon \right) m \right] \right\} \to 0.$$

*The normalized column-Gram statistic*

$$\Lambda_A^{(r)}[k] = \min_{\substack{T\subset[N_m]\\|T|=m+r}} \lambda_{(k)}\big((m+r)^{-1} A_{m,T}^* A_{m,T}\big)$$

*has the same deterministic transfer from the corresponding statistic* $\Lambda_G^{(r)}[k]$ *defined with* $G_m$ *in place of* $A_m$*, and hence has the lower-side base* $2h(\gamma)/(d_{\mathbb{F}}k(k+r))$ *under the same hypotheses. On any common coupling over* $m$*, the lower envelopes also satisfy*

$$\liminf_{m\to\infty}\frac{1}{m}\log M_A^{(r)}[k]\geq -\frac{h(\gamma)}{d_{\mathbb{F}}k(k+r)}\qquad \textit{almost surely,}$$

*and*

$$\liminf_{m\to\infty}\frac{1}{m}\log \Lambda_A^{(r)}[k]\geq -\frac{2h(\gamma)}{d_{\mathbb{F}}k(k+r)}\qquad \textit{almost surely.}$$

*Proof.* For every row set $T$, the min–max singular-value inequalities applied to

$$A_{m,T}=D_T G_{m,T}\Sigma_m^{1/2}$$

give

$$\Big(\min_i d_{im}\Big)\sqrt{\lambda_{\min}(\Sigma_m)}\,\sigma_{m-k+1}(G_{m,T})\leq \sigma_{m-k+1}(A_{m,T})$$
$$\leq \Big(\max_i d_{im}\Big)\sqrt{\lambda_{\max}(\Sigma_m)}\,\sigma_{m-k+1}(G_{m,T}).$$

The constants are independent of $T$. Taking minima, logs, and dividing by $m$ gives the deterministic comparison, and Corollary 5.20 gives the lower side. Squaring the same comparison and multiplying by $(m+r)^{-1}$ gives the normalized column-Gram transfer, using Corollary 5.21. On a common coupling over $m$, Corollary 5.35 gives the Gaussian lower envelopes for $M_G^{(r)}[k]$ and $\Lambda_G^{(r)}[k]$. The deterministic comparison above and its squared Gram version have $o(1)$ logarithmic error after division by $m$, so the two liminf bounds transfer to $A_m$. Absolute extreme eigenvalue control is used here; a subexponential condition number alone would not prevent a global exponential rescaling. □

*Remark* 7.26 (Random diagonal weights at fixed excess). The same min–max argument used in Propositions 7.18 and 7.19 applies to the fixed-excess fixed-$k$ statistics above. Let

$$\widetilde{A}_m=D_m^{\mathrm{row}}A_mE_m,$$

where the diagonal entries are strictly positive and finite, random and possibly dependent on $A_m$, and set

$$\Delta_m^{\mathrm{diag}}=\max_{i\leq N_m}|\log(D_m^{\mathrm{row}})_{ii}|+\max_{j\leq m}|\log(E_m)_{jj}|,$$

omitting the absent term if only one side is weighted. On the event $M_A^{(r)}[k]>0$,

$$\left|\frac{1}{m}\log M_{\widetilde{A}}^{(r)}[k]-\frac{1}{m}\log M_A^{(r)}[k]\right|\leq\frac{\Delta_m^{\mathrm{diag}}}{m},\qquad \left|\frac{1}{m}\log \Lambda_{\widetilde{A}}^{(r)}[k]-\frac{1}{m}\log \Lambda_A^{(r)}[k]\right|\leq\frac{2\Delta_m^{\mathrm{diag}}}{m}.$$

The first comparison is uniform over $|T|=m+r$, and the normalized column-Gram comparison is its square. Hence these lower sides transfer whenever $\Delta_m^{\mathrm{diag}}=o_{\mathbb{P}}(m)$, and the corresponding lower envelopes transfer on a common coupling whenever $\Delta_m^{\mathrm{diag}}=o(m)$ almost surely. This remains only a diagonal lower-side transfer; no upper side, exact law, row-Gram statement for $r>0$, or non-diagonal column-covariance statement is implied.

**Corollary 7.27** (Normalized Gaussian fixed-excess fixed-$k$ lower sides)**.** *Let* $\mathbb{F}\in\{\mathbb{R},\mathbb{C}\}$*, and let* $G_m\in\mathbb{F}^{N_m\times m}$ *have iid standard Gaussian entries with* $N_m/m\to\gamma>1$*. Let* $U_m$ *and* $V_m$ *be the row- and column-normalized matrices of Corollaries 7.20 and 7.22. Fix* $r\geq 0$ *and* $k\geq 1$*. For* $m\geq k$*,* $N_m\geq m+r$*, and either choice* $B_m=U_m$ *or* $B_m=V_m$*, put*

$$M_B^{(r)}[k]=\min_{\substack{T\subset[N_m]\\|T|=m+r}}\sigma_{m-k+1}((B_m)_T),$$

*and*

$$\Lambda_B^{(r)}[k] = \min_{\substack{T\subset[N_m]\\|T|=m+r}} \lambda_{(k)}\big((m+r)^{-1}(B_m)_T^*(B_m)_T\big)\,.$$

*Then, for every $\varepsilon > 0$,*

$$\mathbb{P}\left\{M_B^{(r)}[k] \le \exp\left[-\left(\frac{h(\gamma)}{d_{\mathbb{F}}k(k+r)}+\varepsilon\right)m\right]\right\} \to 0,$$

*and*

$$\mathbb{P}\left\{\Lambda_B^{(r)}[k] \le \exp\left[-\left(\frac{2h(\gamma)}{d_{\mathbb{F}}k(k+r)}+\varepsilon\right)m\right]\right\} \to 0.$$

*On any common coupling over $m$, the corresponding almost-sure lower envelopes hold:*

$$\liminf_{m\to\infty}\frac{1}{m}\log M_B^{(r)}[k] \ge -\frac{h(\gamma)}{d_{\mathbb{F}}k(k+r)}, \qquad \liminf_{m\to\infty}\frac{1}{m}\log \Lambda_B^{(r)}[k] \ge -\frac{2h(\gamma)}{d_{\mathbb{F}}k(k+r)}.$$

*Proof.* Gaussian $(m+r)$-row submatrices have rank $m$ almost surely. For $B_m = U_m$, the normalization is a positive random row diagonal weighting of $G_m$; for $B_m = V_m$, it is a positive random column diagonal weighting. The Gaussian norm estimates in the proofs of Corollaries 7.20 and 7.22 give the corresponding diagonal logarithmic extrema $o_{\mathbb{P}}(m)$. The upper row-norm, upper column-norm, and lower column-norm tail bounds there are summable; the displayed row ball-volume lower tail is summable after the union over $N_m = O(m)$ rows as well. Hence the same diagonal logarithmic extrema are $o(m)$ almost surely on any common coupling. Apply Remark 7.26 with $A_m = G_m$, using Corollaries 5.20 and 5.21 for the lower sides and Corollary 5.35 for the Gaussian almost-sure lower envelopes. □

**Corollary 7.28** (Quaternion diagonal and normalized fixed-excess fixed-$k$ lower sides)**.** *Let $G_m \in \mathbb{H}^{N_m\times m}$ have iid standard quaternion Gaussian entries with $N_m/m \to \gamma > 1$. Fix $r \ge 0$ and $k \ge 1$. Let*

$$\widetilde{G}_m = D_m^{\mathrm{row}} G_m E_m,$$

*where $D_m^{\mathrm{row}}$ and $E_m$ are positive real diagonal matrices, random and possibly dependent on $G_m$, with strictly positive finite entries, and where either diagonal factor may be absent. Assume*

$$\max_i |\log(D_m^{\mathrm{row}})_{ii}| + \max_j |\log(E_m)_{jj}| = o_{\mathbb{P}}(m),$$

*again omitting absent terms. Put*

$$M_{\widetilde{G},\mathbb{H}}^{(r)}[k] = \min_{\substack{T\subset[N_m]\\|T|=m+r}} \sigma_{m-k+1}((\widetilde{G}_m)_T)$$

*and*

$$\Lambda_{\widetilde{G},\mathbb{H}}^{(r)}[k] = \min_{\substack{T\subset[N_m]\\|T|=m+r}} \lambda_{(k)}\Big((m+r)^{-1}(\widetilde{G}_m)_T^*(\widetilde{G}_m)_T\Big)\,,$$

*with quaternionic singular values and Hermitian eigenvalues counted once. Then, for every $\varepsilon > 0$,*

$$\mathbb{P}\left\{M_{\widetilde{G},\mathbb{H}}^{(r)}[k] \le \exp\left[-\left(\frac{h(\gamma)}{4k(k+r)}+\varepsilon\right)m\right]\right\} \to 0$$

*and*

$$\mathbb{P}\left\{\Lambda_{\widetilde{G},\mathbb{H}}^{(r)}[k] \le \exp\left[-\left(\frac{h(\gamma)}{2k(k+r)}+\varepsilon\right)m\right]\right\} \to 0.$$

*If the displayed logarithmic extreme is $o(m)$ almost surely on a common coupling, then the corresponding a.s. lower envelopes also hold with the same bases.*

*In particular, the same conclusions hold when $D_m^{\mathrm{row}}$ row-normalizes the quaternion Gaussian rows, or when $E_m$ column-normalizes the full quaternion Gaussian columns. For the square least-singular-value case $r = 0, k = 1$, the stronger convergence in probability under the same diagonal hypotheses is Corollary 7.6.*

*Proof.* The entries of the diagonal factors are positive real scalars, hence central in $\mathbb{H}$, and the min–max singular-value sandwich in Remark 7.26 applies with the quaternionic conventions fixed above. The source lower sides are Theorem 7.9 and Corollary 7.10; the same remark gives the corresponding common-coupling lower envelopes when the logarithmic extreme is $o(m)$ almost surely.

For row normalization, the squared norm of one row is $(1/4)\chi^2_{4m}$; for column normalization, the squared norm of one column is $(1/4)\chi^2_{4N_m}$. The Markov upper-tail estimates and ball-volume lower-tail estimates used in Corollaries 7.20 and 7.22, with real dimensions $4m$ and $4N_m$, show that the corresponding row or column logarithmic extremes are $o_{\mathbb{P}}(m)$ and $o(m)$ almost surely after the $O(m)$ unions. Gaussian rows and columns are nonzero almost surely, so the normalizations are well defined. □

**Corollary 7.29** (Two-sided diagonal transfer)**.** *Let $G_k \in \mathbb{F}^{N_k \times n_k}$ have iid standard Gaussian entries with $N_k/k \to \gamma_R > 1$ and $n_k/k \to \gamma_C > 1$. Let*

$$A_k = D_k G_k E_k$$

*where $D_k$ and $E_k$ are deterministic positive diagonal matrices. Assume*

$$\frac{1}{k}\log\Big((\min_i D_{k,ii})(\min_j E_{k,jj})\Big) \to 0, \qquad \frac{1}{k}\log\Big((\max_i D_{k,ii})(\max_j E_{k,jj})\Big) \to 0.$$

*For*

$$M_A^{\mathrm{two}} = \min_{|R|=|C|=k} \sigma_{\min}((A_k)_{R,C}), \qquad M_G^{\mathrm{two}} = \min_{|R|=|C|=k} \sigma_{\min}((G_k)_{R,C}),$$

*one has*

$$\frac{1}{k}\log M_A^{\mathrm{two}} - \frac{1}{k}\log M_G^{\mathrm{two}} \to 0$$

*deterministically. Thus the two-sided lower side of Theorem 5.26 transfers to $A_k$.*

*Proof.* For each $R, C$, $(A_k)_{R,C} = D_R (G_k)_{R,C} E_C$. Therefore

$$(\min_i D_{k,ii})(\min_j E_{k,jj})\, \sigma_{\min}((G_k)_{R,C}) \le \sigma_{\min}((A_k)_{R,C}) \le (\max_i D_{k,ii})(\max_j E_{k,jj})\, \sigma_{\min}((G_k)_{R,C}).$$

Take minima over $R, C$, logs, and divide by $k$. □

The same pathwise comparison remains true if the positive diagonal matrices $D_k, E_k$ are random and possibly dependent on $G_k$: the absolute difference of the normalized logarithms is bounded by

$$\frac{1}{k}\left(\max_i |\log D_{k,ii}| + \max_j |\log E_{k,jj}|\right).$$

Consequently the two-sided lower-side transfer holds in probability whenever this bound is $o_{\mathbb{P}}(1)$, and it holds pathwise for almost-sure lower envelopes whenever the bound is $o(1)$ almost surely. This is still only a diagonal statement.

*Remark* 7.30. The two-sided diagonal transfer should not be replaced by a general column covariance statement. Column selection does not commute with multiplication by a non-diagonal $\Sigma^{1/2}$, since $(G\Sigma^{1/2})_{R,C} = G_R(\Sigma^{1/2})_{:,C}$, not a selected Gaussian minor times a $C \times C$ covariance factor.

7.3. **Gram and sample-covariance transfer.** At the exponential scales of this paper, passing from singular values to Gram or sample-covariance eigenvalues is deterministic bookkeeping. It changes logarithmic constants by a factor of two and preserves the proof status of the underlying singular-value statement.

**Corollary 7.31** (Gram and sample-covariance transfer)**.** *Let $B_{m,I}$, $I \in \mathcal{I}_m$, be a finite family of $q_m \times n_m$ matrices over $\mathbb{R}$, $\mathbb{C}$, or $\mathbb{H}$, with $q_m \geq 1$. In the quaternionic case use the adjoint and singular-value conventions fixed above. Put*

$$M_m = \min_{I \in \mathcal{I}_m} \sigma_{\min}(B_{m,I}), \qquad \Lambda_m = \min_{I \in \mathcal{I}_m} \lambda_{\min}\big(q_m^{-1} B_{m,I}^* B_{m,I}\big).$$

*Then pathwise*

$$\Lambda_m = q_m^{-1} M_m^2.$$

*Consequently, if $a_m \to \infty$, $\log q_m = o(a_m)$, and*

$$\frac{1}{a_m} \log M_m \to L$$

*in probability, then*

$$\frac{1}{a_m} \log \Lambda_m \to 2L$$

*in probability. The same implication holds for deterministic comparisons and for almost-sure lower envelopes, with a lower bound $L$ replaced by the lower bound $2L$.*

*Proof.* For every index $I$,

$$\lambda_{\min}(B_{m,I}^* B_{m,I}) = \sigma_{\min}(B_{m,I})^2.$$

Since $x \mapsto q_m^{-1} x^2$ is increasing on $[0, \infty)$, taking the minimum over $I$ gives the pathwise identity. Taking logarithms gives

$$\frac{1}{a_m} \log \Lambda_m = 2 \frac{1}{a_m} \log M_m - \frac{\log q_m}{a_m},$$

on the event $M_m > 0$, and the stated consequences follow from $\log q_m = o(a_m)$. □

For iid standard Gaussian square row submatrices over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, Corollary 7.31 and the source theorem give

$$\frac{1}{m} \log \min_{\substack{T \subset [N_m] \\ |T| = m}} \lambda_{\min}\big(m^{-1} A_{m,T}^* A_{m,T}\big) \xrightarrow{\mathbb{P}} -\frac{2h(\gamma)}{d_{\mathbb{F}}}.$$

The quaternionic theorem above gives, for iid standard quaternion Gaussian entries,

$$\frac{1}{m} \log \min_{\substack{T \subset [N_m] \\ |T| = m}} \lambda_{\min}\big(m^{-1} A_{m,T}^* A_{m,T}\big) \xrightarrow{\mathbb{P}} -\frac{h(\gamma)}{2}.$$

Combining the same identity with Proposition 7.16, the square sample-covariance exponent is unchanged for $A_m = D_m G_m \Sigma_m^{1/2}$ under the subexponential conditioning hypotheses there, in particular for uniformly elliptic Gaussian samples with common population covariance. A subexponential condition number alone is not enough: an exponential scalar multiple of the covariance shifts the logarithmic edge. Fixed-excess, two-sided, and finite-budget lower sides also translate by the same identity, but only with their original status. A lower-side singular-value theorem gives a lower-side Gram theorem; a conditional exact theorem remains conditional.

*Remark* 7.32. This corollary is only a change of variables from singular values to Gram eigenvalues. It does not identify the proportional-excess Wishart lower edge on the $\sqrt{m}$ singular-value scale, nor the corresponding $m$-scale sample-covariance edge. Those require the left-tail large-deviation input described in Section 5.

7.4. **General rows.** For independent rows, entry independence is not needed for the square lower side. Corollary 3.7 gives a finite budget from uniform affine directional anti-concentration of the row laws. The fixed-excess analogue is Corollary 5.7: the hard-edge exponent $r+1$ comes from $r+1$ remaining rows tested against the normal to an $m-1$-row block. Thus iid real isotropic log-concave rows have the square lower-side base $h(\gamma)$, by Corollary 3.8, and for each fixed $r \geq 0$,

$$\mathbb{P}\left\{ M_m^{(r)} \leq \exp\left[ -\left( \frac{h(\gamma)}{r+1} + \varepsilon \right) m \right] \right\} \to 0.$$

The exact laws remain open: the second moment would need simultaneous incompressibility of random normals and local anti-concentration for two projections of a fresh row, now without product coordinates.

**Problem 7.33** (Log-concave row-submatrix hard edge)**.** Let $X_1, \dots, X_N$ be iid isotropic log-concave vectors in $\mathbb{R}^m$, with $N/m \to \gamma > 1$, and let $A$ be the matrix with rows $X_i$. Determine whether

$$\frac{1}{m} \log \min_{|T|=m} \sigma_{\min}(A_T) \to -h(\gamma)$$

in probability. The lower bound at every threshold below $-h(\gamma)$ is proved above; the open part is the matching upper side and the pair-correlation mechanism for dependent coordinates.

**Problem 7.34** (Non-square sample-covariance edge)**.** For iid samples $X_i \in \mathbb{R}^m$, determine the extremal lower edge of $\ell^{-1} A_T^* A_T$ over row sets $|T| = \ell \geq m$, beyond the square Gaussian and elliptic Gaussian cases covered by Corollary 7.31. For fixed $\ell - m = r$, this is the squared fixed-excess hard-edge problem and has only the transferred lower side unless Conjecture 5.10 is proved. For $\ell/m \to \zeta > 1$, the exponential scale is trivial and the remaining question is the constant-order Wishart edge governed by Problem 10.8.

7.5. **Spherical directions and radial weights.** There is a clean spherical-direction extension that does not require a new second-moment proof. Once the row directions are uniform on the sphere, changing the row radii is a diagonal row weighting. Proposition 7.18 therefore transfers the Gaussian row-normalized theorem to all spherical direction ensembles whose radial weights have subexponential global variation.

**Definition 7.35** (Spherical radial model)**.** Let $\Theta_m \in \mathbb{F}^{N_m \times m}$ have iid rows uniform on the unit sphere of $\mathbb{F}^m$, where $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. A *spherical radial weighting* is a matrix

$$A_m = D_m^{\mathrm{rad}} \Theta_m, \qquad D_m^{\mathrm{rad}} = \operatorname{diag}(r_{1m}, \dots, r_{N_m,m}),$$

where the $r_{im}$ are positive finite random variables, possibly dependent on $\Theta_m$. For a deterministic scale $b_m > 0$, put

$$\Delta_m^{\mathrm{rad}}(b_m) = \max_{1 \leq i \leq N_m} \left| \log \frac{r_{im}}{b_m} \right|.$$

This includes rotationally invariant row laws $a_i = R_{im}\theta_i$, with $\theta_i$ uniform on the sphere and $R_{im}$ independent of $\theta_i$, whenever the rows are independent.

**Theorem 7.36** (Spherical directions with radial weights)**.** *Let $N_m/m \to \gamma > 1$, let $A_m = D_m^{\mathrm{rad}} \Theta_m$ be a spherical radial weighting over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and assume that for some deterministic $b_m > 0$,*

$$\frac{1}{m} \log b_m \to \alpha, \qquad \Delta_m^{\mathrm{rad}}(b_m) = o_{\mathbb{P}}(m).$$

*Then*

$$\frac{1}{m} \log \min_{|T|=m} \sigma_{\min}((A_m)_T) \xrightarrow{\mathbb{P}} \alpha - \frac{h(\gamma)}{d_{\mathbb{F}}}.$$

*In particular, if $b_m = e^{o(m)}$, spherical radial weights with subexponential global variation have the same exponential base as the Gaussian row-submatrix model.*

*Proof.* Write

$$M_m(B) = \min_{|T|=m} \sigma_{\min}((B_m)_T)$$

for any matrix sequence $B_m$. Let $G_m$ have iid standard $\mathbb{F}$-Gaussian entries and let $U_m$ be obtained by normalizing every nonzero row of $G_m$ to Euclidean norm one. The rows of $U_m$ are iid uniform on the unit sphere, so $U_m$ has the same law as $\Theta_m$. Corollary 7.20 gives

$$\frac{1}{m}\log M_m(\Theta) = \frac{1}{m}\log \min_{|T|=m} \sigma_{\min}((\Theta_m)_T) \xrightarrow{\mathbb{P}} -\frac{h(\gamma)}{d_{\mathbb{F}}}.$$

For every row set $T$,

$$(A_m)_T = (D_m^{\mathrm{rad}})_T(\Theta_m)_T.$$

On the event $\Delta_m^{\mathrm{rad}}(b_m) \le \delta m$, all row weights satisfy $b_m e^{-\delta m} \le r_{im} \le b_m e^{\delta m}$. Hence

$$b_m e^{-\delta m}\sigma_{\min}((\Theta_m)_T) \le \sigma_{\min}((A_m)_T) \le b_m e^{\delta m}\sigma_{\min}((\Theta_m)_T)$$

uniformly in $T$. All selected spherical square submatrices are nonsingular almost surely; positive finite radii preserve this, so the logarithms below are taken off a null event. Taking minima over $T$, logarithms, and division by $m$ gives

$$\left|\frac{1}{m}\log M_m(A) - \frac{1}{m}\log b_m - \frac{1}{m}\log M_m(\Theta)\right| \le \delta$$

on that event. Since $\delta > 0$ is arbitrary and $\Delta_m^{\mathrm{rad}}(b_m) = o_{\mathbb{P}}(m)$, the claimed limit follows. □

**Corollary 7.37** (Rotationally invariant rows with controlled radii)**.** *Let the rows of $A_m \in \mathbb{F}^{N_m\times m}$ be iid rotationally invariant vectors $a_i = R_{im}\theta_i$, with $\theta_i$ uniform on the unit sphere of $\mathbb{F}^m$, $R_{im} > 0$, and $N_m/m \to \gamma > 1$. If there is a deterministic scale $b_m > 0$ such that*

$$\frac{1}{m}\log b_m \to \alpha, \qquad N_m\,\mathbb{P}\left\{\left|\log\frac{R_{1m}}{b_m}\right| > \varepsilon m\right\} \to 0 \quad (\varepsilon > 0),$$

*then*

$$\frac{1}{m}\log \min_{|T|=m} \sigma_{\min}(A_{m,T}) \xrightarrow{\mathbb{P}} \alpha - \frac{h(\gamma)}{d_{\mathbb{F}}}.$$

*Proof.* The displayed tail condition is exactly the union-bound criterion for $\Delta_m^{\mathrm{rad}}(b_m) = o_{\mathbb{P}}(m)$. Apply Theorem 7.36. □

**Corollary 7.38** (Almost-sure spherical radial lower envelope)**.** *Assume the matrices $A_m = D_m^{\mathrm{rad}}\Theta_m$ are realized on a common probability space, with $N_m/m \to \gamma > 1$. Let $b_m > 0$ be deterministic and put*

$$\alpha_- = \liminf_{m\to\infty} \frac{1}{m}\log b_m.$$

*If*

$$\Delta_m^{\mathrm{rad}}(b_m) = o(m) \qquad \text{almost surely,}$$

*then*

$$\liminf_{m\to\infty} \frac{1}{m}\log \min_{|T|=m} \sigma_{\min}((A_m)_T) \ge \alpha_- - \frac{h(\gamma)}{d_{\mathbb{F}}} \qquad \text{almost surely.}$$

*For iid rotationally invariant rows $a_i = R_{im}\theta_i$, it is enough that, for every $\varepsilon > 0$,*

$$\sum_{m=1}^{\infty} N_m\,\mathbb{P}\left\{\left|\log\frac{R_{1m}}{b_m}\right| > \varepsilon m\right\} < \infty.$$

*Proof.* First consider $\Theta_m$. It has the same law as the row-normalized Gaussian matrix $U_m$. The proof of Corollary 7.20 gives summable failure probabilities at every threshold below $-h(\gamma)/d_{\mathbb{F}}$; hence, by the first Borel–Cantelli lemma, the same bound holds for any common coupling of the spherical matrices $\Theta_m$:

$$\liminf_{m\to\infty} \frac{1}{m} \log \min_{|T|=m} \sigma_{\min}((\Theta_m)_T) \geq -\frac{h(\gamma)}{d_{\mathbb{F}}} \qquad \text{almost surely.}$$

No independence across $m$ is needed.

The radial sandwich in the proof of Theorem 7.36 gives

$$\min_{|T|=m} \sigma_{\min}((A_m)_T) \geq b_m e^{-\Delta_m^{\mathrm{rad}}(b_m)} \min_{|T|=m} \sigma_{\min}((\Theta_m)_T).$$

Taking logarithmic liminf and using $\Delta_m^{\mathrm{rad}}(b_m) = o(m)$ proves the displayed lower envelope. The final summability condition implies $\Delta_m^{\mathrm{rad}}(b_m) = o(m)$ almost surely by the union bound and the first Borel–Cantelli lemma. □

**Corollary 7.39** (Spherical random frames have Balan–Wang base 1/4)**.** *Let $A_m \in \mathbb{R}^{(2m-1)\times m}$ have iid rows uniform on the sphere of deterministic radius $b_m > 0$, where $b_m = e^{o(m)}$. Then $A_m$ is full spark almost surely and*

$$\frac{1}{m} \log \omega(A_m) \xrightarrow{\mathbb{P}} -\log 4, \qquad \omega(A_m) = 4^{-m+o_{\mathbb{P}}(m)}.$$

*On any common coupling over $m$, the lower envelope also satisfies*

$$\liminf_{m\to\infty} \frac{1}{m} \log \omega(A_m) \geq -\log 4 \qquad \textit{almost surely.}$$

*Proof.* For each fixed $m$, iid continuous spherical rows are full spark almost surely. On this event the Balan–Wang statistic at $N = 2m-1$ is

$$\omega(A_m) = \min_{|T|=m} \sigma_{\min}(A_{m,T}).$$

Apply Theorem 7.36 with $\gamma = 2$, $h(2) = \log 4$, and $\alpha = 0$. □

On a common coupling over $m$, the displayed lower envelope follows from Corollary 7.38.

*Remark* 7.40 (Radial extremes). The theorem covers non-degenerate radial mixtures as long as the largest and smallest radii among the $O(m)$ sampled rows remain subexponential after a common deterministic scaling. Such mixtures do not change the exponential base. What remains outside the theorem is a genuinely exponential radial extreme regime, where some available rows have radii $e^{\Theta(m)}b_m$ or $e^{-\Theta(m)}b_m$. Then the global diagonal sandwich gives only crude bounds, and the sharp base becomes a weighted selection problem for the row radii; see Problem 10.11.

## 8. Phase retrieval stability, generalized

For real phase retrieval, the critical number of measurements is $N = 2m-1$ [1, 4]. When $A \in \mathbb{R}^{(2m-1)\times m}$ is full spark, the complement property [1, 4, 7] implies that the Balan–Wang parameter [2] reduces to the extremal square-row-submatrix quantity:

$$\omega(A) = \min_{|T|=m} \sigma_{\min}(A_T).$$

**Theorem 8.1** (Continuous critical frames: lower side)**.** *Let $A_m \in \mathbb{R}^{(2m-1)\times m}$ have independent entries with densities uniformly bounded by $K$. Then for every $\varepsilon > 0$,*

$$\mathbb{P}\{\omega(A_m) \leq e^{-(\log 4+\varepsilon)m}\} \to 0,$$

*provided $A_m$ is interpreted on the full-spark event. For continuous entry laws the full-spark event has probability one.*

*Proof.* At $N = 2m - 1$, Stirling's formula gives

$$\frac{1}{m} \log \binom{2m-1}{m} \to \log 4.$$

Apply Corollary 3.4. Continuous iid entries give full spark almost surely; more generally, independent absolutely continuous entries have this property because each $m \times m$ minor is a nonzero polynomial in the entries, and there are only finitely many minors for each $m$. □

**Corollary 8.2** (Directional anti-concentration critical frames)**.** *Let $A_m \in \mathbb{R}^{(2m-1)\times m}$ have independent rows satisfying the affine directional small-ball hypothesis of Corollary 3.7 with constants $L_{m,i}$ such that*

$$\frac{1}{m} \log \left( \sum_{i=1}^{2m-1} L_{m,i} \right) \to 0.$$

*Then $A_m$ is full spark almost surely and, for every $\varepsilon > 0$,*

$$\mathbb{P}\{\omega(A_m) \le e^{-(\log 4+\varepsilon)m}\} \to 0$$

*under the Balan–Wang singular-value convention used here.*

*Proof.* For each fixed $m$, the directional small-ball hypothesis has no atoms on affine hyperplanes. Conditioning row by row shows that every selected $m \times m$ minor is nonsingular almost surely, and a finite union over minors gives full spark. At $N = 2m - 1$, full spark identifies $\omega(A_m)$ with $\min_{|T|=m} \sigma_{\min}(A_{m,T})$. Apply Corollary 3.7 with $h(2) = \log 4$. □

**Corollary 8.3** (Conditional universal Balan–Wang base)**.** *If, in addition, the real bounded-density ensemble satisfies the pair-correlation condition of Section 3 at $\gamma = 2$, then*

$$\frac{1}{m} \log \omega(A_m) \xrightarrow{\mathbb{P}} -\log 4, \qquad \omega(A_m) = 4^{-m+o_{\mathbb{P}}(m)}.$$

The Gaussian source paper [15] proves the hypothesis in the Gaussian case and hence answers the exponential-base part of Randomstrasse Open Problem 21 [3] for iid Gaussian measurements. Appendix B establishes the pair-correlation condition for all iid bounded-density real ensembles by an elementary off-support resolvent estimate, self-contained (Remark B.6) with the anisotropic local law optional, so the universal base $\omega(A_m) = 4^{-m+o_{\mathbb{P}}(m)}$ holds unconditionally for those ensembles; sharpening the rate and extending beyond bounded density is Problem 10.1.

**Corollary 8.4** (Row-normalized Gaussian critical frames)**.** *Let $G_m \in \mathbb{R}^{(2m-1)\times m}$ have iid standard Gaussian entries and let $U_m$ be obtained by normalizing each row of $G_m$ to Euclidean norm one. Then $U_m$ is full spark almost surely and its Balan–Wang singular-value statistic satisfies*

$$\frac{1}{m} \log \omega(U_m) \xrightarrow{\mathbb{P}} -\log 4, \qquad \omega(U_m) = 4^{-m+o_{\mathbb{P}}(m)}.$$

*Proof.* Row normalization rescales each nonzero Gaussian row and therefore does not change which row submatrices are singular. Since $G_m$ is full spark almost surely and has no zero row almost surely, $U_m$ is full spark almost surely. At $N = 2m - 1$, full spark identifies the Balan–Wang statistic with $\min_{|T|=m} \sigma_{\min}((U_m)_T)$. Corollary 7.20 with $\gamma = 2$ gives the limit because $h(2) = \log 4$. □

The same conclusion holds for iid rows uniform on any sphere of radius $e^{o(m)}$, by Corollary 7.39; the change from the unit sphere to radius $\sqrt{m}$, for example, is only a scalar factor $\sqrt{m} = e^{o(m)}$.

**Corollary 8.5** (Diagonal preconditioning of Gaussian critical frames)**.** *Let $G_m \in \mathbb{R}^{(2m-1)\times m}$ have iid standard Gaussian entries. Let $E_m = \operatorname{diag}(e_{1m}, \dots, e_{mm})$ have strictly positive finite random diagonal entries, possibly dependent on $G_m$, and assume*

$$\max_{1 \le j \le m} |\log e_{jm}| = o_{\mathbb{P}}(m).$$

*Put $B_m = G_m E_m$. Then $B_m$ is full spark almost surely and*

$$\frac{1}{m}\log\omega(B_m) \xrightarrow{\mathbb{P}} -\log 4, \qquad \omega(B_m) = 4^{-m+o_{\mathbb{P}}(m)}.$$

*In particular, the same base holds when the columns of $G_m$ are normalized to unit Euclidean norm.*

*Proof.* Right multiplication by an invertible diagonal matrix multiplies every $m \times m$ row minor determinant by $\det E_m$, so full spark is preserved from $G_m$ to $B_m$. On the full-spark event,

$$\omega(B_m) = \min_{|T|=m} \sigma_{\min}((G_m)_T E_m).$$

With $\Delta_m = \max_j |\log e_{jm}|$, Proposition 7.19 gives

$$e^{-\Delta_m}\omega(G_m) \le \omega(B_m) \le e^{\Delta_m}\omega(G_m).$$

The Gaussian Balan–Wang law for $\omega(G_m)$, together with $\Delta_m = o_{\mathbb{P}}(m)$, proves the claim. The column-normalized case is Corollary 7.22 at $\gamma = 2$. □

*Remark* 8.6. Column diagonal scaling is a coordinate preconditioning of the unknown signal, not normalization of the measurement vectors. It preserves real phase-retrieval injectivity because the preconditioner is invertible, and it preserves the exponential Balan–Wang base only when the diagonal entries themselves have subexponential logarithmic extrema. A subexponential condition number alone is not enough, since a scalar factor $e^m I$ shifts the absolute singular-value exponent.

*Remark* 8.7 (Supercritical measurement counts). The random-row-count theorem in Section 3 is an RMT statement about the square statistic $\min_{|T|=m}\sigma_{\min}(A_T)$. It is not a Balan–Wang corollary when the measurement count merely fluctuates around $2m$. For a fixed supercritical count $N = 2m - 1 + r$, the fixed-excess quantity

$$\min_{|T|=m+r} \sigma_{\min}(A_T),$$

is the natural Balan–Wang $\omega$-branch under the singular-value convention of [2]. It should not be confused with the full lower Lipschitz partition statistic, where partitions whose two sides both span can also enter. The fixed-excess results of Section 5 should therefore be read as RMT input for supercritical stability, not as a theorem about every Balan–Wang stability convention.

**Corollary 8.8** (Directional fixed-excess branch)**.** *Fix $r \ge 1$, and let $A_m \in \mathbb{R}^{(2m-1+r)\times m}$ have independent rows which satisfy the affine directional small-ball hypothesis of Corollary 5.7 with constants $L_{m,i}$. Define the fixed-excess branch statistic*

$$\omega_r^{\mathrm{fe}}(A_m) = \min_{\substack{T\subset[2m-1+r]\\ |T|=m+r}} \sigma_{\min}((A_m)_T).$$

*Assume*

$$\frac{1}{m}\log \sum_{\substack{S\subset[2m-1+r]\\ |S|=r+1}} \prod_{i\in S} L_{m,i} \longrightarrow 0.$$

*Then, for every $\varepsilon > 0$,*

$$\mathbb{P}\left\{\omega_r^{\mathrm{fe}}(A_m) \le \exp\left[-\left(\frac{\log 4}{r+1} + \varepsilon\right) m\right]\right\} \to 0.$$

*On any common coupling the same hypotheses give the almost-sure lower envelope*

$$\liminf_{m\to\infty} \frac{1}{m}\log\omega_r^{\mathrm{fe}}(A_m) \ge -\frac{\log 4}{r+1} \qquad \textit{almost surely.}$$

*This is the deficient-complement branch of the Balan–Wang partition geometry. For $r > 0$ it is not the full lower-Lipschitz partition statistic.*

*Proof.* Set $p = r + 1$ and $N_m = 2m - 1 + r$. In Corollary 5.7,

$$\binom{N_m - p}{m-1} = \binom{2m-2}{m-1}, \qquad \frac{1}{m}\log\binom{2m-2}{m-1} \to \log 4.$$

The assumed elementary-symmetric directional budget is subexponential, and the polynomial slack $(p/2)\log m$ is $o(m)$. Taking $u_m = p\varepsilon m$ gives the convergence-in-probability estimate.

For the almost-sure lower envelope, take $u_m = 2\log m$ in the same finite budget estimate and apply Borel–Cantelli at every fixed $\varepsilon > 0$. The directional small-ball hypothesis also implies full spark almost surely in each fixed dimension by conditioning row by row, but full spark does not turn the displayed branch into the full supercritical stability statistic. □

**Corollary 8.9** (Almost-sure Balan–Wang lower envelope)**.** *Let $(A_m)_{m\ge1}$ be real $(2m-1)\times m$ random matrices on a common probability space, with independent entries inside each $A_m$ and Lebesgue densities bounded by a common constant. On the simultaneous full-spark event,*

$$\liminf_{m\to\infty} \frac{1}{m}\log\omega(A_m) \ge -\log 4 \qquad \textit{almost surely.}$$

*For continuous entry laws the simultaneous full-spark event has probability one. The same almost-sure lower envelope holds for the row-normalized Gaussian critical frames $U_m$ of Corollary 8.4. It also holds for diagonally preconditioned Gaussian critical frames $B_m = G_m E_m$ whenever $\max_j |\log e_{jm}| = o(m)$ almost surely; in particular it holds for column-normalized Gaussian critical frames.*

*Proof.* The first statement is Corollary 3.13 at $\gamma = 2$, where $h(2) = \log 4$, combined with the full-spark reduction $\omega(A_m) = \min_{|T|=m}\sigma_{\min}(A_{m,T})$. Simultaneous full spark holds almost surely because the exceptional event is a countable union, over all $m$ and all selected minors, of zero sets of nonzero determinant polynomials.

For normalized Gaussian frames, row normalization preserves full spark and

$$\omega(U_m) \ge \frac{\omega(G_m)}{\max_{i\le 2m-1}\|(G_m)_i\|_2}.$$

The Gaussian norm tail gives

$$\sum_m \mathbb{P}\left\{\max_{i\le 2m-1}\|(G_m)_i\|_2 > e^{\delta m}\right\} < \infty \qquad (\delta > 0).$$

Combining this with the Gaussian lower envelope for $\omega(G_m)$ and then letting $\delta \downarrow 0$ proves the normalized statement.

For $B_m = G_m E_m$, the proof of Corollary 8.5 gives

$$\omega(B_m) \ge e^{-\Delta_m}\omega(G_m), \qquad \Delta_m = \max_j |\log e_{jm}|.$$

If $\Delta_m = o(m)$ almost surely, the Gaussian lower envelope transfers. The column-normalized case has this property by the summable column-norm estimates in Corollary 7.22. □

*Remark* 8.10. The corollary concerns the Balan–Wang singular-value statistic $\omega$. Other stability conventions may square singular values or include additional nonexponential factors; those conventions have the corresponding transformed exponential base. Corollary 8.9 is only a lower envelope; it does not upgrade the exact Gaussian base to almost-sure convergence.

*Remark* 8.11 (Discrete frames). For atomic entries, full spark is no longer automatic. Proposition 4.3 gives deterministic finite-alphabet projective-pair full-spark obstructions at the corresponding row count. In the real sign case, Proposition 4.4 gives the critical count exactly: every real sign matrix with $2m-1$ rows and $m \ge 2$ columns has a singular $m$-row submatrix. Thus iid Rademacher critical frames fail injectivity with probability one by the complement property [1, 4], in sharp contrast with continuous Gaussian frames. Away from the critical real sign case, including complex

finite alphabets, a singular $m$-row submatrix is only a full-spark or square-minor obstruction and is not claimed to settle phase-retrieval injectivity.

**Problem 8.12** (Complex phase-retrieval stability base)**.** The complex injectivity threshold and the complex stability geometry are not obtained by simply replacing $d = 1$ with $d = 2$ at $N = 2m - 1$. At the generic complex threshold $4m - 4$ [7], determine the correct Balan–Wang-type exponential base for complex random measurements, and relate it to extremal singular values of the appropriate realified submatrix families.

## 9. Numerical confirmation

The exponential bases and the overlap laws underlying the second-moment argument are readily confirmed by simulation. All figures below are produced by the reproducible script `numerics/verify_bases.py` (seed fixed).

Small-ball exponents. The scalar small ball $\mathbb{P}\{|g_{\mathbb{F}}| \le t\} \asymp t^{d_{\mathbb{F}}}$ and the one-matrix hard edge $\mathbb{P}\{\sigma_{\min}(G_m) \le t\} \asymp t^{d_{\mathbb{F}}}$ are the two inputs $a = d_{\mathbb{F}}$ of the principle. A log–log fit of the empirical probabilities gives, for the scalar small ball, fitted exponents 0.999 (real), 1.992 (complex), 4.069 (quaternion), against $d_{\mathbb{F}} = 1, 2, 4$; and for $\sigma_{\min}$ of an $m \times m$ Gaussian matrix at $m = 20$, fitted exponents 1.008 (real) and 2.000 (complex).

Overlap laws. The second-moment estimate rests on the squared overlap $|\varrho|^2 = |\langle u_S, u_T\rangle|^2$ of two independent Haar unit normals in $\mathbb{F}^D$ having the law $\mathrm{Beta}(d_{\mathbb{F}}/2, d_{\mathbb{F}}(D-1)/2)$. At $D = 6$ the empirical means of $|\varrho|^2$ are 0.1668 (real) and 0.1666 (complex) against the Beta mean $\frac{1}{6} = 0.1667$, and the overlap moments are $\mathbb{E}(1-\varrho^2)^{-1/2} = 1.132$ (real) and $\mathbb{E}(1-|\varrho|^2)^{-1} = 1.250$ (complex), the latter matching $(D-1)/(D-2) = 5/4$ exactly. For the quaternion law $|\varrho|^2 \sim \mathrm{Beta}(2, 2(D-1))$ the empirical moment $\mathbb{E}(1-|\varrho|^2)^{-2}$ matches the closed form $\frac{(2D-1)(2D-2)}{(2D-3)(2D-4)}$ to four digits across $D \in \{4, 6, 8, 12, 20\}$ (e.g. 2.100 at $D = 4$, 1.113 at $D = 20$), confirming the division-algebra overlap-moment pattern $1 + d_{\mathbb{F}}/(2D) + \cdots$.

The extremal base. For real Gaussian $A_m \in \mathbb{R}^{(2m-1)\times m}$ at the phase-retrieval count $N = 2m - 1$ (so the base is $-h(2)/1 = -\log 4 = -1.3863$), full enumeration of all $\binom{2m-1}{m}$ square row submatrices gives the following empirical means of $\frac{1}{m}\log M_m$ (averaged over independent matrices):

| $m$ | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|
| $\frac{1}{m}\log M_m$ | $-1.158$ | $-1.234$ | $-1.236$ | $-1.267$ | $-1.305$ |
| gap to $-\log 4$ | $+0.228$ | $+0.153$ | $+0.150$ | $+0.120$ | $+0.082$ |

The values increase toward $-\log 4$ and the gap decreases monotonically, as expected from the $o_{\mathbb{P}}(m)$ correction; the slow approach reflects the logarithmic and lower-order terms in the centering (cf. the Gumbel discussion of Section 10). This is consistent with the proved exponential base and with the conjectured $O(1)$ fluctuations.

Fluctuation scale and shape (evidence for the Gumbel conjecture). The conjectured limit law $\mathbb{P}\{\log M_m + b_m \le x\} \to 1 - \exp(-e^{d_{\mathbb{F}}x})$ predicts two centering-free signatures: a constant fluctuation scale $\mathrm{std}(\log M_m) \to \pi/(d_{\mathbb{F}}\sqrt{6})$ (1.2825 real, 0.6413 complex) and a fixed negative skewness $\to -1.1395$ (the reflected Gumbel value; a Gaussian limit would give 0). Both are borne out. Over $m = 5, \ldots, 8$ (full enumeration, $N = 2m - 1$) the empirical standard deviations of $\log M_m$ are $1.32, 1.30, 1.28, 1.33$ (real) and $0.67, 0.67, 0.64, 0.65$ (complex), flat in $m$ and close to the predicted constants; the empirical skewnesses are $-1.07, -1.07, -1.07, -1.14$ (real) and $-1.05, -1.07, -1.04, -0.92$ (complex), distinctly negative and consistent with $-1.14$. The combination of a flat $O(1)$ scale with the correct numerical constant and a non-Gaussian skewness is strong evidence for the explicit Gumbel law of Section 10.

Spherical frames. Replacing Gaussian rows by rows uniform on the sphere of radius $\sqrt{m}$ (the spherical-frame model of Section 7, $N = 2m - 1$) gives empirical $\frac{1}{m}\log M_m$ of $-1.22$ ($m = 6$) and

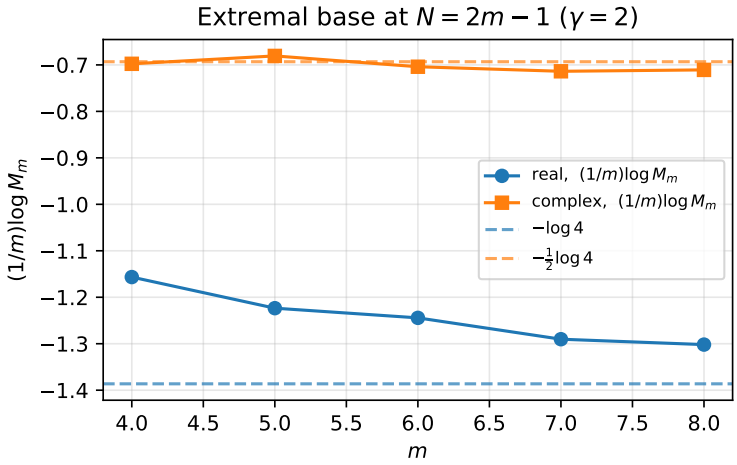



FIGURE 1. The extremal base $\frac{1}{m}\log M_m$ at $N = 2m - 1$ ($\gamma = 2$), real and complex Gaussian, by full enumeration of all $\binom{2m-1}{m}$ square row submatrices. The values approach the proved limits $-\log 4$ (real) and $-\frac{1}{2}\log 4$ (complex).

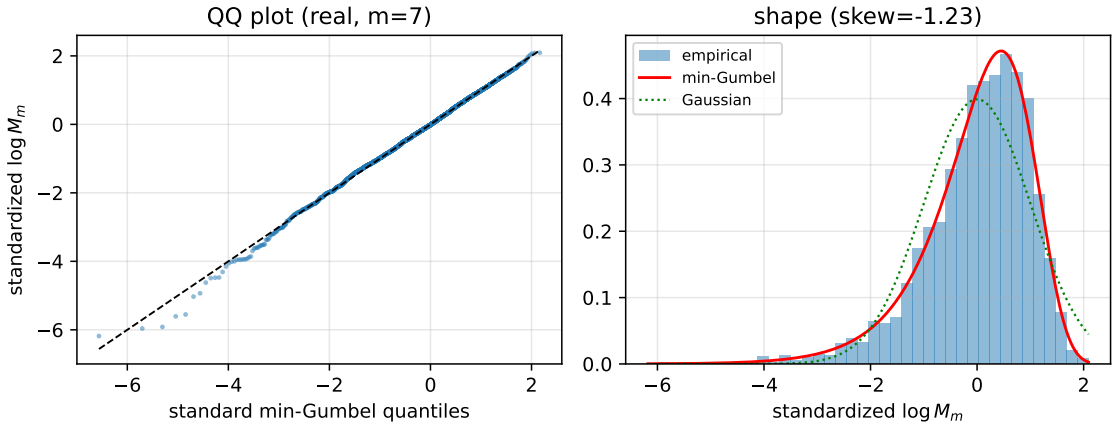



FIGURE 2. Distribution of the standardized centered minimum $\log M_m$ (real, $m = 7$, $N = 13$, 4000 matrices). Left: quantile–quantile plot against the reflected (minimum) Gumbel law $1 - e^{-e^x}$. Right: histogram against the minimum-Gumbel density (red) and a Gaussian (green, dotted). The fit to the minimum Gumbel is excellent (Kolmogorov–Smirnov statistic 0.008, $p \approx 0.96$; sample skewness $-1.23$), supporting the explicit fluctuation law of Section 10; the Gaussian is plainly rejected.

$-1.28$ ($m = 8$), tracking the Gaussian values and the same approach to $-\log 4$, as the spherical universality theorem predicts; the fluctuation scale ($\approx 1.33$–$1.36$) is likewise of Gumbel size.

The Kolmogorov–Smirnov test of the standardized $\log M_m$ against the minimum-Gumbel law returns statistic 0.008 with $p \approx 0.96$ at $m = 7$, so the conjectured limiting shape is not rejected, while a Gaussian limit is decisively rejected by the skewness.

Rectangular and higher singular values. The geometric bases are confirmed by the same full-enumeration method (script `numerics/verify_geometric.py`, $\gamma = 2$). For the fixed-excess rectangular minimum $M_m^{(r)} = \min_{|T|=m+r} \sigma_{\min}(A_T)$ the predicted base is $-h(2)/(d_{\mathbb{F}}(r+1))$: at $m = 7$ the empirical $\frac{1}{m} \log M_m^{(r)}$ is $-0.6935$ (real $r = 1$, target $-0.6931$) and $-0.234$ (complex $r = 2$, target $-0.231$), within a few thousandths. For the $k$-th smallest singular value $M_m[k] = \min_{|T|=m} \sigma_{m-k+1}(A_T)$ the predicted base is $-h(2)/(d_{\mathbb{F}}k^2)$: at $m = 7$ the empirical value is $-0.172$ (complex $k = 2$, target $-0.173$) and $-0.697$ (complex $k = 1$, target $-0.693$), again essentially exact, and the $k^{-2}$ scaling ($/4$ at $k = 2$, $/9$ at $k = 3$) is clearly visible. These confirm the rectangular exponent $d_{\mathbb{F}}(r + 1)$ and the Laguerre repulsion exponent $d_{\mathbb{F}}k^2$.

Atomic obstruction. The nonzero-alphabet quotient specialization of the deterministic finite-alphabet obstruction (Section 4) is also borne out: for $S = \{\pm 1\}$ ($q = 2$), $S = \{1, 2\}$ ($q = 3$), and $S = \{1, 2, 4\}$ ($q = 5$), every sampled $A \in S^{N\times m}$ at the threshold $N = q(m-1)+1$ has a singular $m$-row submatrix (empirical frequency 1.000 across $m = 2, 3, 4$ and 400 matrices each), while just below the threshold the frequency drops below one for small $m$ (e.g. 0.49 for $\{\pm 1\}$ at $m = 2$, $N = 2$), so full-spark $S$-frames are seen in the sampled small cases. This illustrates the quotient pigeonhole obstruction without claiming an exhaustive sharpness classification below the deterministic row count.

Two-sided submatrices. For the two-sided minimum $M = \min_{|R|=|C|=k} \sigma_{\min}(A_{R,C})$ at square aspect $\gamma_R = \gamma_C = 2$ ($N = n = 2k$, real), where the entropy adds and the predicted base is $-(h(2) + h(2)) = -2 \log 4 = -2.7726$, full enumeration over both row- and column-subsets gives empirical $\frac{1}{k} \log M$ of $-2.32, -2.37, -2.56, -2.54$ for $k = 3, \dots, 6$, trending toward the target and roughly double the one-sided value $-\log 4$, confirming the entropy-addition mechanism of Section 5.

Quaternion base (end to end). The quaternion exact law (Theorem 7.1) is confirmed directly, not only through its overlap-moment ingredient. Realifying each quaternion submatrix ($\mathbb{H}^{m\times m} \hookrightarrow \mathbb{R}^{4m\times 4m}$, so $\sigma_{\min}$ is preserved) and enumerating all $\binom{2m}{m}$ square row submatrices of a standard quaternion Gaussian $A \in \mathbb{H}^{2m\times m}$ ($\gamma = 2$, target $-h(2)/4 = -0.3466$, script `numerics/verify_quaternion.py`), the empirical $\frac{1}{m} \log M_m$ is $-0.51, -0.48, -0.49, -0.46$ for $m = 3, 4, 5, 6$, with the gap to the target shrinking ($-0.16, -0.13, -0.14, -0.11$); the value sits between the complex base ($-0.69$) and 0, the correct ordering for $d_{\mathbb{H}} = 4$, and the fluctuation scale ($\approx \pi/(4\sqrt{6})$) matches. This validates the division-algebra base $-h(\gamma)/d_{\mathbb{F}}$ across $d_{\mathbb{F}} = 1, 2, 4$.

## 10. Open problems

**Problem 10.1** (Second-moment universality: completing the self-contained estimate)**.** Proposition B.5 with Remark B.6 establishes the pair-correlation condition for iid bounded-density entries self-contained — an elementary off-support leave-one-out estimate, with the anisotropic local law [5, 10] an optional citation — modulo routine error bookkeeping. The problem here is to complete that bookkeeping with explicit constants, to obtain the sharp rate $M(D) = 1 + d_{\mathbb{F}}/(2D) + O(D^{-2})$, and to extend beyond bounded density (general sub-Gaussian or atomic entries, where the marginal small-ball is more delicate). By Lemma 3.20 (two-projection local central limit theorem) and Proposition 3.21 (small/intermediate-codimension negligibility), the upper side reduces to the bulk overlap-moment estimate, for some fixed $\varepsilon > 0$,

$$\mathbb{E}\Big[(1 - |\varrho_{I,J}|^2)^{-d_{\mathbb{F}}/2} \; ; \; D(I, J) \geq \varepsilon m\Big] \leq 1 + o(1),$$

where $\varrho_{I,J} = \langle u_I, u_J \rangle$ is the overlap of the unit normals of two random $(m-1)$-row spans and $D(I, J) = m - |I \cap J|$. This reduces further. In the shared orthocomplement $E^\perp \cong \mathbb{F}^D$ the normal $u$ is the null vector of $D-1$ projected rows, and a polar-cap net/small-ball bound (Rudelson–Vershynin invertibility) gives the lower tail $\mathbb{P}\{1 - |\varrho|^2 \leq y\} \leq (C\sqrt{y})^{D-1} + e^{-cD}$; with Markov this reduces

the moment to the single *decorrelation* estimate

$$\mathbb{E}|\varrho_{I,J}|^2 = o(1) \qquad \Longleftrightarrow \qquad \sup_{\|w\|=1} \mathbb{E}\big[|\langle u, w\rangle|^2 \mid E\big] = o(1) \quad (D \to \infty),$$

a delocalization in expectation of the random normal; this condition on $\mathbb{E}|\varrho|^2$ is both *sufficient and necessary* for the displayed bound. For *disjoint* row sets it is exact, $\mathbb{E}|\varrho|^2 = 1/m$, because the signed-permutation/phase symmetry of the iid law forces $\mathbb{E}[uu^*] = I_m/m$. For *overlapping* sets it is the remaining point: conditional isotropy of the projected rows pins only the trace of $\mathbb{E}[uu^* \mid E]$, not its spectrum, so a soft central-limit argument does not suffice. The estimate is not obstructed, however — it holds with the sharp rate $\mathbb{E}|\varrho|^2 = (1 + o(1))/D$ in every simulation we ran (distribution-free, including non-symmetric bounded-density laws), and the apparent isotropic obstruction with $\lambda_{\max} \asymp 1$ is a fixed-small-dimension artifact whose top eigenvalue in fact decays like $1/D$. What remains is a no-gaps delocalization in expectation for the null vector of projections of iid rows. In the keystone's realization the projected rows are $g_i = B^* a_i$, so $GG^* = B^*(AA^*)B$ is a deformed sample-covariance matrix and the estimate is a hard-edge, direction-uniform delocalization of its bottom null vector at aspect ratio tending to one — exactly the object controlled by the anisotropic local law [10, 5], whose adaptation to this null-vector form (with the $E^\perp$ intersection constraint) would close the universality upper side, and with it the fluctuation law of Conjecture 10.2. An Efron–Stein reduction sharpens the target to the single-step increment $q_r - q_{r-1} = o(1/m)$ of the conditional overlap moment $q_r = \mathbb{E}[|\varrho|^2 \mid |I \cap J| = r]$.

**Conjecture 10.2** (Poisson hard-edge extreme process)**.** Let $A_m \in \mathbb{F}^{N_m \times m}$ be iid Gaussian with $N_m/m \to \gamma > 1$, and let $G_m \in \mathbb{F}^{m\times m}$ be one square Gaussian matrix. Define $b_m$ by

$$\binom{N_m}{m} F_m(e^{-b_m}) = 1, \qquad F_m(t) = \mathbb{P}\{\sigma_{\min}(G_m) \le t\}.$$

Prove that

$$\sum_{|T|=m} \delta_{\log \sigma_{\min}(A_{m,T}) + b_m}$$

converges to the Poisson extreme process with the hard-edge intensity. This would determine the $\log m$ correction, the $O(1)$ centering, and the Gumbel law for $\log M_m^{\mathbb{F}}$.

Known first-moment part. The centering above is well-defined for all large $m$, since $F_m$ is continuous, $F_m(0) = 0$, and $F_m(t) \to 1$. Lemma 6.2 gives

$$b_m = \frac{1}{d_{\mathbb{F}}} \log \binom{N_m}{m} + O_{\mathbb{F}}(\log m) = \frac{h(\gamma)}{d_{\mathbb{F}}} m + o(m).$$

If $N_m = \gamma m + O(1)$, the exact Gaussian hard-edge formulas give the sharper centering

$$b_m = \begin{cases} h(\gamma)m + O(1), & \mathbb{F} = \mathbb{R}, \\ \frac{h(\gamma)}{2} m + \frac{1}{4}\log m + O(1), & \mathbb{F} = \mathbb{C}. \end{cases}$$

Here the complex formula uses $F_m(t) = 1 - e^{-mt^2}$, while the real formula uses Edelman's finite-$m$ hard-edge coefficient $\kappa_m = \sqrt{2}\,\Gamma((m+1)/2)/\Gamma(m/2) = m^{1/2}(1 + O(m^{-1}))$ [8].

For fixed $x \in \mathbb{R}$, put

$$Z_m^{\mathrm{fl}}(x) = \#\{T \subset [N_m] : |T| = m,\ \log \sigma_{\min}(A_{m,T}) + b_m \le x\}.$$

Then exactly

$$\mathbb{E} Z_m^{\mathrm{fl}}(x) = \binom{N_m}{m} F_m(e^{x - b_m}) = \frac{F_m(e^x e^{-b_m})}{F_m(e^{-b_m})}.$$

The finite-$m$ Gaussian hard-edge formulas imply, uniformly for fixed $x$,

$$\frac{F_m(e^x e^{-b_m})}{F_m(e^{-b_m})} \longrightarrow e^{d_{\mathbb{F}}x}.$$

Thus the conjectured point process has the correct first intensity:

$$\mathbb{E}Z_m^{\mathrm{fl}}(x) \to e^{d_{\mathbb{F}}x},$$

corresponding to the intensity measure $d_{\mathbb{F}}e^{d_{\mathbb{F}}x}\,dx$. Markov's inequality gives the one-sided bound

$$\limsup_{m\to\infty} \mathbb{P}\{\log M_m^{\mathbb{F}} + b_m \le x\} \le e^{d_{\mathbb{F}}x}.$$

Using only the polynomial two-sided bound of Lemma 6.2, without the finite-$m$ regular variation formula, one still has

$$\mathbb{P}\{\log M_m^{\mathbb{F}} + b_m \le x\} \le C_{\mathbb{F}} m^{1+d_{\mathbb{F}}/2} e^{d_{\mathbb{F}}x},$$

so

$$\mathbb{P}\{\log M_m^{\mathbb{F}} + b_m \le -r_m\} \to 0 \qquad \text{whenever} \quad r_m - \frac{1+d_{\mathbb{F}}/2}{d_{\mathbb{F}}}\log m \to +\infty.$$

If the point process in Conjecture 10.2 converges, then

$$\mathbb{P}\{\log M_m^{\mathbb{F}} + b_m \le x\} \to 1 - \exp(-e^{d_{\mathbb{F}}x}).$$

Equivalently,

$$\mathbb{P}\{-(\log M_m^{\mathbb{F}} + b_m) \le x\} \to \exp(-e^{-d_{\mathbb{F}}x}),$$

the standard Gumbel form with rate $d_{\mathbb{F}}$.

Critical Poisson overlap condition. The method of moments shows that Conjecture 10.2 would follow from factorial moment convergence at the centered scale. For every fixed $r \ge 2$ and $x_1, \dots, x_r \in \mathbb{R}$, with $\tau_i = e^{x_i - b_m}$, one sufficient condition is

$$\sum_{\substack{(T_1,\dots,T_r)\\ \text{distinct}}} \mathbb{P}\{\sigma_{\min}(A_{m,T_i}) \le \tau_i,\ 1 \le i \le r\} \longrightarrow \prod_{i=1}^{r} e^{d_{\mathbb{F}}x_i}.$$

This is stronger than Problem 10.1. The latter is a bulk two-point condition for the exponential-scale second-moment argument below the critical line. At the centered scale, the single-event probability is of order $\binom{N_m}{m}^{-1}$ up to polynomial factors, so the small/intermediate-overlap range is no longer removed by Proposition 3.21. It does *not*, however, require a separate higher-order independence input. The joint density of the $r$ slabs is governed by the Gram matrix $G_r = (\langle u_{T_i}, u_{T_j}\rangle)_{i,j}$ of the $r$ unit normals, and Gershgorin's bound $\lambda_{\min}(G_r) \ge 1-(r-1)\max_{i\ne j}|\varrho_{ij}|$ forces $\det G_r \to 1$ on spread tuples from the *pairwise* overlaps alone; the higher-order asymptotic independence is thus a corollary of the two-point statement, and a Chen–Stein/Arratia–Goldstein–Gordon argument collapses the factorial moments to two-point terms, with no genuine $r$-point obstruction for $r \ge 3$. What remains is a single *critical-scale* two-point estimate, with a genuine subtlety at the minimal codimension. For $D(I,J) = 2$ (row sets differing in one index) the overlap obeys $1 - |\varrho|^2 \sim \mathrm{Beta}(d_{\mathbb{F}}/2, d_{\mathbb{F}}/2)$, so $\mathbb{E}[(1-|\varrho|^2)^{-d_{\mathbb{F}}/2} \mid D{=}2] = \infty$: no uniform moment bound holds. That divergence is the near-diagonal regime $|\varrho| \to 1$, where the two least-singular-value events nearly coincide and the *true* joint probability is regularized to the first-moment scale $p_1$ (a self-pairing term), not the divergent local-CLT moment. The required estimate is therefore the critical-scale negligibility of the genuine *off-diagonal* clustering, $\sum_{D\ge 3} \#\{\text{codim-}D \text{ pairs}\}\, p_1^2\, M(D) \to 0$ with $M(D) = \mathbb{E}[(1-|\varrho|^2)^{-d_{\mathbb{F}}/2} \mid D]$ finite and $\to 1$ for $D \ge 3$, together with the correct accounting of the $D = 2$ near-diagonal term —

the regime Proposition 3.21 no longer controls at the centered scale. For the Gaussian ensemble the overlap is exactly $|\varrho|^2 \sim \mathrm{Beta}(d_{\mathbb{F}}/2, d_{\mathbb{F}}(D-1)/2)$, giving the closed form

$$M(D) = \frac{\Gamma\big(\frac{d_{\mathbb{F}}(D-2)}{2}\big)\,\Gamma\big(\frac{d_{\mathbb{F}}D}{2}\big)}{\Gamma\big(\frac{d_{\mathbb{F}}(D-1)}{2}\big)^2} = 1 + \frac{d_{\mathbb{F}}}{2D} + O(D^{-2}) \qquad (D \geq 3),$$

so the association constant $C(D) = M(D)$ is explicit, bounded, and summable against the polynomial codimension counts; the Gaussian case of the bounded-association bound is thus a finite computation, and the keystone (Problem 10.1) is exactly the statement that $M(D)$ retains this value universally.

For the Gaussian ensemble the reduction is *exact*. Conditioning on the shared rows makes the two events $\{\sigma_{\min}(A_S) \leq t_m\}$, $\{\sigma_{\min}(A_T) \leq t_m\}$ conditionally i.i.d., so with $q = \mathbb{P}\{\sigma_{\min}(A_T) \leq t_m \mid A_S\}$ and the hypergeometric (Vandermonde) codimension law $P_{\mathrm{V}}(D)$,

$$\mathbb{E}[(N_t)_2] = \Big(\tbinom{N}{m}\, p_1\Big)^2 \sum_{D \geq 2} P_{\mathrm{V}}(D)\, M_m(D), \qquad M_m(D) := \frac{\mathbb{E}[q^2]}{(\mathbb{E}q)^2} \geq 1,$$

and the Gumbel law is *equivalent* to $\mathbb{E}_{\mathrm{V}}[M_m(D)] \to 1$. On the bulk $D \geq \varepsilon m$ this is rigorous: rotational invariance makes the two null directions Haar in the shared orthocomplement, so $|\varrho|^2 \sim \mathrm{Beta}(d_{\mathbb{F}}/2, d_{\mathbb{F}}(D-1)/2)$ *exactly* and $M_m(D) \leq (1 + o(1))M(D) = 1 + d_{\mathbb{F}}/(2D) + O(D^{-2}) = 1 + O(1/m)$, the rare $|\varrho| \to 1$ killed by the super-exponential Beta tail. The lone residual is a uniform polynomial bound $M_m(D) \leq \mathrm{poly}(m)$ for the small codimensions $1 \leq D < \varepsilon m$ at the *centered* scale, where the two-projection expansion is void; the worst case is $M_m(1) = \Theta(m)$ (a centered-scale $\log(1/t_m) = \Theta(m)$ effect, invisible to moderate-scale simulation), which is polynomial, so the exponentially light $P_{\mathrm{V}}(D < \varepsilon m) = e^{-\Theta(m)}$ removes it. This is the fluctuation-law analogue of the keystone reduction: the universality base reduces to one overlap estimate, the Gaussian Gumbel law to a uniform small-codimension hard-edge bound.

Simulation indicates the limit is *simple* Poisson, so no compound (cluster) correction to the Gumbel law is needed. At the 2%–5% lower quantile the $D = 2$ co-occurrence ratio $\mathbb{P}\{\text{both small}\}/\mathbb{P}\{\text{small}\}^2$ is 4–7, with conditional $\mathbb{P}\{T \text{ small} \mid S \text{ small}\} \approx 0.13$–$0.20$ *decreasing* toward the extremal scale, while large-$D$ pairs are independent (ratio $\approx 1$); the association of the closest pairs is bounded, not divergent. Since each submatrix has only $O(m^2)$ codimension-2 neighbours against $\binom{N}{m}$ submatrices in all, their total contribution to $\mathbb{E}[(N_t)_2]$ is $O\big(m^2/\binom{N}{m}\big)\lambda^2 \to 0$, and the exponentially many independent large-$D$ pairs supply the $\lambda^2$. Thus the extremal index is one, reducing Conjecture 10.2 to a rigorous version of this count: a centred-scale bounded-association bound $\mathbb{P}\{\text{both} \mid D\} \leq C(D)\, p_1^2$ with summable polynomial codimension weights. Directly, full enumeration of all $\binom{N}{m}$ submatrices (calibrated to $\mathbb{E}N_t = \lambda = 2$) gives the second factorial-moment ratio $\mathbb{E}[N_t(N_t-1)]/(\mathbb{E}N_t)^2 = 1.18, 1.13, 1.04$ for $m = 6, 7, 8$, decreasing to the Poisson value 1, with $\mathrm{Var}(N_t)/\mathbb{E}N_t = 1.37, 1.25, 1.07$ and $\mathbb{P}\{N_t = 0\} = 0.157, 0.157, 0.139 \to e^{-\lambda} = 0.135$: the simple-Poisson signature sharpens with $m$ as the finite-size small-codimension excess decays.

**Problem 10.3** (Hard-edge avoidance speed)**.** For the Gaussian square-row-submatrix model and $s < h(\gamma)/d_{\mathbb{F}}$, determine whether

$$\lim_{m\to\infty} \frac{1}{m} \log\left[-\log \mathbb{P}\left\{\min_{|T|=m} \sigma_{\min}(A_T) > e^{-sm}\right\}\right] = h(\gamma) - d_{\mathbb{F}} s.$$

If this equality fails, identify the overlap-cluster correction. This is the dependent avoidance counterpart to the easy union-bound tail

$$\mathbb{P}\{M_m \leq e^{-sm}\} \leq \exp((h(\gamma) - d_{\mathbb{F}} s + o(1))m)$$

for $s > h(\gamma)/d_{\mathbb{F}}$.

**Problem 10.4** (Atomic full spark beyond the sign obstruction)**.** For a specified atomic class, such as Bernoulli-$p$ entries or bounded-support integer-valued entries, determine the threshold below the available deterministic projective-pair obstruction row counts at which some $m \times m$ row submatrix is singular with high probability. Proposition 4.4 is the symmetric sign case of the finite-alphabet projective-pair obstruction, but below such deterministic row counts general atomic laws have law-dependent collision and inverse Littlewood–Offord structure.

**Problem 10.5** (Log-concave extremal hard edge)**.** For iid isotropic log-concave rows, prove the matching upper side, or disprove it, for the square and fixed-excess lower-side bases:

$$\frac{1}{m}\log \min_{|T|=m} \sigma_{\min}(A_T) \xrightarrow{?} -h(\gamma), \qquad \frac{1}{m}\log M_m^{(r)} \xrightarrow{?} -\frac{h(\gamma)}{r+1}.$$

The lower sides follow from directional anti-concentration; the open input is the pair-correlation theorem for dependent-coordinate rows.

**Problem 10.6** (Frame-design optimization)**.** Among iid continuous real row ensembles at $N = 2m-1$ with isotropy $\mathbb{E}aa^T = I_m$, bounded row-norm moments, and uniformly bounded one-dimensional densities, determine which laws minimize or maximize the Balan–Wang exponential base. The problem should specify whether rows are normalized to the sphere or allowed to have exponential radial extremes; subexponential radial variation is covered by Theorem 7.36 and does not change the base.

**Problem 10.7** (Complex stability base)**.** At the generic $4m-4$ complex phase-retrieval threshold, determine the exponential base for the complex Balan–Wang stability constant

$$a(\Phi) = \inf_{x \not\sim e^{i\theta}y} \frac{\||Ax| - |Ay|\|}{\operatorname{dist}(x, \mathbb{T}y)} = \min_{S=S^*,\ \operatorname{rank} S \le 2,\ S\neq 0} \frac{\|(\phi_i^* S \phi_i)_{i=1}^N\|}{\|S\|},$$

the second equality because $xx^* - yy^*$ is Hermitian of rank $\le 2$. The rank-$\le 2$ Hermitian cone has real dimension $2(2m-2) = 4m-4$, matching the threshold $N$, so $S \mapsto (\phi_i^* S \phi_i)$ is a *square* sensing map and $a(\Phi)$ is its extremal hard edge — the exact analogue of the real case (cone dimension $2m-1 = N$). Two ingredients differ from the real frame. *Small-ball (here proved):* for the signature-$(1,1)$ perturbations $S = xx^* - yy^*$, $\phi_i^* S \phi_i = \lambda_1 |\langle \phi_i, e_1\rangle|^2 - \lambda_2 |\langle \phi_i, e_2\rangle|^2$ is a difference of two independent exponentials with strictly positive density at the origin, so $\mathbb{P}\{|\phi_i^* S \phi_i| \le t\} \asymp t$ with exponent 1 — *not* the naive complex value $d_{\mathbb{C}} = 2$, since an indefinite quadratic form crosses zero transversally. *Entropy:* the realified configuration count is $\frac{1}{m}\log \binom{4m-4}{2m-2} \to \log 16$. The principle of Section 2 then predicts a finite *candidate base*

$$a(\Phi) = c^{-m+o(m)} \qquad \text{(conjectured, } c \text{ finite)},$$

with $c = 16$ (the square of the real base $4^{-m}$) suggested by the realification count $\binom{4m-4}{2m-2}$; the exact value is open, since the two-direction stratification below counts $\binom{N}{k}^2$ and admits $c$ up to 256. The lower side is partially settled. The constant reduces *exactly* to a Grassmannian least-singular-value statistic,

$$a(\Phi) = \min_{U \in \mathrm{Gr}_{\mathbb{C}}(2,m)} \sigma_{\min}\big(M(U)\big),$$

where $M(U)$ is the $(4m-4) \times 4$ real matrix of the evaluation $H \mapsto (w_i^* H w_i)_i$, $w_i = U^* \phi_i$, over $2 \times 2$ Hermitian $H$; and a first-moment/net argument over this Grassmannian (real dimension $4m-8$), fed the exponent-1 small-ball, proves the sub-exponential bound

$$a(\Phi) \ \ge\ m^{-(2+o(1))m} \qquad \text{with high probability.}$$

The gap to a finite base is the net's Lipschitz cost $e^{\Theta(m \log m)}$: closing it needs a Segre/bilinear least-singular-value small-ball (constant exponent, subexponential constant) for $\Re(\bar{p}_i q_i)$, which is irreducible (no exact finite combinatorial family as in the real case). That single estimate is the

missing RMT input, and a net analysis shows it is unavoidable: at the critical count every split of the $N$-dimensional sensing into netted and exactly-treated coordinates leaves union-deficit one, while the eigenvectors enter $|\langle\phi_i, u\rangle|^2$ nonlinearly, so $\Theta(m)$ directions must be netted and the Lipschitz cost is paid $\Theta(m)$ times. The exponential decay is moreover a *critical-point* phenomenon: at $N = (4+\eta)m$, $\eta > 0$, the same argument gives only the polynomial bound $a(\Phi) \geq m^{-2/\eta}$, so genuine exponential decay of the stability constant occurs precisely at the injectivity threshold $N = 4m - 4$.

**Problem 10.8** (Proportional rectangular / sample-covariance edge)**.** For $\ell/m \to \zeta \in (1, \gamma)$, define

$$M_{m,\ell} = \min_{|T|=\ell} \sigma_{\min}(A_T)/\sqrt{m}.$$

Equivalently,

$$\min_{|T|=\ell} \lambda_{\min}\big(\ell^{-1} A_T^* A_T\big) = \frac{m}{\ell} M_{m,\ell}^2.$$

Prove convergence of $M_{m,\ell}$, and identify its limit through the left-tail large-deviation rate $I^-_{\zeta,d_{\mathbb{F}}}$ for the smallest rectangular Wishart eigenvalue:

$$I^-_{\zeta,d_{\mathbb{F}}}(x_{\gamma,\zeta}) = \gamma \log \gamma - \zeta \log \zeta - (\gamma - \zeta) \log(\gamma - \zeta).$$

The fixed-excess formula $-h(\gamma)/(d_{\mathbb{F}}(r+1))$ does not apply in this regime.

**Problem 10.9** (Trimmed projection concentration)**.** For iid isotropic rows $a_i$, develop sharp concentration and large deviations for

$$\inf_{\|x\|=1} \sum_{i=1}^{\lfloor N/2 \rfloor} \big(\langle a_i, x\rangle^2\big)_{(i)},$$

where the order statistics are increasing. This statistic is the fixed-split analogue of the worst half-frame conditioning problem in phase retrieval.

**Problem 10.10** (Smoothed bad frames)**.** Let $F_m \in \mathbb{R}^{N_m \times m}$, $N_m/m \to \gamma > 1$, be a deterministic frame with normalized row norms and poor extremal row-submatrix conditioning. Put $A_m = F_m + \sigma_m G_m$, where $G_m$ is Gaussian. Determine the threshold scale of $\sigma_m$ at which

$$\min_{|T|=m} \sigma_{\min}((A_m)_T)$$

recovers the Gaussian exponential scale. The answer should depend on the worst deterministic row-submatrix deficiency of $F_m$.

**Problem 10.11** (Exponential radial extremes)**.** Theorem 7.36 shows that spherical row directions with subexponential global radial variation have the same base as the Gaussian row-submatrix model, up to the deterministic radial scale. Determine the sharp base when the sampled radii themselves have exponential extremes: for instance, when the empirical lower tail of

$$m^{-1} \log(R_{im}/b_m)$$

has a nontrivial large-deviation rate among the $N_m \asymp m$ available rows. The answer should combine the hard-edge entropy $h(\gamma)$ with a weighted selection problem for the row radii, rather than only the worst global diagonal-sandwich bound.

## Appendix A. Proof of the two-projection anti-concentration lemma

We give the complete proof of Lemma 3.20. Throughout $\mathbb{F} = \mathbb{R}$ (the complex case is identical with $d_{\mathbb{F}} = 2$, replacing each slab by a disk and $\sqrt{1-\varrho^2}$ by $1 - |\varrho|^2$). Recall $x \in \mathbb{R}^m$ has i.i.d. entries with mean 0, variance 1, density $f \leq K$; $u, v$ are deterministic unit vectors with $\|u\|_\infty, \|v\|_\infty \leq \beta := m^{-1/2+\delta}$, $\varrho = \langle u, v\rangle$, $|\varrho| \leq 1 - \kappa_0$; and $S_1 = \langle x, u\rangle$, $S_2 = \langle x, v\rangle$, with $p(t) = \mathbb{P}\{|S_1| \leq t\}$, $q(t) = \mathbb{P}\{|S_1| \leq t, |S_2| \leq t\}$.

Let $\psi(\theta) = \mathbb{E}e^{i\theta x_1}$ denote the (common) entry characteristic function.

**Lemma A.1** (Characteristic-function bound)**.** *There are constants $\eta > 0$ and $C_\psi < \infty$, depending only on the entry law, with*

$$|\psi(\theta)| \le e^{-\eta \min(\theta^2,1)} \quad (\theta \in \mathbb{R}), \qquad \|\psi\|^2_{L^2(\mathbb{R})} = 2\pi \int f^2 \le 2\pi K =: C_\psi.$$

*Proof.* Let $x_1'$ be an independent copy and $Y = x_1 - x_1'$, so $\mathbb{E}Y^2 = 2$ and $|\psi(\theta)|^2 = \mathbb{E}\cos(\theta Y)$. From $1 - \cos s \ge \frac{1}{4}s^2$ for $|s| \le 1$, choosing $\theta_0 \in (0,1]$ with $\mathbb{E}[Y^2 \mathbf{1}_{|Y|>1/\theta_0}] \le 1$ gives, for $|\theta| \le \theta_0$, $1 - |\psi(\theta)|^2 \ge \frac{1}{4}\theta^2 \mathbb{E}[Y^2 \mathbf{1}_{|\theta Y|\le 1}] \ge \frac{1}{4}\theta^2$, so $|\psi(\theta)| \le e^{-\theta^2/8}$ (only finite variance is used). Since $f$ is a density, $x_1$ is non-lattice, $|\psi| < 1$ off 0, and $\psi \to 0$ at infinity (Riemann–Lebesgue), so $\kappa := \sup_{|\theta|\ge\theta_0} |\psi| < 1$. Taking $\eta = \min\{\frac{1}{8}, \log(1/\kappa)\}$ gives the pointwise bound on the three ranges $|\theta| \le \theta_0$, $\theta_0 \le |\theta| \le 1$, $|\theta| \ge 1$. Plancherel gives $\|\psi\|_2^2 = 2\pi \int f^2 \le 2\pi K$. □

Set $w_j = (u_j, v_j) \in \mathbb{R}^2$ and $c_j(\xi) = \langle \xi, w_j \rangle$ for $\xi \in \mathbb{R}^2$, so that $\xi_1 S_1 + \xi_2 S_2 = \sum_j c_j(\xi) x_j$ and, since $\|u\| = \|v\| = 1$, $\langle u, v \rangle = \varrho$,

$$\sum_j w_j w_j^\top = \Sigma := \begin{pmatrix} 1 & \varrho \\ \varrho & 1 \end{pmatrix}, \quad \sum_j c_j(\xi)^2 = \xi^\top \Sigma \xi \ge \kappa_0 |\xi|^2, \quad |c_j(\xi)| \le \sqrt{2}\,\beta |\xi|, \tag{14}$$

using $\lambda_{\min}(\Sigma) = 1 - |\varrho| \ge \kappa_0$ and $\det \Sigma = 1 - \varrho^2 \ge \kappa_0$. The joint characteristic function factorizes as $\phi_m(\xi) = \prod_j \psi(c_j(\xi))$, and the target Gaussian has $\widehat{\Gamma_\Sigma}(\xi) = e^{-\frac{1}{2}\xi^\top \Sigma \xi}$.

**Lemma A.2** (Transverse pairs and integrability)**.** *For $m$ large there are two disjoint index pairs $\{p_i, q_i\}$ $(i = 1, 2)$ with $|\det[w_{p_i} \,|\, w_{q_i}]| \ge \Delta := \sqrt{\kappa_0}/m$. Consequently $\phi_m \in L^1(\mathbb{R}^2)$ with $\int_{\mathbb{R}^2} |\phi_m| \le \|\psi\|_2^4/\Delta \le m\, C_\psi^2/\sqrt{\kappa_0}$.*

*Proof.* With $W = [w_1 \,|\, \cdots \,|\, w_m]$, Cauchy–Binet gives $\sum_{p<q} \det[w_p|w_q]^2 = \det(WW^\top) = \det \Sigma \ge \kappa_0$; as there are $< m^2/2$ pairs, some pair has $|\det| \ge \sqrt{2\kappa_0}/m$. Removing it changes $\Sigma$ by $\le 4\beta^2$ in operator norm, leaving determinant $\ge \kappa_0/2$, so a second disjoint pair exists with $|\det| \ge \sqrt{\kappa_0}/m$. Bounding the other $m-4$ factors of $|\phi_m|$ by 1, Cauchy–Schwarz over the two pairs and the linear change of variables $\xi \mapsto (c_{p_i}, c_{q_i})$ (Jacobian $\ge \Delta$) with Lemma A.1 give $\int |\phi_m| \le \prod_{i=1,2} (\int |\psi(c_{p_i})\psi(c_{q_i})|^2)^{1/2} \le \|\psi\|_2^4/\Delta$. □

**Lemma A.3** (Energy bound)**.** $F(\xi) := \sum_j \min(c_j(\xi)^2, 1) \ge \frac{\kappa_0}{4} \min(|\xi|^2, \beta^{-2})$*, hence by Lemma A.1,* $|\phi_m(\xi)| \le \exp\{-\frac{\eta\kappa_0}{4} \min(|\xi|^2, \beta^{-2})\}$.

*Proof.* If $|\xi| \le (\sqrt{2}\beta)^{-1}$ then all $|c_j| \le 1$ and $F = \sum c_j^2 \ge \kappa_0 |\xi|^2$. If $|\xi| > (\sqrt{2}\beta)^{-1}$, split into $\mathcal{B} = \{|c_j| > 1\}$ and its complement. If $|\mathcal{B}| \ge \frac{\kappa_0}{4}\beta^{-2}$ then $F \ge |\mathcal{B}| \ge \frac{\kappa_0}{4}\beta^{-2}$; otherwise $\sum_{j\notin\mathcal{B}} c_j^2 \ge \xi^\top \Sigma \xi - |\mathcal{B}|\, 2\beta^2 |\xi|^2 \ge \frac{\kappa_0}{2}|\xi|^2 > \frac{\kappa_0}{4}\beta^{-2}$. In all cases $F \ge \frac{\kappa_0}{4} \min(|\xi|^2, \beta^{-2})$. □

**Proposition A.4** (Local CLT)**.** $(S_1, S_2)$ *has a bounded continuous density $g_m$, and $\varepsilon_m := \sup_w |g_m(w) - \Gamma_\Sigma(w)| \le (2\pi)^{-2} \int_{\mathbb{R}^2} |\phi_m - \widehat{\Gamma_\Sigma}| \to 0$, uniformly over delocalized unit $u, v$ with $|\varrho| \le 1 - \kappa_0$.*

*Proof.* $\phi_m \in L^1$ (Lemma A.2) and $\widehat{\Gamma_\Sigma} \in L^1$ ($\det \Sigma \ge \kappa_0$) justify Fourier inversion and the displayed bound. Fix $\varepsilon > 0$ and split $\int |\phi_m - \widehat{\Gamma_\Sigma}|$ over $|\xi| \le R$, $R < |\xi| \le \beta^{-1}$, $|\xi| > \beta^{-1}$. Choose $R = R(\kappa_0, \varepsilon)$ with $\int_{|\xi|>R} e^{-\kappa_0|\xi|^2/2} \le \varepsilon$. *(I)* $|\xi| \le R$*:* since $\max_j |c_j| \le \sqrt{2}\beta R \to 0$, the expansion $\psi(\theta) = 1 - \frac{1}{2}\theta^2 + \theta^2 \omega(|\theta|)$ $(\omega \to 0)$ gives $\log \phi_m(\xi) = -\frac{1}{2}\xi^\top \Sigma \xi + E_m(\xi)$ with $|E_m| \le \omega^*(\sqrt{2}\beta R)\, \xi^\top \Sigma \xi \le 2R^2 \omega^*(\sqrt{2}\beta R) \to 0$; hence the integrand $\to 0$ uniformly on the ball and term I $\to 0$. *(II)* $R < |\xi| \le \beta^{-1}$*:* here $\min(|\xi|^2, \beta^{-2}) = |\xi|^2$, so by Lemma A.3 the integrand is $\le e^{-\frac{\eta\kappa_0}{4}|\xi|^2} + e^{-\frac{\kappa_0}{2}|\xi|^2}$, with integral $\le \varepsilon$ for $R$ large. *(III)* $|\xi| > \beta^{-1}$*:* the residual $m-4$ indices have $\lambda_{\min} \ge \kappa_0/2$, so Lemma A.3 gives $\sum_{j\notin Q} \min(c_j^2, 1) \ge \frac{\kappa_0}{8}\beta^{-2}$, and with Lemma A.2, $\int_{|\xi|>\beta^{-1}} |\phi_m| \le e^{-\frac{\eta\kappa_0}{8} m^{1-2\delta}} m C_\psi^2/\sqrt{\kappa_0} \to 0$; the Gaussian far tail $\to 0$ likewise. Thus $\limsup_m \int |\phi_m - \widehat{\Gamma_\Sigma}| \le 2\varepsilon$. □

*Proof of Lemma 3.20.* The one-dimensional specialization of Proposition A.4 (single form $S_1$, coefficients $u_j$, integrability from the two largest coefficients $u_{(1)}^2 \ge 1/m$, $u_{(2)}^2 \ge 1/(2m)$) gives $\varepsilon'_m := \sup_w |g_{1,m}(w) - \Gamma_1(w)| \to 0$, where $\Gamma_1$ is the standard Gaussian density and $g_{1,m}$ the density of $S_1$. Write $V_t = 2t$. Since the Gaussian densities peak at the origin, with $\Gamma_\Sigma(0) = \Gamma_1(0)^2/\sqrt{1-\varrho^2}$,

$$q(t) \le (\Gamma_\Sigma(0) + \varepsilon_m)V_t^2, \qquad p(t) \ge \big(\Gamma_1(0)(1 - \tfrac{1}{2}t^2) - \varepsilon'_m\big)V_t.$$

As $|\varrho| \le 1 - \kappa_0$ keeps $\Gamma_\Sigma(0) \le \Gamma_1(0)^2\kappa_0^{-1/2}$ bounded, both $\varepsilon_m/\Gamma_\Sigma(0) \to 0$ and $\varepsilon'_m/\Gamma_1(0) \to 0$ uniformly, so for $t \to 0$

$$\frac{q(t)}{p(t)^2} \le \frac{(\Gamma_\Sigma(0) + \varepsilon_m)V_t^2}{(\Gamma_1(0)(1 - \frac{1}{2}t^2) - \varepsilon'_m)^2 V_t^2} = \frac{\Gamma_\Sigma(0)}{\Gamma_1(0)^2}(1 + o(1)) = \frac{1 + o(1)}{\sqrt{1-\varrho^2}}.$$

For $\mathbb{F} = \mathbb{C}$ the argument is the same with the following substitutions, all routine. Coefficients $c_j(\xi) \in \mathbb{C}$ ($\xi \in \mathbb{C}^2$); the entry characteristic function is $\psi(\theta) = \mathbb{E}e^{i\Re(\bar\theta x_1)}$, and Lemma A.1 holds verbatim from $\mathbb{E}x_1 = 0$, $\mathbb{E}|x_1|^2 = 1$ (properness $\mathbb{E}x_1^2 = 0$ makes the quadratic term $\frac{1}{2}|\theta|^2$ isotropic) and the bounded $\mathbb{R}^2$-density, with $\|\psi\|^2_{L^2(\mathbb{R}^2)} = (2\pi)^2 \int f^2 \le (2\pi)^2 K$. The frame lives in $\mathbb{C}^2 \simeq \mathbb{R}^4$: $w_j = (u_j, v_j)$, $\sum_j w_j w_j^* = \Sigma$, and Cauchy–Binet (now over the $4 \times 2m$ real form of $W$) again yields two transverse pairs with $|\det| \gtrsim m^{-1}$, so $\phi_m \in L^1(\mathbb{C}^2)$. The energy bound and three-region inversion are unchanged, giving $\|g_m - \Gamma_\Sigma\|_\infty \to 0$ for the proper complex Gaussian $\Gamma_\Sigma$ on $\mathbb{C}^2$ with $\Gamma_\Sigma(0) = \Gamma_1(0)^2/(1 - |\varrho|^2)$. Converting with disks ($V_t = \pi t^2$, $d_\mathbb{F} = 2$) gives $q(t) \le (1 + o(1))p(t)^2/(1 - |\varrho|^2)$. □

## Appendix B. Reduction of the pair-correlation condition to one decorrelation estimate

Throughout, $A_m \in \mathbb{F}^{N_m \times m}$ has iid entries that are mean zero, unit variance, with a symmetric (real) / circular (complex) law of bounded Lebesgue density $\le K$ and sub-Gaussian. Fix the distinguished row $x = a_{N_m}$. For an $(m-1)$-set $S$ let $u_S$ be a unit normal to $\operatorname{span}\{a_i : i \in S\}$; for $S \ne T$ with $|S \cap T| = m - D$ write $\varrho = \langle u_S, u_T\rangle$ for their overlap and, conditioning on the shared rows, work in the $D$-dimensional orthocomplement $E^\perp$ of their span, where $u_S, u_T$ are the unit null vectors of the $D-1$ projected disjoint rows. Put $X = 1 - |\varrho|^2 \in (0,1]$ and $a = d_\mathbb{F}/2$. The pair-correlation condition of Section 3 requires $\operatorname{Var} Z_m(s)/(\mathbb{E}Z_m(s))^2 \to 0$, which (after Lemma 3.20 and Proposition 3.21) is the bulk overlap moment $\mathbb{E}[X^{-a};\, D \ge \varepsilon m] \le 1 + o(1)$. We reduce this to a single decorrelation estimate.

**Lemma B.1** (Polar-cap lower tail)**.** *There are $C, c > 0$ depending only on $(K, \mathbb{F})$ such that, for the null vector $u$ of $D-1$ iid isotropic sub-Gaussian bounded-density vectors in $\mathbb{F}^D$ and any fixed unit $e \in \mathbb{F}^D$,*

$$\mathbb{P}\{\,|\langle e, u\rangle|^2 \ge 1 - y\,\} \le (C\sqrt{y}\,)^{\,d_\mathbb{F}(D-1)} + 2e^{-cD}, \qquad 0 < y < 1,$$

*Proof (real case; complex replaces $D-1$ by $2(D-1)$).* Let $G = [g_1 \,|\, \cdots \,|\, g_{D-1}]$ and $A = G^*$, so $u$ is the unit right-null vector of $A$, and $\|A\| = \|G\|$. For iid isotropic sub-Gaussian columns, the operator-norm bound [16] gives $\mathbb{P}\{\|A\| > R\sqrt{D}\} \le 2e^{-c_1 R^2 D}$ for $R \ge C_1(K)$. The polar cap $\mathrm{Cap} = \{v \in S^{D-1} : |\langle e, v\rangle| \ge \sqrt{1-y}\}$ has transverse radius $\sin\theta = \sqrt{y}$, so it admits a $\delta$-net $\mathcal{N}$ with $|\mathcal{N}| \le (C_0\sqrt{y}/\delta)^{D-1}$. If $u \in \mathrm{Cap}$ and $\|A\| \le R\sqrt{D}$, choose $z \in \mathcal{N}$ with $\|z - u\| \le \delta$; since $Au = 0$, $\|Az\| = \|A(z-u)\| \le R\sqrt{D}\,\delta$. Hence

$$\mathbb{P}\{u \in \mathrm{Cap}\} \le 2e^{-c_1 R^2 D} + \sum_{z \in \mathcal{N}} \mathbb{P}\{\|Az\| \le R\sqrt{D}\,\delta\}.$$

For fixed unit $z$, $\|Az\|^2 = \sum_{i=1}^{D-1} |\langle g_i, z\rangle|^2$ is a sum of $D-1$ iid nonnegative variables with unit mean and marginal small-ball $\le Kr$ (bounded density); the tensorized Laplace bound [14] gives

$\mathbb{P}\{\|Az\|^2 \le R^2 D\delta^2\} \le (C_2 K R\delta)^{D-1}$. Therefore

$$\mathbb{P}\{u \in \mathrm{Cap}\} \le 2e^{-c_1 R^2 D} + (C_0\sqrt{y}/\delta)^{D-1}(C_2 K R\delta)^{D-1} = 2e^{-c_1 R^2 D} + (C_0 C_2 K R\sqrt{y}\,)^{D-1};$$

the $\delta$ cancels — the cap's smallness $\sqrt{y}$ is exactly the per-coordinate small-ball gain — so no $\sqrt{D}$ factor survives. Take $R = C_1$. □

**Proposition B.2** (Assembly: the bar is $\mathbb{E}|\varrho|^2 = o(1)$)**.** *Let $\mu_1 = \mathbb{E}|\varrho|^2 = \mathbb{E}[1-X]$. Using Lemma B.1 (tuned $R = y^{-1/4}$ for the singular part),*

$$\mathbb{E}[X^{-a}] \le 1 + O(\sqrt{\mu_1}) + o(1),$$

*where $o(1) \to 0$ as $D \to \infty$ uniformly. In particular $\mathbb{E}[X^{-a}] = 1 + o(1)$ if and only if $\mu_1 = o(1)$.*

*Proof.* Split $\mathbb{E}[X^{-a}] = \mathbb{E}[X^{-a}; X \le y_1] + \mathbb{E}[X^{-a}; y_1 < X \le \frac12] + \mathbb{E}[X^{-a}; X > \frac12]$ with $y_1$ a small constant. (i) By Lemma B.1, $\mathbb{E}[X^{-a}; X \le y_1] \le \sum_{k \ge k_1} 2^{(k+1)a}\mathbb{P}\{X \le 2^{-k}\} = o(1)$, independent of $\mu_1$ (the geometric/tuned-tail series converges). (ii) Markov on $1 - X$: $\mathbb{P}\{X \le \frac12\} \le 2\mu_1$, so $\mathbb{E}[X^{-a}; y_1 < X \le \frac12] \le y_1^{-a} \cdot 2\mu_1 = O(\mu_1)$. (iii) On $X > \frac12$ write $X^{-a} = 1 + (X^{-a} - 1)$; splitting at $1 - \eta$, the part $X > 1 - \eta$ has $X^{-a} - 1 \le C_a\eta$, and the part $\frac12 < X \le 1 - \eta$ has $X^{-a} - 1 \le 2^a$ with $\mathbb{P}\{X \le 1 - \eta\} \le \mu_1/\eta$, so $\mathbb{E}[X^{-a}; X > \frac12] \le 1 + C_a\eta + 2^a\mu_1/\eta$. Optimizing $\eta = \sqrt{\mu_1}$ gives $1 + O(\sqrt{\mu_1})$. Summing the three parts gives the claim; necessity follows from $\mathbb{E}[X^{-a}] \ge \mathbb{E}[1 + aX^{-1}(1 - X)] \ge 1 + a\mu_1$ via $(1 - t)^{-a} \ge 1 + at$. □

**Proposition B.3** (Exact decorrelation for disjoint sets)**.** *For every $(m-1)$-set $S$, $\mathbb{E}[u_S u_S^*] = \frac1m I_m$. Consequently, for disjoint $S, T$, $\mathbb{E}|\varrho|^2 = \frac1m$.*

*Proof.* The iid entry law is invariant under coordinate permutations $\pi$ and under sign flips (real) / coordinate phase rotations (complex); the normal direction is equivariant, $u_S \mapsto P_\pi u_S$ resp. $D_\phi u_S$, under the corresponding action on all rows. Hence the projector $\Pi_S = u_S u_S^*$ has a law invariant under conjugation by the signed-permutation (resp. phase-permutation) group, so $\mathbb{E}[\Pi_S]$ commutes with that group; the only such matrices are scalars, and $\operatorname{tr}\mathbb{E}[\Pi_S] = 1$ forces $\mathbb{E}[\Pi_S] = \frac1m I_m$. For disjoint $S, T$ the normals $u_S, u_T$ are independent, so $\mathbb{E}|\varrho|^2 = \mathbb{E}[\,u_T^*\mathbb{E}[u_S u_S^*]\,u_T] = \frac1m\mathbb{E}\|u_T\|^2 = \frac1m$. □

**Theorem B.4** (Reduction of the pair-correlation condition)**.** *For the iid bounded-density sub-Gaussian ensemble above, the pair-correlation condition — hence the universality upper side of Theorem 3.18 — holds provided*

$$\mathbb{E}\big[|\langle u_S, u_T\rangle|^2 \;\big|\; D(S,T)\big] = o(1) \qquad (D(S,T) \ge \varepsilon m), \tag{15}$$

*equivalently the delocalization in expectation $\sup_{\|w\|=1}\mathbb{E}[|\langle u, w\rangle|^2 \mid E] = o(1)$ of the null vector $u$ in $E^\perp$. The estimate (15) is exact ($= 1/m$) for disjoint sets (Proposition B.3), and is the only remaining input: Lemmas B.1–B.2 are unconditional.*

*Proof.* By Lemma 3.20 and Proposition 3.21 the second-moment ratio reduces to $\mathbb{E}[X^{-a}; D \ge \varepsilon m]$. Proposition B.2 bounds this by $1 + O(\sqrt{\mu_1}) + o(1)$ with $\mu_1 = \mathbb{E}[|\varrho|^2; D \ge \varepsilon m]$, which is $o(1)$ under (15) (and $\mathbb{E}|\varrho|^2 \mid D = \mathbb{E}[\,u_T^*(\mathbb{E}[u_S u_S^* \mid E])u_T\,] \le \sup_w \mathbb{E}[|\langle u, w\rangle|^2 \mid E]$). Thus the ratio $\to 1$ and the pair-correlation condition holds. □

The residual (15) is met by a single, standard application of the local law, through a resolvent inequality that sidesteps the hard edge.

**Proposition B.5** (Resolvent reduction; conditional closure of (15))**.** *Conditionally on $E$ the projected rows are isotropic, $\mathbb{E}[g_i g_i^* \mid E] = I_{E^\perp}$; let $Q = \frac{1}{D-1}GG^*$ on $E^\perp \cong \mathbb{F}^D$, the normalized sample-covariance matrix of $D - 1$ isotropic vectors (aspect ratio $\to 1$), of which $u$ is the null eigenvector. For every $\eta > 0$,*

$$uu^* \;\preceq\; \eta\,(Q + \eta)^{-1}, \qquad \textit{hence} \qquad \lambda_{\max}\big(\mathbb{E}[uu^* \mid E]\big) \le \eta\,\big\|\mathbb{E}[(Q + \eta)^{-1} \mid E]\big\|,$$

*since $uu^*$ and $\eta(Q+\eta)^{-1}$ are simultaneously diagonal in the eigenbasis of $Q$ and their difference is the positive-semidefinite* $\operatorname{diag}(0, \frac{\eta}{\lambda_2+\eta}, \dots)$. *Conditionally on $E$, $(D-1)Q$ equals the compression $P_{E^\perp}(AA^*)P_{E^\perp}$ of an iid-entry sample covariance, realized through the separable model to which the anisotropic local law [10] (isotropic [5] in the Gaussian sub-case) applies with population $P_{E^\perp}$; the rank-deficient kernel decouples, and off support the resolvent is $\eta$-Lipschitz in the population so the projection is admissible. Its direction-uniform form at the off-support point $z=-\eta$, together with the deterministic bound $\|(Q+\eta)^{-1}\| \le \eta^{-1}$ on the local law's exceptional event (of probability $o(\eta^{1/2})$), gives the operator-norm-in-expectation estimate $\|\mathbb{E}[(Q+\eta)^{-1} \mid E]\| = (1+o(1))\, m(-\eta)$, where $m$ is the Stieltjes transform of the Marčenko–Pastur law at aspect one; as $m(-\eta) = (1+o(1))\eta^{-1/2}$ for $\eta \to 0$,*

$$\lambda_{\max}\big(\mathbb{E}[uu^* \mid E]\big) \le (1+o(1))\sqrt{\eta} \xrightarrow[\eta = 1/\log D]{} 0.$$

*This is (15). The hard edge is never resolved: the resolvent is evaluated off the spectrum at distance $\eta \gg D^{-1}$, the robust regime of the local law.*

*Remark* B.6 (The off-support bound is self-contained). The estimate $\|\mathbb{E}[(Q+\eta)^{-1} \mid E]\| \le (1+o(1))m(-\eta)$ does not require the anisotropic local law as a black box; in the off-support regime $\eta = 1/\log D \gg \lambda_{\min}(Q) \sim D^{-2}$ it is elementary, and Lemma B.7 gives a complete self-contained proof (the local law an optional alternative). The mechanism is that $\lambda_{\max}(R) \approx \eta^{-1}$ (a spike on the random null vector $u$) while $\mathbb{E}[R]$ is near-scalar, so $\lambda_{\max}(\mathbb{E}[R]) = (1+o(1))m(-\eta) = (1+o(1))\eta^{-1/2}$ — confirmed distribution-free in simulation (Rademacher entries, $D = 40$–$320$: $\lambda_{\max}(\mathbb{E}R)/m(-\eta) \approx 1.03$, against $\eta^{-1}/m(-\eta) \approx 2.5$–$3$).

**Lemma B.7** (Off-support resolvent law)**.** *Let $g_i = B^*a_i$ $(i = 1, \dots, n = D-1)$ be the projected rows, $a_i \in \mathbb{F}^N$ having independent mean-zero unit-variance sub-Gaussian entries with $\max_k \|a_{1,k}\|_{\psi_2} \le K$; put $Q = \frac{1}{n}\sum_i g_i g_i^*$, $\eta = 1/\log D$, $R = (Q+\eta)^{-1}$, $m_D = \frac{1}{D}\operatorname{tr} R$, and let $m(-\eta) = \frac{\sqrt{\eta^2+4\eta}-\eta}{2\eta} \sim \eta^{-1/2}$ be the aspect-one Marčenko–Pastur transform. Then there is $C = C(K)$ with*

$$\sup_{\|w\|=1} \big|\, \mathbb{E}[\langle w, Rw\rangle] - m(-\eta) \,\big| \le C\, \frac{(\log D)^{7/4}}{\sqrt{D}}\, m(-\eta),$$

*so $\|\mathbb{E}[R]\| = \lambda_{\max}(\mathbb{E}[R]) = (1+o(1))m(-\eta)$.*

*Proof. Leave-one-out.* With $Q^{(i)} = \frac{1}{n}\sum_{j\neq i} g_j g_j^*$, $R^{(i)} = (Q^{(i)}+\eta)^{-1}$ and $\beta_i = (1+\frac{1}{n}g_i^* R^{(i)} g_i)^{-1} \in (0,1]$, Sherman–Morrison gives $Rg_i = \beta_i R^{(i)} g_i$, and from $\eta R = I - \frac{1}{n}\sum_i g_i g_i^* R$ the exact identity $\eta R = I - \frac{1}{n}\sum_i \beta_i g_i g_i^* R^{(i)}$. Deterministically $0 \preceq R, R^{(i)} \preceq \eta^{-1} I$ and $\|R^{(i)}\|_F \le \sqrt{D}\,\eta^{-1}$; applying $\frac{1}{D}\operatorname{tr}$ gives $\eta m_D = 1 - \frac{n}{D} + \frac{1}{D}\sum_i \beta_i$.

*Quadratic forms.* By Hanson–Wright in variance form [13], $\operatorname{Var}(a^*Ma) \le b_{\mathbb{F}}\|M\|_F^2$ with $b_{\mathbb{F}} \le CK^4$ for independent-entry $a$, Hermitian $M$. Since $B^*B = I_D$, $g_i^* M g_i = a_i^*(BMB^*)a_i$ with $\|BMB^*\|_F = \|M\|_F$, so this transfers to $g_i$ with no loss — the one place the projected-iid structure is used, a bare isotropic $g_i$ need not have independent coordinates. Thus $s_i := \frac{1}{n} g_i^* R^{(i)} g_i$ has $\mathbb{E}[s_i \mid R^{(i)}] = t_i := \frac{1}{n}\operatorname{tr} R^{(i)}$ and $\operatorname{Var}(s_i \mid R^{(i)}) \le 2b_{\mathbb{F}}\eta^{-2}/D$.

*Trace law.* Rank-one interlacing gives the deterministic $0 \le \operatorname{tr} R^{(j)} - \operatorname{tr} R \le \eta^{-1}$, so $\operatorname{Var}(m_D) \le \eta^{-2}/(2D)$ by Efron–Stein. Let $\bar m$ solve $\eta\bar m^2 + \eta\bar m - 1 = 0$ (so $\bar m = m(-\eta)$), i.e. $(\eta+\bar\beta)\bar m = 1$, $\bar\beta = (1+\bar m)^{-1} \asymp \eta^{1/2}$. With $\Phi(x) = 1 - \frac{n}{D} + \frac{n}{D}(1+\phi x)^{-1} - \eta x$ $(\phi = D/n)$, Taylor-expanding $\beta_i$ around $t_i$ (via $\operatorname{Var}(s_i \mid R^{(i)})$) and $t_i$ around $\phi\bar m$ gives $|\Phi(\mathbb{E}m_D)| \le Cb_{\mathbb{F}}\eta^{-2}/D$; as $\Phi' \le -\eta$, the mean value theorem yields $|\mathbb{E}m_D - \bar m| \le Cb_{\mathbb{F}}\eta^{-3}/D = O((\log D)^{5/2}/D)\bar m$. (Stability uses $\inf|\Phi'| \ge \eta$, not contraction of the iteration, whose derivative is $\asymp 1$.)

*Direction-uniform law.* Fix unit $w$, $T_w = \mathbb{E}[w^*Rw] \le \eta^{-1}$, and $\beta_i' = (1+t_i)^{-1}$ (a function of $R^{(i)}$ only). Testing the identity against $w$ and taking $\mathbb{E}$, $\eta T_w = 1 - \frac{1}{n}\sum_i \mathbb{E}[\beta_i (w^*g_i)(g_i^* R^{(i)} w)]$. Split $\beta_i =$

$\beta_i' + (\beta_i - \beta_i')$: for the main term, exact isotropy $\mathbb{E}[g_i g_i^* \mid \{g_j\}_{j\neq i}] = I$ gives $\mathbb{E}[\beta_i'(w^* g_i)(g_i^* R^{(i)} w)] = \mathbb{E}[\beta_i' w^* R^{(i)} w]$, removing the $w$-dependence, with $\beta_i' \to \bar\beta$ costing $O(\eta^{-3/2} T_w^{1/2}/\sqrt{D})$; the cross term is, by Cauchy–Schwarz, $\le (\mathbb{E}|\beta_i - \beta_i'|^2)^{1/2} (\mathbb{E}|w^* g_i|^2 |g_i^* R^{(i)} w|^2)^{1/2} \le CK^2 \sqrt{b_{\mathbb{F}}}\, \eta^{-3/2} T_w^{1/2}/\sqrt{D}$ (using $\mathrm{Var}(s_i \mid R^{(i)})$ and the transferred sub-Gaussian fourth moment). With the removal bound $T_w \le T_w^{(i)} \le T_w(1 + O(\frac{\log D}{D}))$, $(\eta + \bar\beta) T_w = 1 + \mathcal{E}_w$, $|\mathcal{E}_w| \le C(\frac{\log D}{D} \bar\beta T_w + K^2 \sqrt{b_{\mathbb{F}}} \eta^{-3/2} T_w^{1/2}/\sqrt{D})$. The a priori $T_w \le \eta^{-1}$ makes $\mathcal{E}_w = o(1)$, so $T_w = (1 + o(1))\bar m \asymp \eta^{-1/2}$; re-inserting sharpens $|\mathcal{E}_w| \le CK^4 (\log D)^{7/4}/\sqrt{D}$ uniformly in $w$. With $\bar m = m(-\eta)(1 + O(\sqrt{\log D}/D))$ this is the claim, and $\sup_w T_w = \lambda_{\max}(\mathbb{E}R)$. □

**Corollary B.8.** *By Proposition B.5, whose off-support resolvent estimate is supplied either self-contained (Remark B.6) or by citing [5, 10], the non-Gaussian universality upper side of Theorem 3.18 holds for iid bounded-density sub-Gaussian entries. The same estimate supplies the second-moment (pair-correlation) input to the fluctuation analysis of Section 10, hence the one-sided Gumbel bound; the full Gumbel law of Conjecture 10.2 requires in addition the higher k-point correlations and is not implied by the pair-correlation alone.*

*Remark* B.9. Estimate (15) is a no-gaps delocalization *in expectation* for the null vector of projections of iid rows. Isotropy of the projected rows pins only $\mathrm{tr}\,\mathbb{E}[uu^* \mid E] = 1$, so a soft central-limit argument does not establish the spectral bound directly; this is why the resolvent route of Proposition B.5 — which trades the spectral bound for an off-support resolvent norm controlled by the local law — is the natural one. The estimate holds with the sharp rate $(1 + o(1))/D$ in simulation, distribution-free. We state the clean local-law input separately as Problem 10.1 for readers who wish to avoid invoking [5, 10].

## References


[1] Radu Balan, Pete Casazza, and Dan Edidin. On signal reconstruction without phase. *Applied and Computational Harmonic Analysis*, 20(3):345–356, 2006.

[2] Radu Balan and Yang Wang. Invertibility and robustness of phaseless reconstruction. *Applied and Computational Harmonic Analysis*, 38(3):469–488, 2015.

[3] Afonso S. Bandeira. Injectivity and stability of phase retrieval (problems 19–21). Randomstrasse 101, 2025.

[4] Afonso S. Bandeira, Jameson Cahill, Dustin G. Mixon, and Aaron A. Nelson. Saving phase: Injectivity and stability for phase retrieval. *Applied and Computational Harmonic Analysis*, 37(1):106–125, 2014.

[5] A. Bloemendal, L. Erdős, A. Knowles, H.-T. Yau, and J. Yin. Isotropic local laws for sample covariance and generalized Wigner matrices. *Electron. J. Probab.*, 19(33):1–53, 2014.

[6] Silouanos Brazitikos, Apostolos Giannopoulos, Petros Valettas, and Beatrice-Helen Vritsiou. *Geometry of Isotropic Convex Bodies*, volume 196 of *Mathematical Surveys and Monographs*. American Mathematical Society, Providence, RI, 2014.

[7] Aldo Conca, Dan Edidin, Milena Hering, and Cynthia Vinzant. An algebraic characterization of injectivity in phase retrieval. *Applied and Computational Harmonic Analysis*, 38(2):346–356, 2015.

[8] Alan Edelman. Eigenvalues and condition numbers of random matrices. *SIAM Journal on Matrix Analysis and Applications*, 9(4):543–560, 1988.

[9] Peter J. Forrester. *Log-Gases and Random Matrices*. Princeton University Press, 2010.

[10] A. Knowles and J. Yin. Anisotropic local laws for random matrices. *Probab. Theory Related Fields*, 169(1–2):257–352, 2017.

[11] Vladimir A. Marchenko and Leonid A. Pastur. Distribution of eigenvalues for some sets of random matrices. *Matematicheskii Sbornik*, 72(4):507–536, 1967.

[12] V. V. Petrov. *Sums of Independent Random Variables*. Springer-Verlag, 1975.

[13] M. Rudelson and R. Vershynin. Hanson–wright inequality and sub-gaussian concentration. *Electron. Commun. Probab.*, 18(82):1–9, 2013.

[14] Mark Rudelson and Roman Vershynin. The littlewood–offord problem and invertibility of random matrices. *Advances in Mathematics*, 218(2):600–633, 2008.

[15] Yitzchak Shmalo. Extreme least singular values of gaussian row submatrices and a phase retrieval stability problem. ResearchGate preprint, 2026.

[16] R. Vershynin. *High-Dimensional Probability: An Introduction with Applications in Data Science*. Cambridge University Press, 2018.

[17] Fuzhen Zhang. Quaternions and matrices of quaternions. *Linear Algebra and its Applications*, 251:21–57, 1997.

Einstein Institute of Mathematics, The Hebrew University of Jerusalem, Givat Ram, Jerusalem, Israel

*Email address*: yitzchak.shmalo@gmail.com